

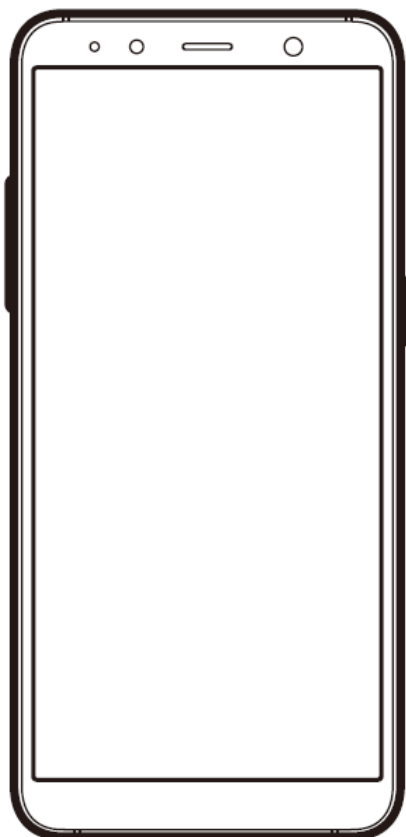
SAMSUNG

Mobile Device SM-A530D

SERVICE ***Manual***

Mobile Device

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1. Safety Precautions

1-1. Repair Precaution

Before attempting any repair or detailed tuning, shield the device from RF noise or static electricity discharges.

Use only demagnetized tools that are specifically designed for small electronic repairs, as most electronic parts are sensitive to electromagnetic forces.

Use only high quality screwdrivers when servicing products. Low quality screwdrivers can easily damage the heads of screws.

Use only conductor wire of the properly gauge and insulation for low resistance, because of the low margin of error of most testing equipment.

We recommend 22-gauge twisted copper wire.

Hand-soldering is not recommended, because printed circuit boards (PCBs) can be easily damaged, even with relatively low heat. Never use a soldering iron with a power rating of more than 100 watts and use only lead-free solder with a melting point below 250°C (482°F).

Prior to disassembling the battery charger for repair, ensure that the AC power is disconnected.

Always use the replacement parts that are registered in the SEC system. Third-party replacement parts may not function properly.

1. Safety Precautions

1-2. ESD(Electrostatically Sensitive Devices) Precaution

Many semiconductors and ESDs in electronic devices are particularly sensitive to static discharge and can be easily damaged by it. We recommend protecting these components with conductive anti-static bags when you store or transport them.

Always use an anti-static strap or wristband and remove electrostatic buildup or dissipate static electricity from your body before repairing ESDs.

Ensure that soldering irons have AC adapter with ground wires and that the ground wires are properly connected.

Use only desoldering tools with plastic tips to prevent static discharge.

Properly shield the work environment from accidental electrostatic discharge before opening packages containing ESDs.

The potential for static electricity discharge may be increased in low humidity environments, such as air-conditioned rooms. Increase the airflow to the working area to decrease the chance of accidental static electricity discharges.

2. Specification

2-1. GSM General Specification

Item		GSM 850	EGSM 900	DCS1800	PCS1900
Freq. Band[MHz] Uplink/Downlink		824~849	880~915	1710~1785	1850~1910
		869~894	925~960	1805~1880	1930~1990
ARFCN range		128~251	0~124 & 975~1023	512~885	512~810
Tx/Rx spacing		45MHz	45MHz	95MHz	80MHz
Mod. Bit rate/ Bit Period		270.833kbps	270.833kbps	270.833kbps	270.833kbps
		3.692us	3.692us	3.692us	3.692us
Time Slot Period/ Frame Period		576.9us	576.9us	576.9us	576.9us
		4.615ms	4.615ms	4.615ms	4.615ms
Modulation	GSM/ EGPRS	GMSK/ 8PSK	GMSK/ 8PSK	GMSK/ 8PSK	GMSK/ 8PSK
MS Power		33dBm~5dBm	33dBm~5dBm	30dBm~0dBm	30dBm~0dBm
Power Class		4(GMSK) E2(8PSK)	4(GMSK) E2(8PSK)	1(GMSK) E2(8PSK)	1(GMSK) E2(8PSK)
Sensitivity		-102dBm	-102dBm	-100dBm	-100dBm
TDMA Mux		8	8	8	8

2. Specification

2-2. GSM Tx Power Class

TX Power control level	GSM850	TX Power control level	EGSM900	TX Power control level	DCS1800	TX Power control level	PCS1900
5	33±2 dBm	5	33±2 dBm	0	30±3 dBm	0	30±3 dBm
6	31±2 dBm	6	31±2 dBm	1	28±3 dBm	1	28±3 dBm
7	29±2 dBm	7	29±2 dBm	2	26±3 dBm	2	26±3 dBm
8	27±2 dBm	8	27±2 dBm	3	24±3 dBm	3	24±3 dBm
9	25±2 dBm	9	25±2 dBm	4	22±3 dBm	4	22±3 dBm
10	23±2 dBm	10	23±2 dBm	5	20±3 dBm	5	20±3 dBm
11	21±2 dBm	11	21±2 dBm	6	18±3 dBm	6	18±3 dBm
12	19±2 dBm	12	19±2 dBm	7	16±3 dBm	7	16±3 dBm
13	17±2 dBm	13	17±2 dBm	8	14±3 dBm	8	14±3 dBm
14	15±2 dBm	14	15±2 dBm	9	12±4 dBm	9	12±4 dBm
15	13±2 dBm	15	13±2 dBm	10	10±4 dBm	10	10±4 dBm
16	11±3 dBm	16	11±3 dBm	11	8±4 dBm	11	8±4 dBm
17	9±3 dBm	17	9±3 dBm	12	6±4 dBm	12	6±4 dBm
18	7±3 dBm	18	7±3 dBm	13	4±4 dBm	13	4±4 dBm
19	5±3 dBm	19	5±3 dBm	14	2±5 dBm	14	2±5 dBm
-	-	-	-	15	0±5 dBm	15	0±5 dBm

2. Specification

2-3. WCDMA General Specification

Item	WCDMA2100(B1)	WCDMA850(B5)	WCDMA800(B6)	WCDMANew800(B19)
Freq. Band[MHz] Uplink/Downlink	1920~1980 2110~2170	824~849 869~894	830~840 875~885	830~845 875~890
ARFCN range	UL: 9612~9888 DL: 10562~10838	UL: 4132~4233 DL: 4357~4458	UL: 4162~4188 DL: 4387~4413	UL: 312~363 DL: 712~763
Tx/Rx spacing	190MHz	45MHz	45MHz	45MHz
Mod. Bit rate/ Bit Period	14.4Mbps(DL) 5.76Mbps(UL)	14.4Mbps(DL) 5.76Mbps(UL)	14.4Mbps(DL) 5.76Mbps(UL)	14.4Mbps(DL) 5.76Mbps(UL)
Time Slot Period/ Frame Period	WCDMA 10ms/0.667ms HSPA 2ms/0.667ms	WCDMA 10ms/0.667ms HSPA 2ms/0.667ms	WCDMA 10ms/0.667ms HSPA 2ms/0.667ms	WCDMA 10ms/0.667ms HSPA 2ms/0.667ms
Modulation	QPSK 16QAM 64QAM	QPSK 16QAM 64QAM	QPSK 16QAM 64QAM	QPSK 16QAM 64QAM
MS Power (dBm)	25.7 ~ -49(↓)	25.7 ~ -49(↓)	25.7 ~ -49(↓)	25.7 ~ -49(↓)
Power Class	3(max+24dBm)	3(max+24dBm)	3(max+24dBm)	3(max+24dBm)
Sensitivity	-106.7dBm	-104.7dBm	-106.7dBm	-106.7dBm

2. Specification

2-4. LTE General Specification

Item	LTE Band1	LTE Band3	LTE Band5	LTE Band12	LTE Band17
Freq. Band[MHz] Uplink/Downlink	1920~1980 2110~2170	1710~1785 1805~1880	824~849 869~894	699~716 729~746	704~716 734~746
ARFCN range	UL:18000~18599 DL:0~599	UL:19200~19949 DL:1200~1949	UL:20400~20649 DL:2400~2649	UL:23010~23179 DL:5010~5179	UL:23730~23849 DL:5730~5849
Tx/Rx spacing (MHz)	190	95	45	30	30
Channel Bandwidth (MHz)	5/10/15/20	1.4/3/5/10/15/20	1.4/3/5/10	1.4/3/5/10	5/10
Modulation	QPSK,16/64QAM 256QAM(DL only)	QPSK,16/64QAM 256QAM(DL only)	QPSK,16/64QAM 256QAM(DL only)	QPSK,16/64QAM 256QAM(DL only)	QPSK,16/64QAM 256QAM(DL only)
MS Power (dBm)	25.7~-39(↓)	25.7~-39(↓)	25.7~-39(↓)	25.7~-39(↓)	25.7~-39(↓)
Sensitivity (QPSK, BW 10MHz) (dBm)	-96.3	-93.3	-94.3	-93.3	-93.3

2. Specification

Item	LTE Band19	LTE Band21
Freq. Band[MHz] Uplink/Downlink	830~845 875~890	1447.9~1462.9 1495.9~1510.9
ARFCN range	UL:24000~24149 DL:6000~6149	UL:24450~24599 DL:6450~6599
Tx/Rx spacing (MHz)	45	45
Channel Bandwidth (MHz)	5/10/15	5/10/15
Modulation	QPSK,16/64QAM 256QAM(DL only)	QPSK,16/64QAM 256QAM(DL only)
MS Power (dBm)	25.7~-39(↓)	25.7~-39(↓)
Sensitivity (QPSK, BW 10MHz) (dBm)	-96.3	-96.3

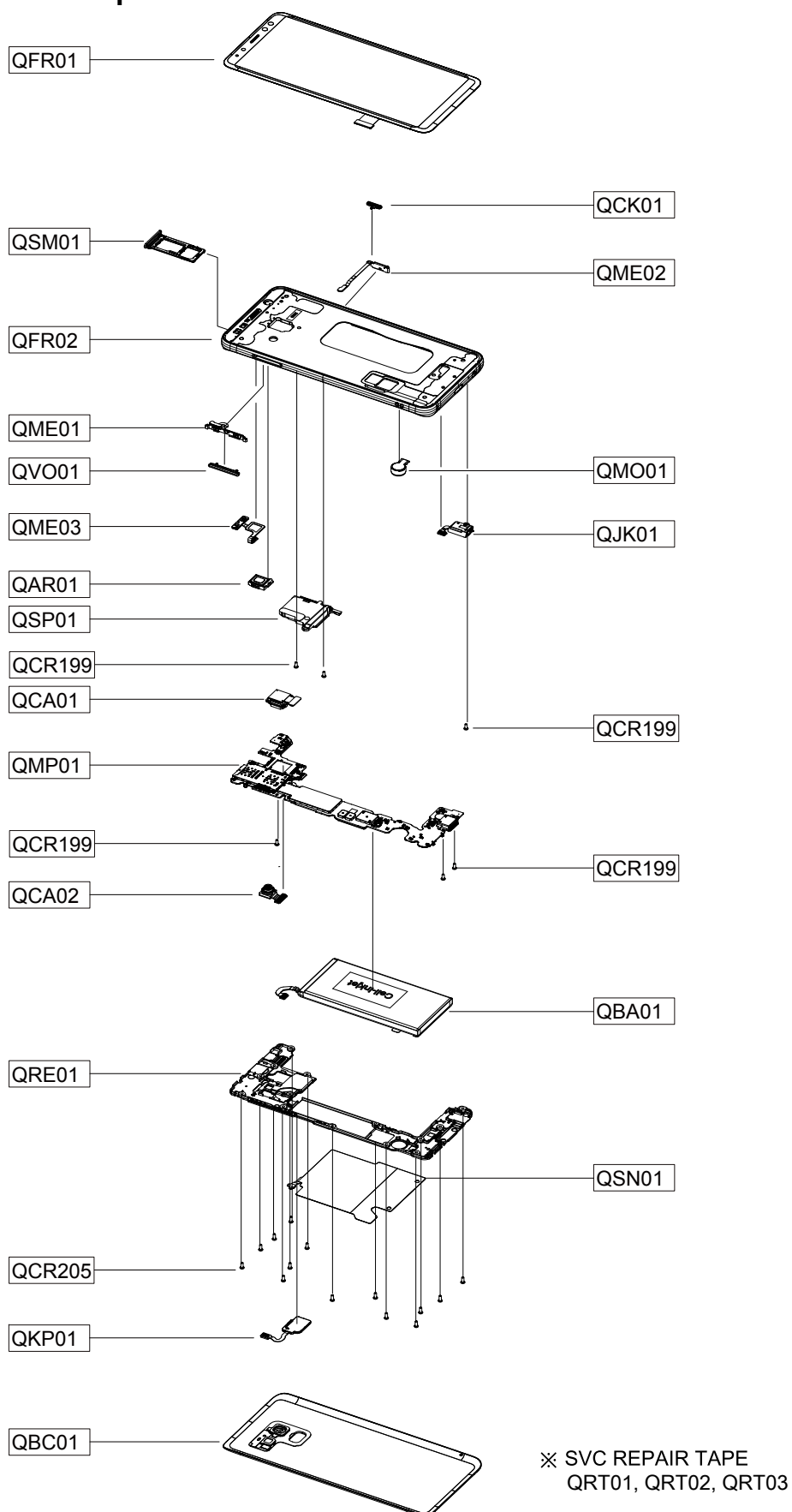
3. Product Function

Main Function

Item	Description
OS	Android V8.1
RF	GSM850 / GSM900 / DCS1800 / PCS1900 WCDMA: B1/ B5/ B6 / B19 LTE: B1/ B3/ B5/ B12/ B17/ B19/ B21
Battery	3000mAh
Base Band	2.2Ghz Quad + 1.6GHz Quad
Other RF	A-GPS, Glonass, BEIDOU, BT5.0, USB 2.0 Type-C, WIFI 802.11 a/b/g/n/ac MIMO, NFC
Camera	Front Camera (16M, F1.9), REAR Camera (16MP A/F, F1.7)
LCD	5.6", HD+, OCTA, 1480x720
RAM / ROM	4GB / 32GB
Sensor	Accelerometer, Barometer, Fingerprint Sensor, Gyro Sensor, Geomagnetic Sensor, Hall Sensor, Proximity Sensor, RGB Light Sensor
Accessory	Data cable: 2.75pi, 1.5m(USB Type B) Gender : (A type -> C type) Ear phone Ant : 13cm

4. Exploded View and Parts List

4-1. Cellular phone Exploded View



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5. MAIN Electrical Parts List

Parts Code	Design LOC	Description
0404-001250	D5000	DIODE
0406-001561	ZD6000,ZD6003,ZD6008	DIODE
0406-001561	ZD6013,ZD6015,ZD6019	DIODE
0406-001561	ZD6026	DIODE
0406-001592	ZD5013	DIODE
0406-001675	ZD5002	DIODE
0406-001728	ZD1000,ZD5000	DIODE
0406-001743	ZD6007,ZD6014,ZD6021	DIODE
0406-001743	ZD6022,ZD6023,ZD6024	DIODE
0406-001743	ZD6025	DIODE
0406-001780	ZD5007,ZD5008	DIODE
0406-001781	ZD6016,ZD6017,ZD6018	DIODE
0406-001789	ZD6006	DIODE
0406-001797	ZD5011,ZD5012	DIODE
0406-001808	D5001	DIODE
0505-003618	Q5000	FET-SILICON
0601-003667	LED7014	LED
0903-002171	U3004	IC
1001-001911	U1017,U2004	IC
1001-001970	U1022	IC
1001-001975	U3003	IC
1001-002042	U1002,U1009	IC
1001-002050	U3033	IC
1003-002802	U2008	IC
1009-001070	U3011,U3018	IC
1107-002552	UME5000	IC
1201-003786	U3006	IC
1201-003952	PAM1002	IC
1201-003976	U1001	IC
1201-004036	U3014	IC
1201-004129	PAM1000	IC
1201-004156	U6002	IC
1202-001120	U6004	IC
1203-007884	U7012	IC
1203-008249	U3002,U7017	IC
1203-008250	U3020	IC
1203-008326	U3021	IC
1203-008475	U5004	IC

5. MAIN Electrical Parts List

1203-008693	U1020	IC
1203-008719	U3031	IC
1203-008751	U7022	IC
1203-008867	U3000,U6003,U7016	IC
1203-008870	U7007,U7009	IC
1203-008872	U7015	IC
1203-008904	U5002	IC
1203-008948	U6000	IC
1203-008955	U7005,U7008	IC
1203-008989	U7006	IC
1203-009087	U5001	IC
1204-003629	U3019	IC
1204-003685	U6001	IC
1205-005791	U3032	IC
1205-005859	U2010	IC
1205-005860	U3005	IC
1205-005870	U400	IC
1209-002428	U3017	IC
1209-002450	U3016	IC
1209-002513	U3013	IC
1404-001664	TH5000	THERMISTOR
1404-001724	TH4000,TH4001,TH5001	THERMISTOR
1404-001734	TH2000	THERMISTOR
1405-001395	VR1005,VR1009,VR1011	VARISTOR
1405-001395	VR1014,VR3002,VR3003	VARISTOR
1405-001395	VR3005,VR6006,VR6008	VARISTOR
1405-001395	VR6009,VR6012,VR6013	VARISTOR
1405-001395	VR6014,VR6016	VARISTOR
1405-001403	VR1006,VR1007,VR1008	VARISTOR
1405-001403	VR1013,VR2002,VR3000	VARISTOR
1405-001403	VR6007,VR6010	VARISTOR
1405-001404	VR6005	VARISTOR
1405-001412	VR1015	VARISTOR
1405-001415	VR1010,VR1012,VR2000	VARISTOR
1405-001416	VR1001,VR1004	VARISTOR
1405-001431	VR2001	VARISTOR
2007-007142	R7006	R-CHIP
2007-007190	R6023,R6026	R-CHIP
2007-007314	R3015,R3023	R-CHIP

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5. MAIN Electrical Parts List

2007-007318	R3024,R5019	R-CHIP
2007-007741	R2001,R2002,R2006	R-CHIP
2007-007741	R3009,R5020,R5027	R-CHIP
2007-007741	R5028,R5029,R5030	R-CHIP
2007-007741	R5031,R5035,R5037	R-CHIP
2007-007741	R6022,R7005	R-CHIP
2007-007798	R5040,R5041	R-CHIP
2007-007942	R5024	R-CHIP
2007-007946	R5025	R-CHIP
2007-008043	R5044	R-CHIP
2007-008296	R6034	R-CHIP
2007-008531	R3011,R3012,R3013	R-CHIP
2007-008531	R5018,R6001,R7000	R-CHIP
2007-008531	R7001	R-CHIP
2007-008800	R6013	R-CHIP
2007-009111	R3010,R5036	R-CHIP
2007-009157	R3001,R3014,R4028	R-CHIP
2007-009157	R4042,R4043,R5015	R-CHIP
2007-009158	R5011	R-CHIP
2007-009171	R5008,R5034	R-CHIP
2007-009201	R5017,R6007,R6011	R-CHIP
2007-009201	R6012	R-CHIP
2007-009212	R4051,R5009,R5013	R-CHIP
2007-009212	R5021,R6004,R6008	R-CHIP
2007-009212	R6027	R-CHIP
2007-009314	R6003	R-CHIP
2007-009352	R3004,R3005,R4017	R-CHIP
2007-009352	R4023,R4024,R5014	R-CHIP
2007-009408	R3002,R3003,R3006	R-CHIP
2007-009408	R3007,R3008,R4025	R-CHIP
2007-009408	R4026,R4030,R4031	R-CHIP
2007-009408	R4032,R4033,R4034	R-CHIP
2007-009408	R4035,R4036,R4037	R-CHIP
2007-009408	R4047,R4049,R5010	R-CHIP
2007-009408	R6009,R7009	R-CHIP
2007-009410	R6017	R-CHIP
2007-009766	R3000,R6006	R-CHIP
2007-009793	R6016	R-CHIP
2007-009801	R5001	R-CHIP

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5. MAIN Electrical Parts List

2007-009805	R1000,R1001,R1002	R-CHIP
2007-009805	R1004,R2000	R-CHIP
2007-009807	R6031	R-CHIP
2007-009920	R4004,R4005,R4012	R-CHIP
2007-009920	R4013	R-CHIP
2007-009969	R6010	R-CHIP
2007-010202	R3017,R5032,R5033	R-CHIP
2007-010685	R6035	R-CHIP
2007-010690	R3026	R-CHIP
2007-010691	R3019	R-CHIP
2007-011043	R4003,R4006,R4007	R-CHIP
2007-011043	R4009,R4010	R-CHIP
2007-011532	R6018	R-CHIP
2007-011648	R6032	R-CHIP
2007-011991	R3028	R-CHIP
2007-012033	R6000	R-CHIP
2007-012038	R5012	R-CHIP
2007-012068	R5022	R-CHIP
2007-012202	R6033	R-CHIP
2007-012416	R3016	R-CHIP
2007-012417	R3027	R-CHIP
2203-000233	C3085,C3093	C-CERAMIC
2203-000386	C3088	C-CERAMIC
2203-000995	C3087	C-CERAMIC
2203-001239	C5134,C5135	C-CERAMIC
2203-002668	C3106	C-CERAMIC
2203-005682	C3040,C5108,C5109	C-CERAMIC
2203-005682	C5118,C5130,C5131	C-CERAMIC
2203-005682	C6036,C6039	C-CERAMIC
2203-005719	C3098	C-CERAMIC
2203-005725	C1050,C1102,C2058	C-CERAMIC
2203-005725	C3026	C-CERAMIC
2203-005726	C6041,C6043	C-CERAMIC
2203-005727	C3082,C3083	C-CERAMIC
2203-005729	C1134,C6000,C6001	C-CERAMIC
2203-005729	C6002,C6003,C7006	C-CERAMIC
2203-005729	C7009	C-CERAMIC
2203-005731	C2056,C6051,C6052	C-CERAMIC
2203-005732	C5124,C5138,C5140	C-CERAMIC

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5. MAIN Electrical Parts List

2203-005734	C1033,C6049,C6050	C-CERAMIC
2203-005734	C7036,C7043,C7058	C-CERAMIC
2203-005736	C1000,C1001,C1007	C-CERAMIC
2203-005736	C1025,C1026,C1027	C-CERAMIC
2203-005736	C1028,C1047,C1055	C-CERAMIC
2203-005736	C1056,C1061,C1062	C-CERAMIC
2203-005736	C1065,C1066,C1077	C-CERAMIC
2203-005736	C1083,C1094,C1103	C-CERAMIC
2203-005736	C1104,C1108,C1109	C-CERAMIC
2203-005736	C1110,C1119,C1129	C-CERAMIC
2203-005736	C1136,C1141,C1149	C-CERAMIC
2203-005736	C2000,C2012,C2013	C-CERAMIC
2203-005736	C2030,C2031,C2046	C-CERAMIC
2203-005736	C2050,C2066,C3001	C-CERAMIC
2203-005736	C3002,C3005,C3006	C-CERAMIC
2203-005736	C3012,C3014,C3016	C-CERAMIC
2203-005736	C3038,C3099,C3110	C-CERAMIC
2203-005736	L1044,L1047,L1054	C-CERAMIC
2203-005736	L1080	C-CERAMIC
2203-005739	C3049,C3050	C-CERAMIC
2203-005777	C2064	C-CERAMIC
2203-005789	C1105,C1111,C1127	C-CERAMIC
2203-005789	C1128	C-CERAMIC
2203-005792	C1046,C2049	C-CERAMIC
2203-005806	C1010,C2018,C2019	C-CERAMIC
2203-006048	C3068	C-CERAMIC
2203-006121	C6048	C-CERAMIC
2203-006123	C1012,C3056	C-CERAMIC
2203-006187	C1087,C1090,C3023	C-CERAMIC
2203-006187	L1083	C-CERAMIC
2203-006194	C2021,C2026,C2028	C-CERAMIC
2203-006194	C2033,C2038,C2040	C-CERAMIC
2203-006194	C2042,C2044,C2048	C-CERAMIC
2203-006194	C2061,C3025,C3030	C-CERAMIC
2203-006194	C3054,C3062,C6022	C-CERAMIC
2203-006194	C6054	C-CERAMIC
2203-006305	C1124	C-CERAMIC
2203-006318	C1022,C1113	C-CERAMIC
2203-006400	C1003,C1030,C2059	C-CERAMIC

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5. MAIN Electrical Parts List

2203-006400	C2065,C3029,C3041	C-CERAMIC
2203-006400	C3064,C3079,C3084	C-CERAMIC
2203-006400	C3100,C4009,C5136	C-CERAMIC
2203-006400	C5137,C5141,C5142	C-CERAMIC
2203-006400	C5143,C5144,C6055	C-CERAMIC
2203-006410	C1017	C-CERAMIC
2203-006462	C6067	C-CERAMIC
2203-006556	C1032,C5121,C6038	C-CERAMIC
2203-006556	C6045	C-CERAMIC
2203-006604	C3043	C-CERAMIC
2203-006648	C6044	C-CERAMIC
2203-006665	C1133	C-CERAMIC
2203-006668	C2053,C4099,C5115	C-CERAMIC
2203-006668	C5116	C-CERAMIC
2203-006674	C6012,C6014,L2040	C-CERAMIC
2203-006839	C1085,C1130,C1143	C-CERAMIC
2203-006839	C2008,C2045,C3008	C-CERAMIC
2203-006839	C3009,C3010,C3011	C-CERAMIC
2203-006839	C3022,C3061,C3065	C-CERAMIC
2203-006839	C3066,C3067,C3069	C-CERAMIC
2203-006839	C4010,C4030,C4036	C-CERAMIC
2203-006839	C4100,C5081,C5097	C-CERAMIC
2203-006839	C5111,C5112,C6004	C-CERAMIC
2203-006839	C6007,C6008,C6015	C-CERAMIC
2203-006839	C6017,C6040,C6042	C-CERAMIC
2203-006846	C4000,C4001,C4004	C-CERAMIC
2203-006846	C4005,C6011	C-CERAMIC
2203-006872	C5000,C5004,C5007	C-CERAMIC
2203-006872	C5021,C5022,C5023	C-CERAMIC
2203-006872	C5027,C5031,C5080	C-CERAMIC
2203-006872	C7004	C-CERAMIC
2203-006968	C1038,C1039,C1045	C-CERAMIC
2203-006968	L1024	C-CERAMIC
2203-006978	C6047	C-CERAMIC
2203-007194	C2068	C-CERAMIC
2203-007210	C4101,C4102,C4103	C-CERAMIC
2203-007210	C4104,C4105,C5084	C-CERAMIC
2203-007210	C5126,C5127	C-CERAMIC
2203-007317	C3021,C3046,C3070	C-CERAMIC

5. MAIN Electrical Parts List

2203-007317	C3073,C3074,C4006	C-CERAMIC
2203-007317	C4040,C4046,C4064	C-CERAMIC
2203-007317	C4083,C4086,C5032	C-CERAMIC
2203-007317	C5034,C5035,C5036	C-CERAMIC
2203-007317	C5125,C5128,C6006	C-CERAMIC
2203-007317	C6016	C-CERAMIC
2203-007391	C1142	C-CERAMIC
2203-007392	C5037,C5046,C5092	C-CERAMIC
2203-007393	C1096,C3060,C3115	C-CERAMIC
2203-007393	C3117,C3118,C3119	C-CERAMIC
2203-007393	C5044,C5050,C5053	C-CERAMIC
2203-007393	C5054,C5055,C5083	C-CERAMIC
2203-007393	C5095,C5101,C5102	C-CERAMIC
2203-007393	C5103,C5105,C5110	C-CERAMIC
2203-007393	C5117,C7010,C7046	C-CERAMIC
2203-007456	C5038,C5119	C-CERAMIC
2203-007781	C7038,C7053,C7054	C-CERAMIC
2203-007796	C1095,C2027,C2041	C-CERAMIC
2203-007796	C2047,C3013,C3017	C-CERAMIC
2203-007796	C3019,C3034,C3035	C-CERAMIC
2203-007796	C3059,C3071,C3075	C-CERAMIC
2203-007796	C3080,C3094,C3095	C-CERAMIC
2203-007796	C3096,C4007,C4013	C-CERAMIC
2203-007796	C4014,C4017,C4018	C-CERAMIC
2203-007796	C4019,C4020,C4021	C-CERAMIC
2203-007796	C4022,C4023,C4024	C-CERAMIC
2203-007796	C4025,C4026,C4027	C-CERAMIC
2203-007796	C4028,C4029,C4031	C-CERAMIC
2203-007796	C4032,C4033,C4034	C-CERAMIC
2203-007796	C4035,C4037,C4038	C-CERAMIC
2203-007796	C4041,C4042,C4043	C-CERAMIC
2203-007796	C4044,C4045,C4047	C-CERAMIC
2203-007796	C4048,C4059,C4060	C-CERAMIC
2203-007796	C4062,C4063,C4065	C-CERAMIC
2203-007796	C4066,C4067,C4068	C-CERAMIC
2203-007796	C4072,C4073,C4074	C-CERAMIC
2203-007796	C4075,C4076,C4077	C-CERAMIC
2203-007796	C4078,C4081,C4082	C-CERAMIC
2203-007796	C4088,C4089,C4090	C-CERAMIC

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5. MAIN Electrical Parts List

2203-007796	C4091,C4093,C4094	C-CERAMIC
2203-007796	C4095,C4096,C4097	C-CERAMIC
2203-007796	C4098,C5008,C5009	C-CERAMIC
2203-007796	C5017,C5018,C5019	C-CERAMIC
2203-007796	C5020,C5024,C5025	C-CERAMIC
2203-007796	C5026,C5029,C5030	C-CERAMIC
2203-007796	C5078,C5087,C5088	C-CERAMIC
2203-007796	C5096,C5106,C5107	C-CERAMIC
2203-007796	C5129,C6005,C6009	C-CERAMIC
2203-007796	C6019,C6020,C6023	C-CERAMIC
2203-007796	C6027,C6033,C6034	C-CERAMIC
2203-007796	C7000,C7002,C7003	C-CERAMIC
2203-007796	C7021,C7023,C7024	C-CERAMIC
2203-007796	C7034,C7047,C7051	C-CERAMIC
2203-008097	C1020,C1031,C1034	C-CERAMIC
2203-008097	C2035,C6030	C-CERAMIC
2203-008158	C5056,C5070	C-CERAMIC
2203-008242	C2036,C3000,C3003	C-CERAMIC
2203-008242	C3004,C3015,C4039	C-CERAMIC
2203-008242	C4084,C4087,C4092	C-CERAMIC
2203-008242	C5066,C5067,C5068	C-CERAMIC
2203-008242	C5069	C-CERAMIC
2203-008312	C4008,C4016,C4057	C-CERAMIC
2203-008312	C5040,C5042,C5047	C-CERAMIC
2203-008312	C5051,C5059,C5072	C-CERAMIC
2203-008312	C5073,C5076,C5079	C-CERAMIC
2203-008572	C5049,C5061,C5062	C-CERAMIC
2203-008654	C5123	C-CERAMIC
2203-008860	C2060,C3018,C3033	C-CERAMIC
2203-008860	C3072,C3076,C3089	C-CERAMIC
2203-008860	C5086,C5093,C5094	C-CERAMIC
2203-008860	C5098,C6029,C7014	C-CERAMIC
2203-008860	C7016,C7018,C7020	C-CERAMIC
2203-008860	C7022,C7033,C7035	C-CERAMIC
2203-008860	C7048,C7049	C-CERAMIC
2203-008876	C1021,C1140,C2054	C-CERAMIC
2203-008876	C3007,C3048,C5002	C-CERAMIC
2203-008876	C5012,C5041,C5063	C-CERAMIC
2203-008876	C5064,C5065,C5085	C-CERAMIC

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5. MAIN Electrical Parts List

2203-008876	C5091,C7025,C7026	C-CERAMIC
2203-008876	C7028,C7029,C7050	C-CERAMIC
2203-009167	C3086,C3090	C-CERAMIC
2203-009220	C3101	C-CERAMIC
2203-009537	C5003	C-CERAMIC
2203-009618	C5045,C6032	C-CERAMIC
2203-009733	C1035,C1093,C2020	C-CERAMIC
2203-009733	C2025,C2034,C2037	C-CERAMIC
2203-009733	C2039,C2043,C3045	C-CERAMIC
2203-009733	C3078,C3081,C3102	C-CERAMIC
2203-009733	C3107,C4049,C4052	C-CERAMIC
2203-009733	C4055,C5039,C5043	C-CERAMIC
2203-009733	C5048,C5052,C5060	C-CERAMIC
2203-009733	C5071,C5074,C5077	C-CERAMIC
2203-009733	C5082,C5090,C7001	C-CERAMIC
2203-009733	C7007,C7008,C7042	C-CERAMIC
2203-009858	C5058,C6025,C6026	C-CERAMIC
2203-009859	C7044,C7045	C-CERAMIC
2203-009969	C6031	C-CERAMIC
2703-002649	C3111,C6013,L1059	INDUCTOR
2703-002649	L1072,L2014,L6005	INDUCTOR
2703-002900	L2038,L3016	INDUCTOR
2703-002907	C1121,L1073	INDUCTOR
2703-002953	C1063,L1037,L2036	INDUCTOR
2703-002955	L3028	INDUCTOR
2703-002958	L1021,L3000	INDUCTOR
2703-002959	C1041,L1089,L2008	INDUCTOR
2703-003921	L3007	INDUCTOR
2703-003970	L1075	INDUCTOR
2703-004001	L1048	INDUCTOR
2703-004012	C2022,L1027,L1040	INDUCTOR
2703-004012	L2004	INDUCTOR
2703-004013	C1107	INDUCTOR
2703-004032	L1004,L1006,L1077	INDUCTOR
2703-004034	L1020,L1078,L2003	INDUCTOR
2703-004034	L2006,L3002,L3029	INDUCTOR
2703-004035	C1125,L1009,L1017	INDUCTOR
2703-004035	L1030	INDUCTOR
2703-004036	L1038,L1079,L3005	INDUCTOR

5. MAIN Electrical Parts List

2703-004038	L1025,L1064,L3006	INDUCTOR
2703-004259	L2010	INDUCTOR
2703-004286	L2029	INDUCTOR
2703-004287	L2028	INDUCTOR
2703-004302	L1015	INDUCTOR
2703-004317	L1081,L3030	INDUCTOR
2703-004328	C1005,L1042,L1063	INDUCTOR
2703-004328	L1082,L2034,L3031	INDUCTOR
2703-004362	L3015	INDUCTOR
2703-004363	L2016,L2030	INDUCTOR
2703-004368	L3004,L3027	INDUCTOR
2703-004703	L1074	INDUCTOR
2703-004763	L2013	INDUCTOR
2703-004976	L6004	INDUCTOR
2703-005058	C2063,L2037	INDUCTOR
2703-005066	L1070	INDUCTOR
2703-005067	L5001,L5006,L5007	INDUCTOR
2703-005067	L5009,L5010	INDUCTOR
2703-005085	L1014,L1046,L2023	INDUCTOR
2703-005088	L1031	INDUCTOR
2703-005089	C2024,L1043,L2001	INDUCTOR
2703-005089	L2035	INDUCTOR
2703-005098	L1036,L1045,L1084	INDUCTOR
2703-005104	L3025,L3026	INDUCTOR
2703-005117	L7006,L7009	INDUCTOR
2703-005118	L7011	INDUCTOR
2703-005201	L5002,L5004,L5005	INDUCTOR
2703-005233	R6024	INDUCTOR
2703-005295	L1010,L1029,L1085	INDUCTOR
2703-005295	L2015,L3012	INDUCTOR
2703-005296	C1060,C1122,L3034	INDUCTOR
2703-005296	L3035	INDUCTOR
2703-005505	L6006,L7004	INDUCTOR
2703-005581	L5003	INDUCTOR
2703-005660	L5008	INDUCTOR
2801-005393	OSC5000	CRYSTAL
2801-005476	OSC3001	OSCILLATOR
2802-001287	OSC3002	OSCILLATOR
2805-001118	TCX2000	OSCILLATOR

5. MAIN Electrical Parts List

2901-001635	F5000	FILTER
2901-001690	C4011,C4015,C4050	FILTER
2901-001690	C4054,C4056,C4058	FILTER
2901-001690	C4070	FILTER
2904-002010	F3004	FILTER
2904-002134	F1000	FILTER
2904-002139	F2002	FILTER
2904-002211	F2000	FILTER
2904-002224	F2001	FILTER
2904-002355	F3003	FILTER
2904-002381	F3000	FILTER
2910-000368	F1005	FILTER
2911-000430	U2006	FILTER
2911-000452	U1005	FILTER
3003-001210	MIC6000,MIC6001	MIC-CONDENSOR
3301-001885	L6011,L6014	CORE-FERRITE BEAD
3301-002078	L6007,L6009,L6012	CORE-FERRITE BEAD
3301-002078	L6013,L6015	CORE-FERRITE BEAD
3301-002085	L6001,L6002	CORE-FERRITE BEAD
3301-002122	L1068,L1069,L2025	CORE-FERRITE BEAD
3301-002122	L6000,L6003	CORE-FERRITE BEAD
3301-002223	L7000,L7002,L7003	CORE-FERRITE BEAD
3301-002236	L3019,L3020,L6008	CORE-FERRITE BEAD
3301-002236	L6010,L6017,L6018	CORE-FERRITE BEAD
3301-002237	L1000,L1067,L2031	CORE-FERRITE BEAD
3301-002237	L2032,L3009,L7001	CORE-FERRITE BEAD
3301-002238	L2026,L2027	CORE-FERRITE BEAD
3301-002239	L1001,L1060,L1071	CORE-FERRITE BEAD
3301-002243	L2017,L2018	CORE-FERRITE BEAD
3301-002286	L3023,L3024	CORE-FERRITE BEAD
3301-002287	L3018,L3033	CORE-FERRITE BEAD
3301-002312	L2019,L2020,L2021	CORE-FERRITE BEAD
3301-002312	L2024,L7007,L7008	CORE-FERRITE BEAD
3301-002312	L7010	CORE-FERRITE BEAD
3301-002331	L5000,L5011	CORE-FERRITE BEAD
3705-001708	RFS1000	CONNECTOR-CARD EDGE
3709-001892	SIM6000	CONNECTOR-HEADER
3711-007107	HDC7001	CONNECTOR-HEADER
3711-008182	HDC7002	CONNECTOR-HEADER

5. MAIN Electrical Parts List

3711-008508	HDC7000	CONNECTOR-HEADER
3711-008824	HDC3000,HDC3001	CONNECTOR-HEADER
3711-008847	HDC5000	CONNECTOR-HEADER
3711-008997	HDC6000	CONNECTOR-HEADER
3712-001604	ANT1006,ANT1012	CONNECTOR-HEADER
3712-001604	ANT3004,ANT3005	CONNECTOR-HEADER
3712-001604	ANT5000,ANT5001	CONNECTOR-HEADER
3712-001604	ANT6000,ANT6002	CONNECTOR-HEADER
3712-001621	ANT1009,ANT6001	CONNECTOR-HEADER
3712-001621	ANT6003	CONNECTOR-HEADER
3712-001626	ANT3001,ANT5005	CONNECTOR-HEADER
3712-001626	ANT5006	CONNECTOR-HEADER
3712-001634	ANT5002,ANT5003	CONNECTOR
3712-001634	ANT5004	CONNECTOR
3712-001666	ANT1003,ANT1005	CONNECTOR
3712-001666	ANT1007,ANT1008	CONNECTOR
3712-001666	ANT1010,ANT1011	CONNECTOR
3712-001666	ANT1013,ANT1014	CONNECTOR
3712-001666	ANT2002,ANT2004	CONNECTOR
3712-001666	ANT2005,ANT3002	CONNECTOR
3712-001731	ANT1001,ANT1002	CONNECTOR
3712-001731	ANT3000	CONNECTOR
3722-004110	IFC5000	CONNECTOR
4709-002284	F3001	RF-MODULE
4709-002360	F1001,F2003	RF-MODULE
4709-002496	F1003	RF-MODULE
4709-002589	CPL1000	RF-MODULE
GH61-12118A	SUS6000,SUS6001	sus
GH61-12118A	SUS6002	sus
GH61-12423A	SUS6003	sus
GH62-00042A	GA6000	Gasket
GH63-13321A	SC6001	Shield can
GH63-13700A	SC6005	Shield can
GH63-13704A	SC6004	Shield can
GH63-14237A	SC6006	Shield can
GH63-14521A	SC6002,SC6011	Shield can
GH63-14633A	SC6007	Shield can
GH63-14915A	SC6003	Shield can
GH63-16128A	SC6009	Shield can

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5. MAIN Electrical Parts List

GH63-16129A	SC6010	Shield can
1105-002942	U400UP	IC
GH02-16956A	SC6005SP	TAPE

Please consult the GSPN website (Samsung Portal) for the most recent version of the product's part list.

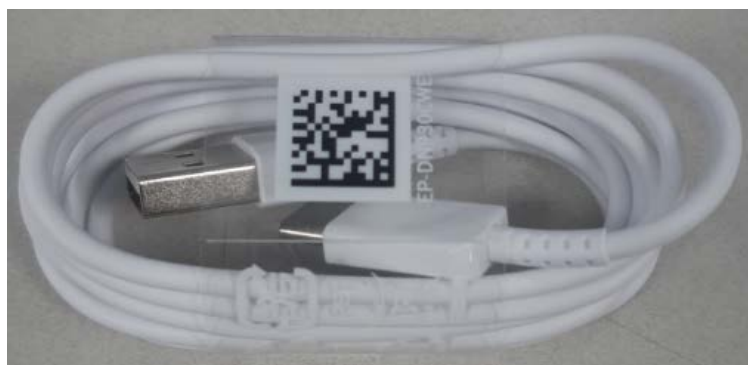
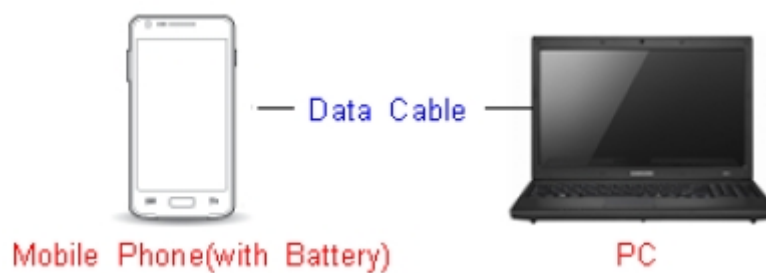
6. Level 1 Repair

6-1. S/W Update

6-1-1. Preparation

- S/W Update program : [Fenrir 5.17.xxxx](#)
- Mobile Phone
- Data Cable

※ Settings



Data Cable : [GH39-01886A](#)

6. Level 1 Repair

6-1-2. How to use 'Fenrir' S/W update program.

1) Launch Fenrir by clicking on the icon on the desktop



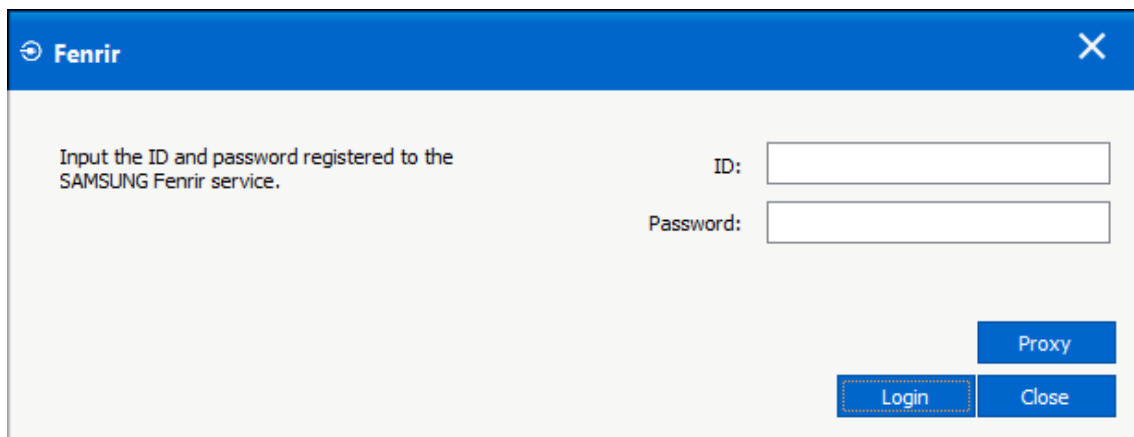
- SVH (Fenrir_Home) : It uses Home binary which does not have user data area in the memory when flashed to a device. (Keep user data)

- SVC (Fenrir_Factory) : It uses Factory binary which erases all user data in the memory when flashed to a device. (Clear user data)

- SVA (Fenrir_All) : It uses Factory and Home binaries. you can download Home and Factory binary in a PC(but requires double HDD storage and NW traffic)

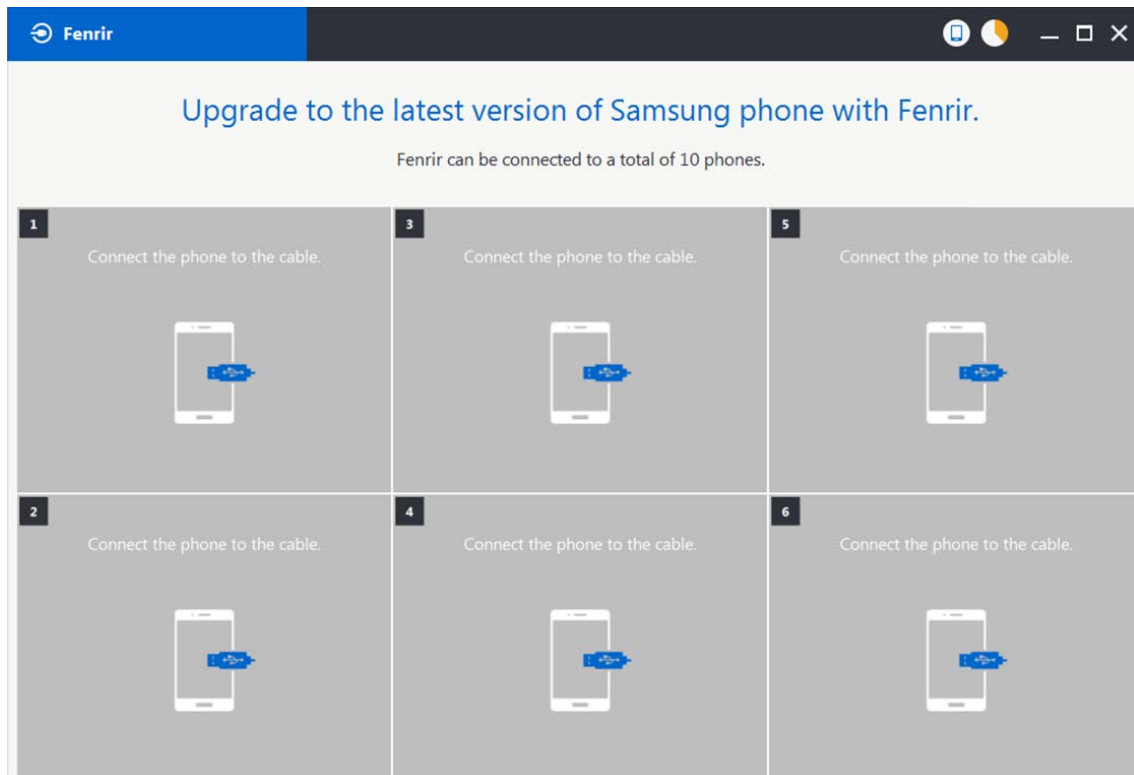
2) Input ID & password

※ You need to reset the ID information in case of PC change and format and repair, hard disk change

A screenshot of the Fenrir software login window. The window has a blue title bar with the 'Fenrir' logo and a close button. The main area is light gray. On the left, it says 'Input the ID and password registered to the SAMSUNG Fenrir service.' On the right, there are two input fields: 'ID:' and 'Password:'. Below the 'Password:' field, there are three buttons: 'Proxy' (blue), 'Login' (blue with a dashed border), and 'Close' (blue).

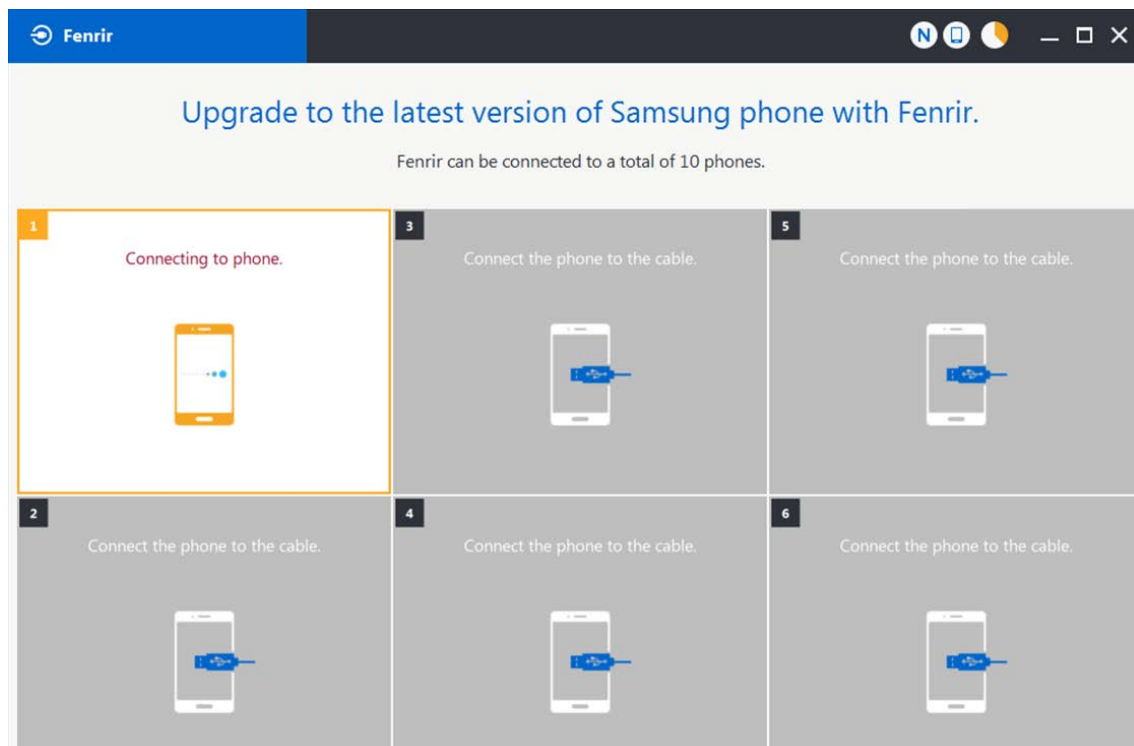
6. Level 1 Repair

3) Ensure device has sufficient charge (at least 20%) to start firmware update.



4) Connect the device to PC via data cable.

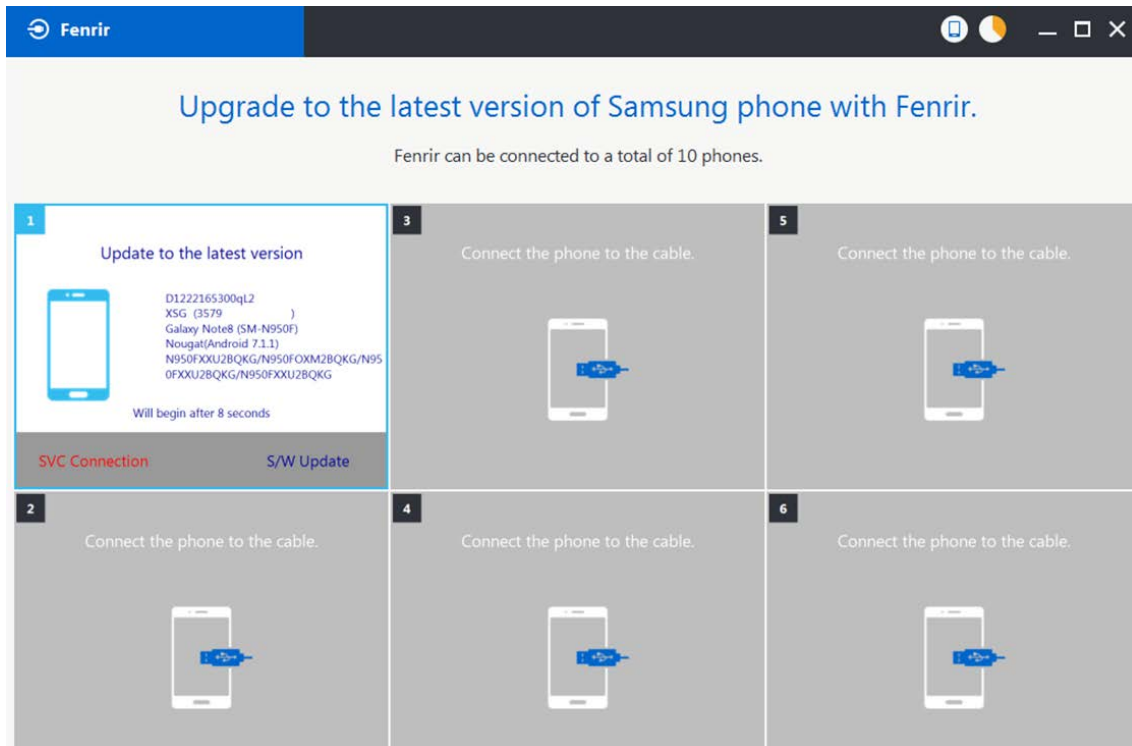
5) Upon USB connection, you will be presented with below screen.



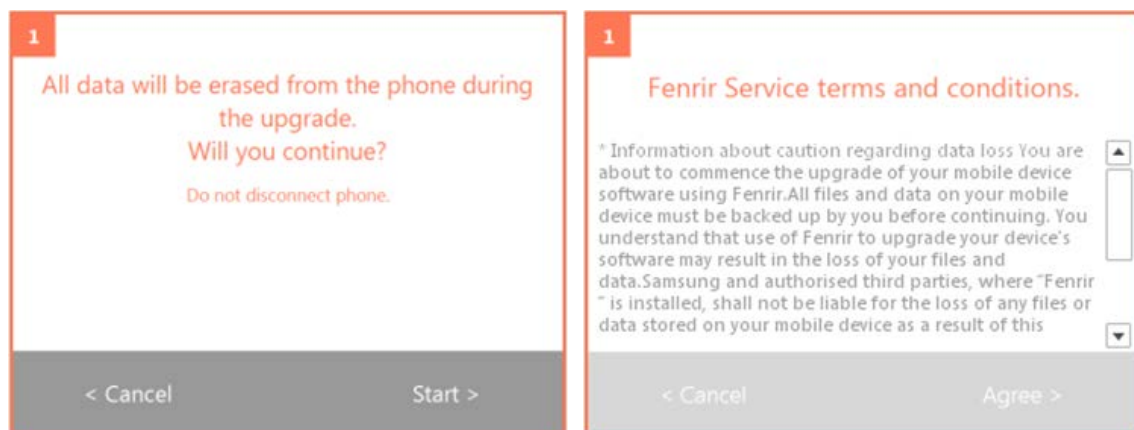
6. Level 1 Repair

=C

6) Once device is detected, you will be presented with below screen. To update S/W, select “S/W Update” or to exit select “SVC Connection”. If you select “SVC Connection”, only Fenrir connection history (record) will be stored in the FUS server to support warranty validation. (This is known as “Service Connection” history)

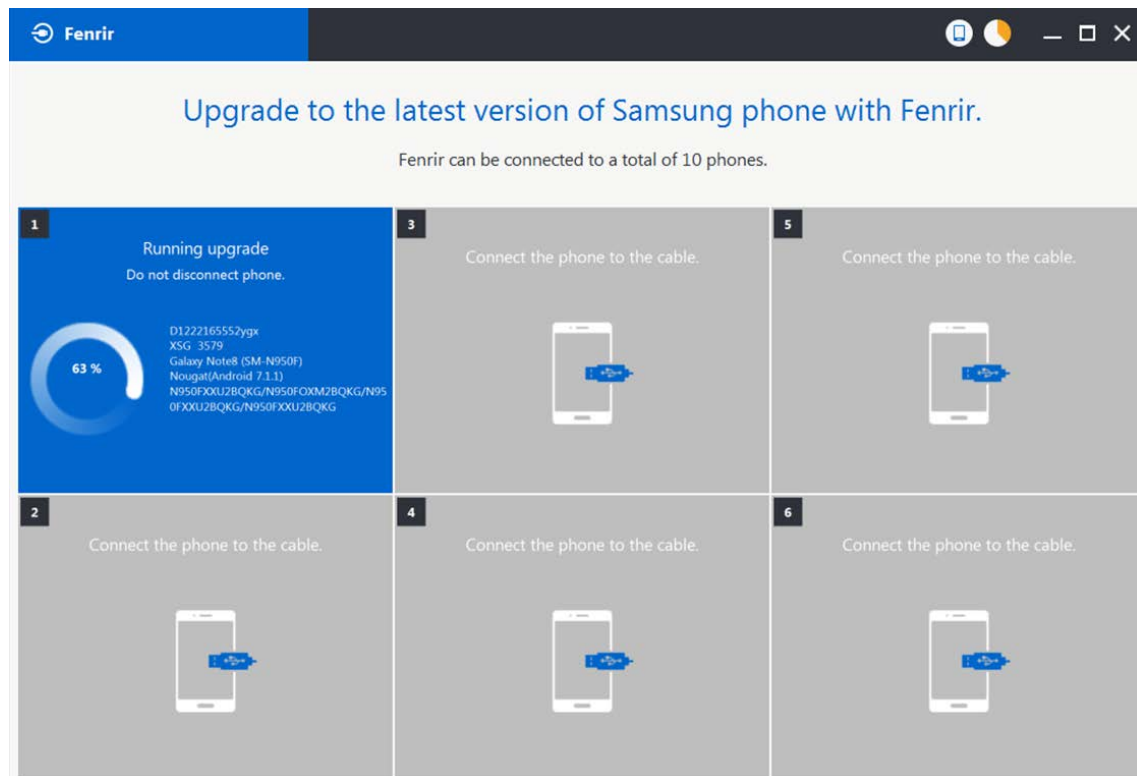


7) Once Fenrir starts, application will display the below screen. And select the Start button & Agree button.

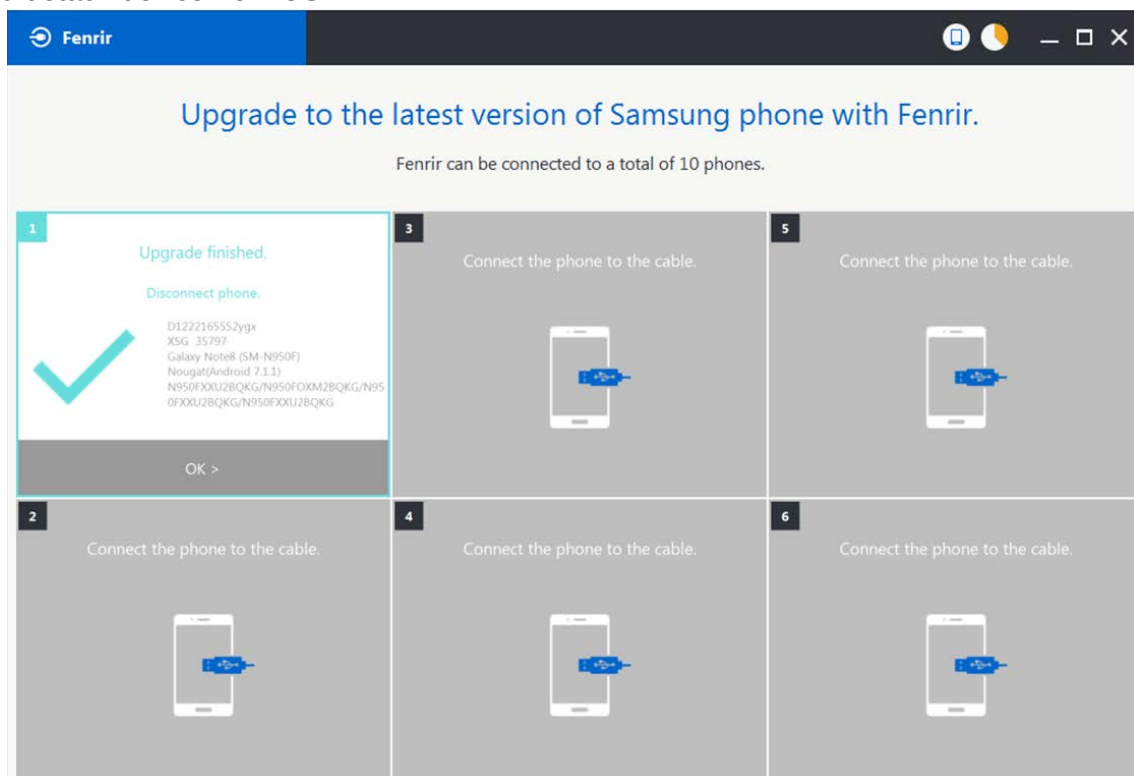


6. Level 1 Repair

8) The status circle increases as the update installs. The update process takes approximately 5-10 minutes to complete. Do not disconnect the device from USB during processing.



9) Once complete, application will present the below screen indicating update complete. Click Ok and detach device from USB.



6. Level 1 Repair

6-2. How to use 'Odin' program

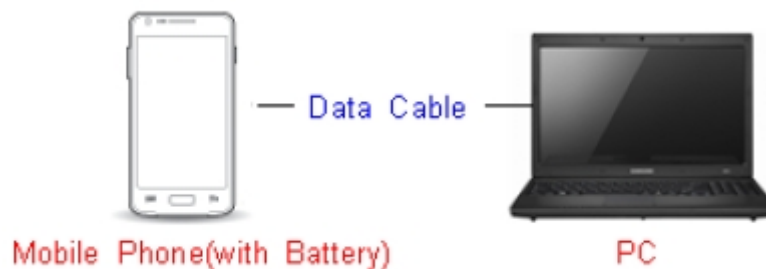
※ S/W Update via Fenrir is mandatory.

Below is the method to use 'Odin' program in any specific case.

6-2-1. Preparation

- Installation program : [Odin3 v3.13.2.exe or above](#)
- Mobile Phone
- Data Cable
- S/W Binary files (downloaded from GSPN)

※ Settings

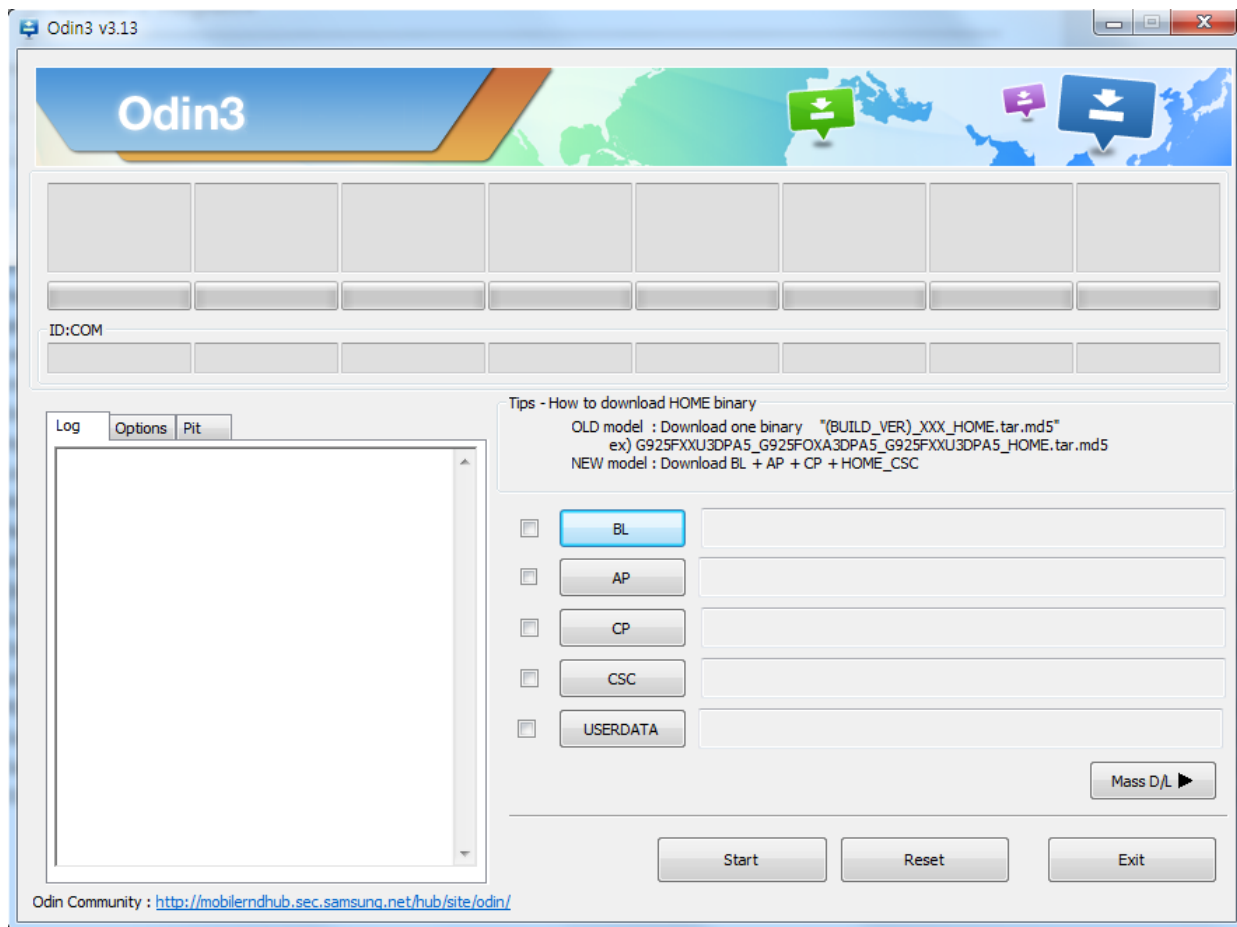


Data Cable : [GH39-01886A](#)

6. Level 1 Repair

6-2-2. S/W Installation Program (Downloader program)

Open up the S/W Installation Program by executing the "**Odin3 v3.13.2.exe**"

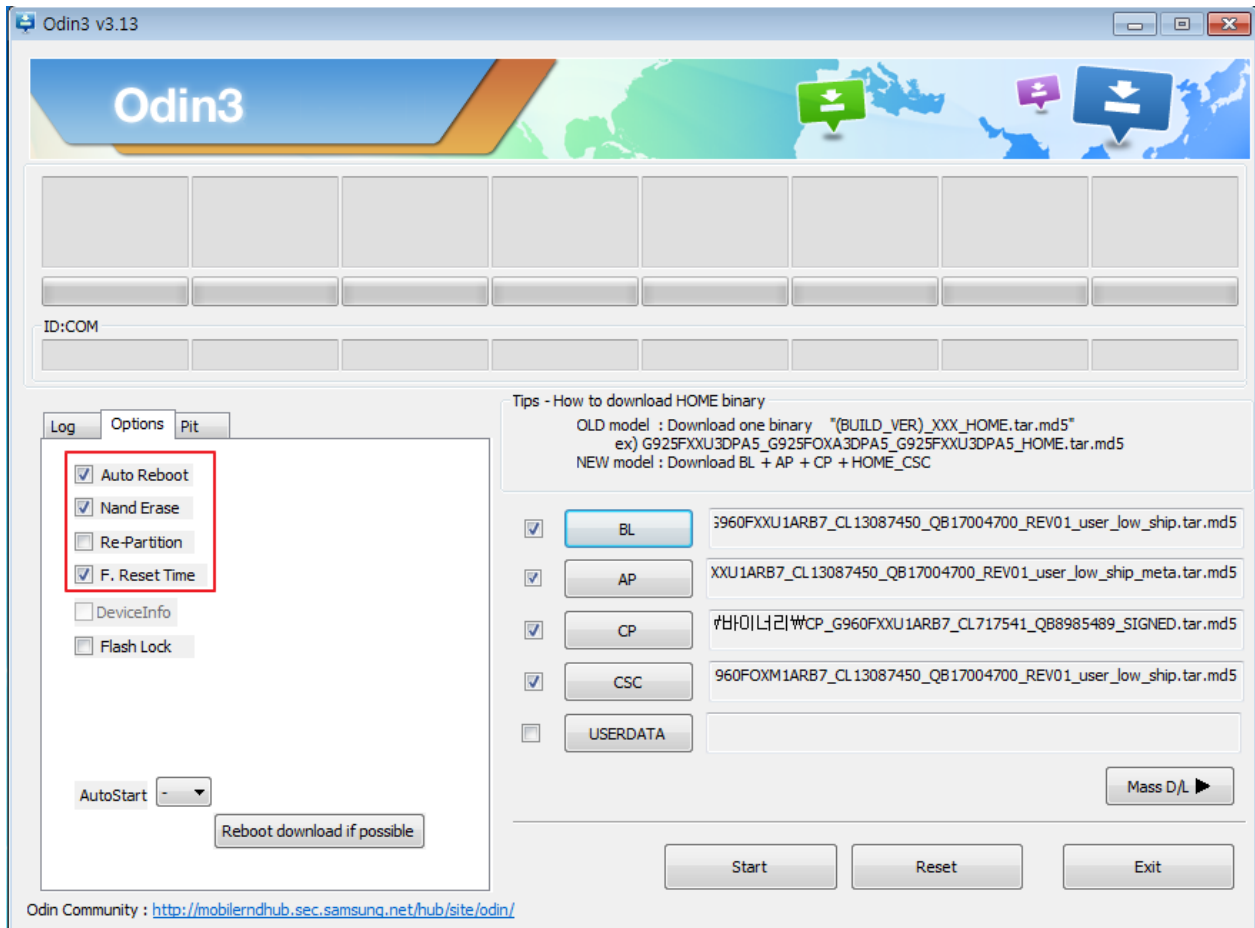


6. Level 1 Repair

1. Enable the check mark by click on the following options

- Check Auto Reboot, F. Reset Time, Nand Erase
- Check BL, AP, CP, CSC Files

* Note : "Odin v3.13.2 or above" checks MD5 checksum just after file selection.



6. Level 1 Repair

2. Enter into Download Mode

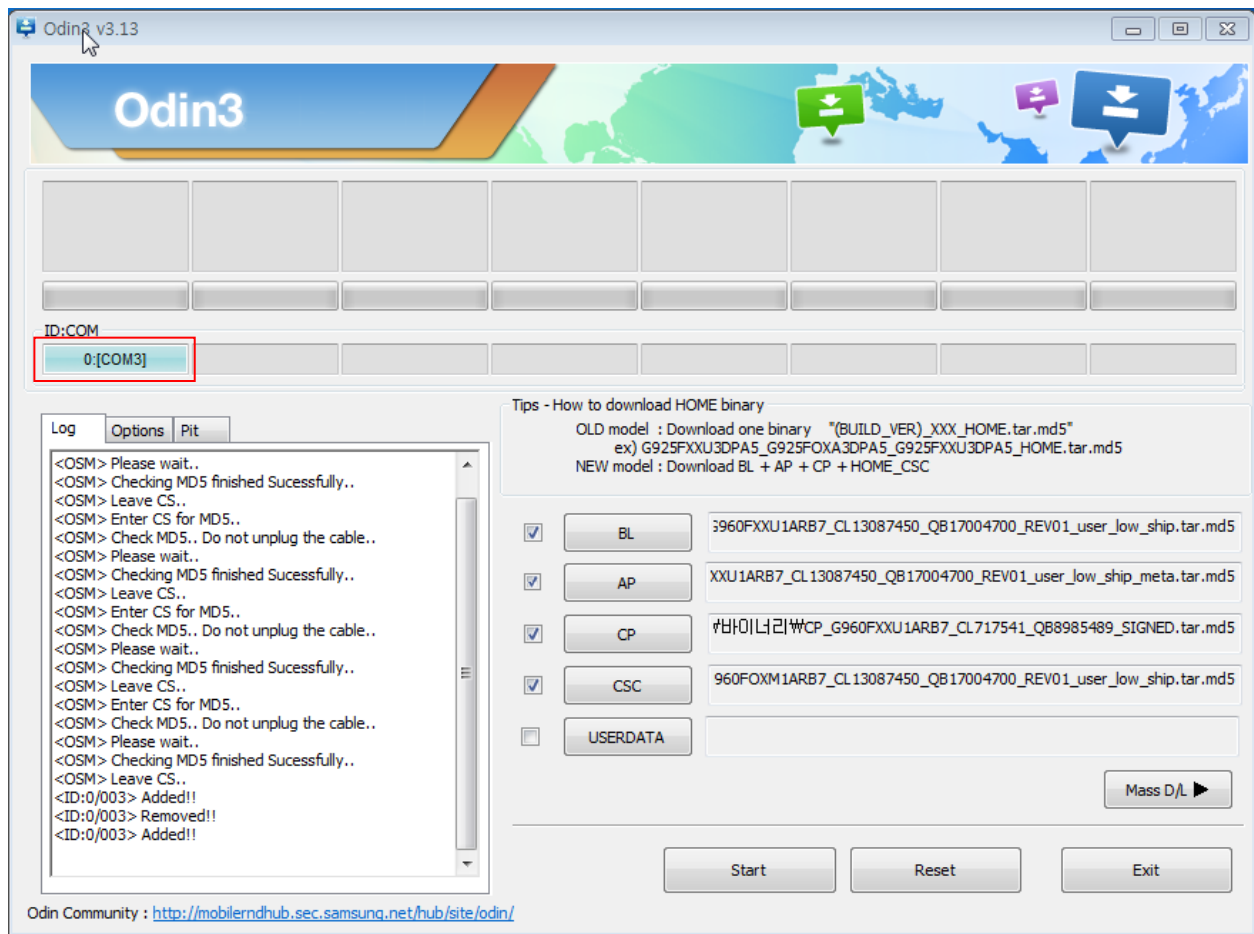
- Enter into Download Mode by pressing Volume Down button, Intelligence button and ON/OFF Button simultaneously followed by pressing Volume up button as a direction of the phone.



6. Level 1 Repair

3. Connect the device to PC via Data Cable.

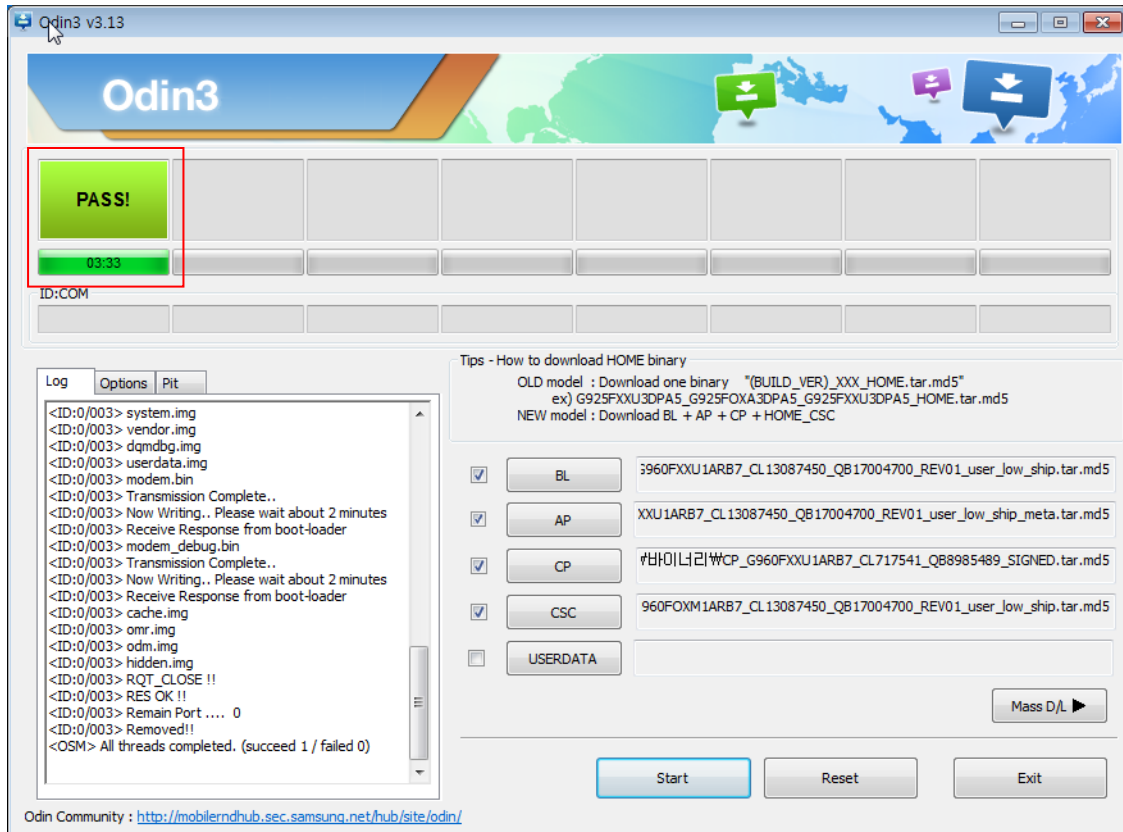
Make sure that the one of communication ports [ID:COM] box is highlighted in sky blue.
The device is now connected with the PC and ready to download the binary files in it.



6. Level 1 Repair

4. Start downloading the binary files into the device by clicking Start button on the screen.

The green colored "PASS!" sign will appear on the upper-left box if the binary files have been successfully downloaded into the device.



5. Disconnect the device from the Data cable.

6. Once the device boots up, you can check the version of the binary file or name by pressing the following code in sequence; ***#1234#**

You can perform Factory data Reset by Settings → General Management → Reset

※ Caution. Never disconnect during the S/W downloading.

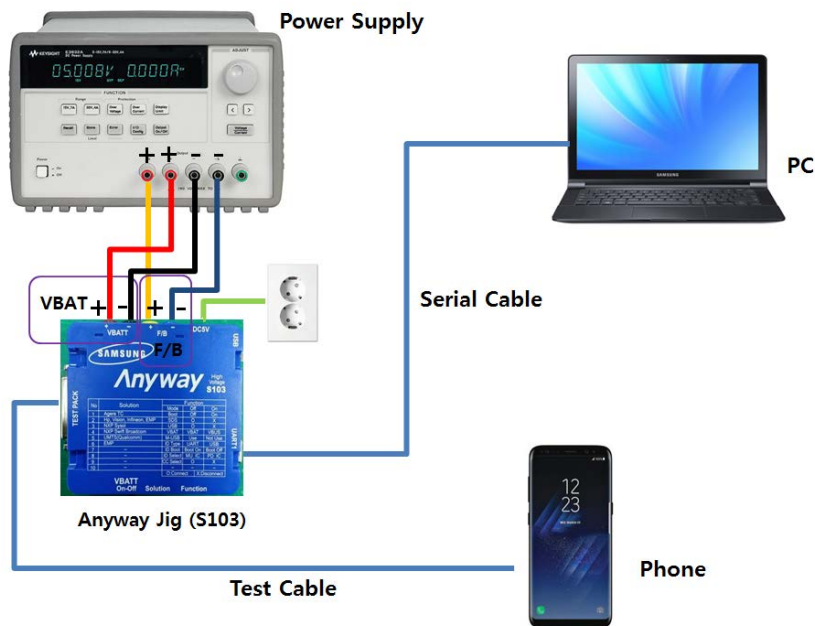
6. Level 1 Repair

6-3. IMEI writing

6-3-1. Preparation

- New IMEI writing Program has been released.
- Supported Model : Models which CAB files are uploaded on HHPsvc INI File category, instead of ini file.
- Refer to below IMEI writing procedure.

- H/W



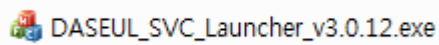
- S/W

① Library Install	To use Daseul, library files should be installed. Refer to SVC Bulletin “(11-82) Daseul (New IMEI writing Program) Library Install guide_rev1.0”
② Launcher	DASEUL_SVC_Launcher_v3.0.12 or higher -Uploaded on HHPsvc Notice
③ Runtime File	1. DASEUL_IMEI_ALL_Runtime_3.1.348.0_r00519.CAB or higher -Uploaded on HHPsvc Notice 2. Make 'ModelName' folder at the same position with launcher & Runtime file. DASEUL_IMEI_ALL_Runtime_3.1.348.0_r00519.CAB DASEUL_Launcher_v4.0.0.exe SM-G960F_SS(CSC)_IMEI_Ver_3.1.343.10.CAB
④ Model File	Copy Model File under the 'SM-A600FN' folder

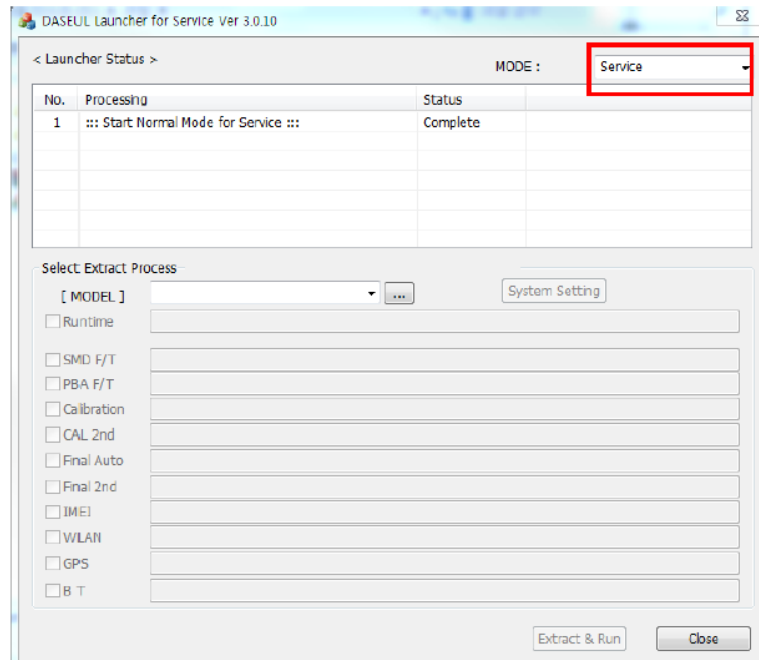
6. Level 1 Repair

6-3-2. IMEI writing Process

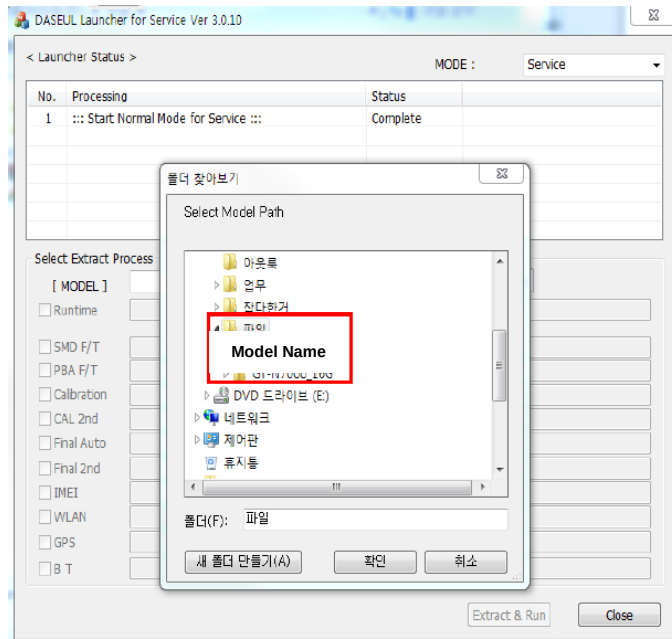
1. Run DASEUL_SVC_Launcher_v3.0.12.exe



2. Select Service Mode

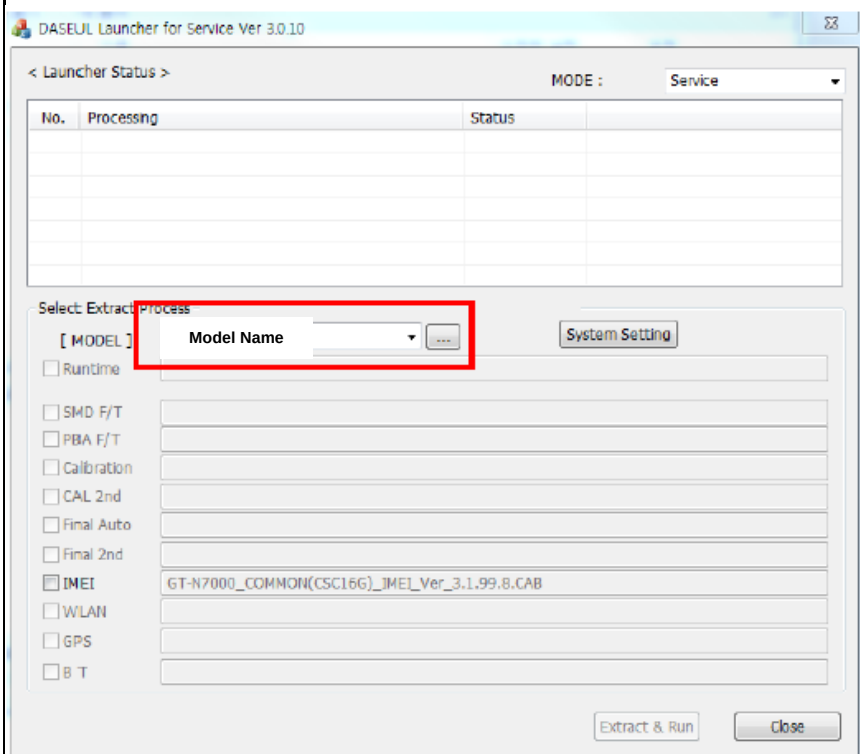


3. Click  and Select folder where the Launcher exists



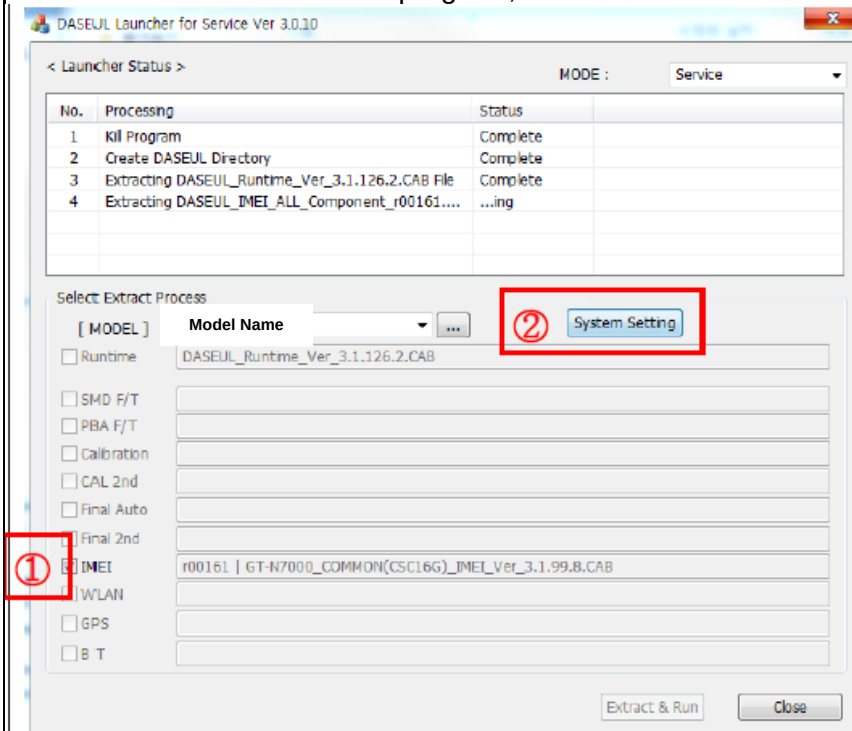
6. Level 1 Repair

4. Select Model



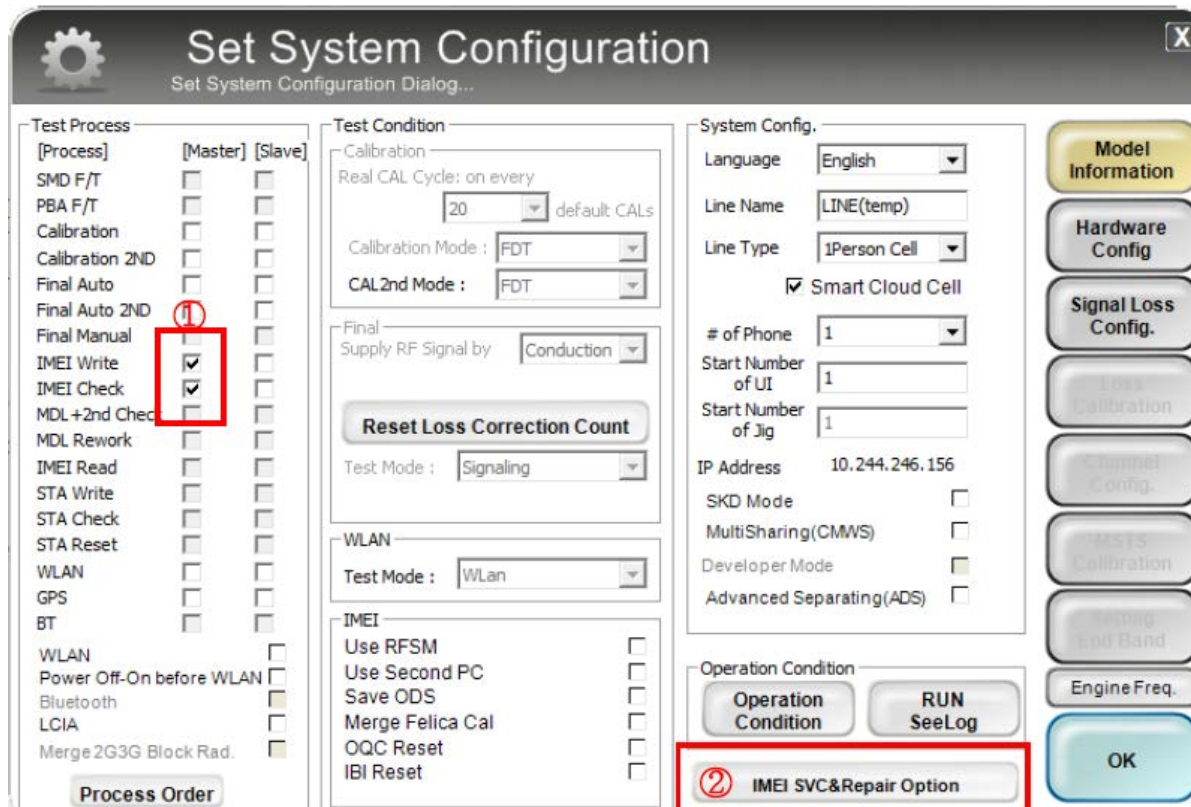
5. Check IMEI and click System Setting

※ Once you setup the setting, you don't have to do it again, unless there is change.
From second run of the IMEI program, check IMEI and click Extract & Run.



6. Level 1 Repair

6. Check IMEI Write / IMEI Check and click IMEI SVC & Repair Option.



The 'Set System Configuration' dialog box is shown. It has a title bar with a gear icon and a close button. The main area is divided into several sections: 'Test Process' with a table of checkboxes for various tests; 'Test Condition' with dropdowns for calibration and RF signal; 'System Config.' with various settings like language, line name, and IP address; and a right sidebar with buttons for 'Model Information', 'Hardware Config', 'Signal Loss Config.', 'Loss Calibration', 'Channel Config.', 'MMS Calibration', 'Setting End Band', and 'Engine Freq.'. At the bottom, there are buttons for 'Operation Condition', 'RUN SeeLog', and 'IMEI SVC&Repair Option' (which is circled in red with a '2'). A red box with a '1' highlights the 'IMEI Write' and 'IMEI Check' checkboxes in the 'Test Process' table.

[Process]	[Master]	[Slave]
SMD F/T	☐	☐
PBA F/T	☐	☐
Calibration	☐	☐
Calibration 2ND	☐	☐
Final Auto	☐	☐
Final Auto 2ND	☐	☐
Final Manual	☐	☐
IMEI Write	☒	☐
IMEI Check	☒	☐
MDL+2nd Check	☐	☐
MDL Rework	☐	☐
IMEI Read	☐	☐
STA Write	☐	☐
STA Check	☐	☐
STA Reset	☐	☐
WLAN	☐	☐
GPS	☐	☐
BT	☐	☐
WLAN	☐	☐
Power Off-On before WLAN	☐	☐
Bluetooth	☐	☐
LCIA	☐	☐
Merge 2G3G Block Rad.	☐	☐

Test Condition

Calibration

Real CAL Cycle: on every 20 default CALs

Calibration Mode: FDT

CAL2nd Mode: FDT

Final

Supply RF Signal by: Conduction

Reset Loss Correction Count

Test Mode: Signaling

WLAN

Test Mode: WLAN

IMEI

Use RFSM ☐

Use Second PC ☐

Save ODS ☐

Merge Felica Cal ☐

OQC Reset ☐

IBI Reset ☐

System Config.

Language: English

Line Name: LINE(temp)

Line Type: 1Person Cell

☒ Smart Cloud Cell

of Phone: 1

Start Number of UI: 1

Start Number of Jig: 1

IP Address: 10.244.246.156

SKD Mode ☐

MultiSharing(CMWS) ☐

Developer Mode ☐

Advanced Separating(ADS) ☐

Operation Condition

Operation Condition

IMEI SVC&Repair Option

Model Information

Hardware Config

Signal Loss Config.

Loss Calibration

Channel Config.

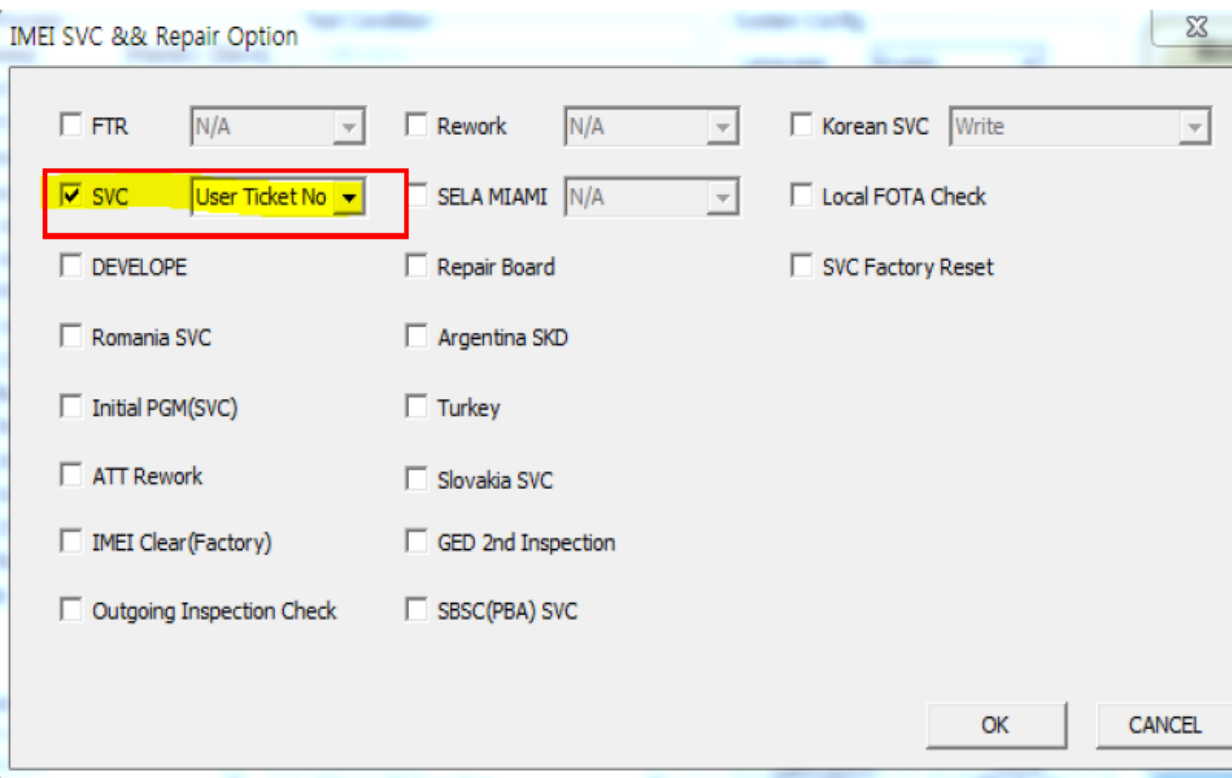
MMS Calibration

Setting End Band

Engine Freq.

OK

7. Check 'SVC , User Ticket No' and click OK



The 'IMEI SVC && Repair Option' dialog box is shown. It has a title bar with a close button. The main area contains various checkboxes and dropdown menus for different service options. A red box highlights the 'SVC' checkbox and the 'User Ticket No' dropdown menu. At the bottom, there are 'OK' and 'CANCEL' buttons.

IMEI SVC && Repair Option

☐ FTR N/A ☐ Rework N/A ☐ Korean SVC Write

☒ SVC User Ticket No ☐ SELA MIAMI N/A ☐ Local FOTA Check

☐ DEVELOPE ☐ Repair Board ☐ SVC Factory Reset

☐ Romania SVC ☐ Argentina SKD

☐ Initial PGM(SVC) ☐ Turkey

☐ ATT Rework ☐ Slovakia SVC

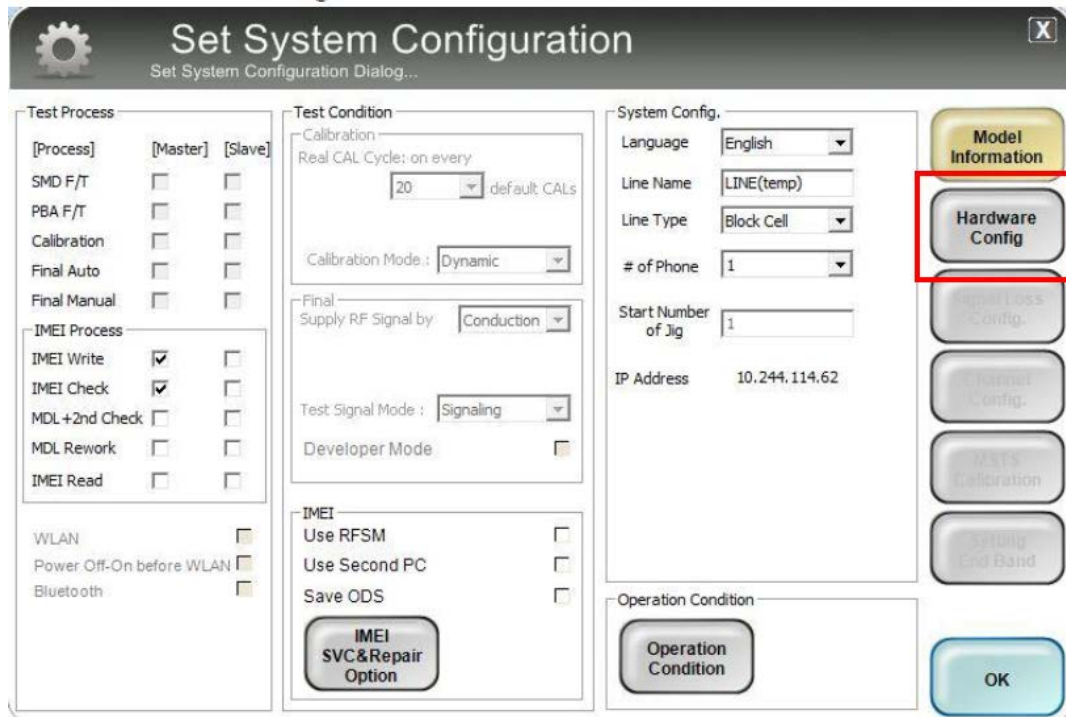
☐ IMEI Clear(Factory) ☐ GED 2nd Inspection

☐ Outgoing Inspection Check ☐ SBSC(PBA) SVC

OK CANCEL

6. Level 1 Repair

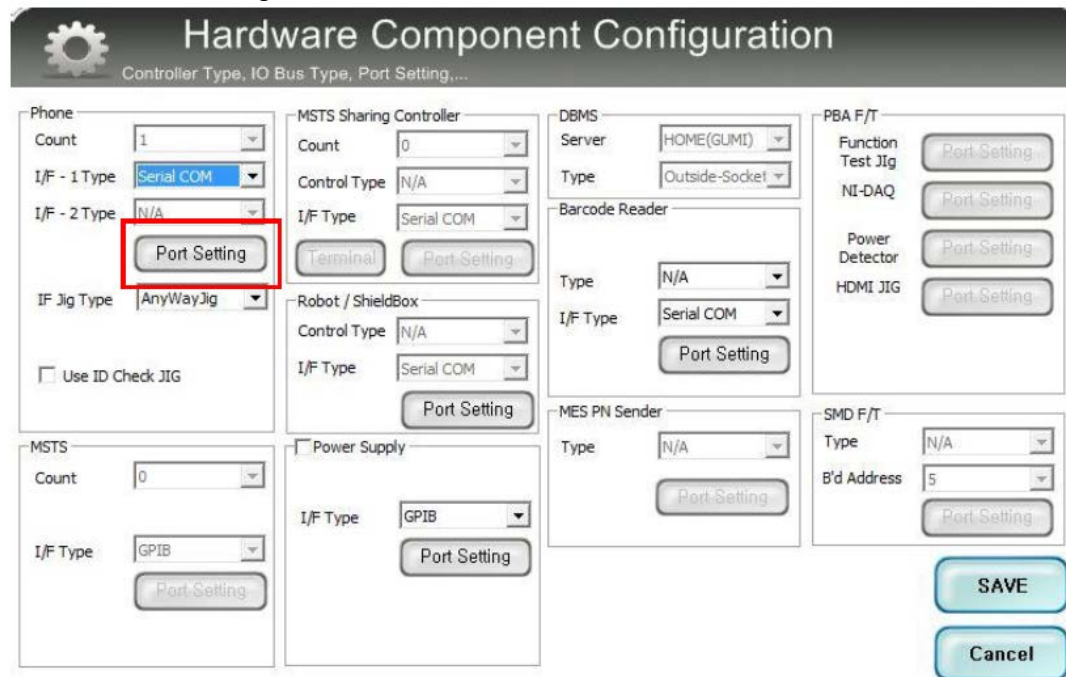
8. Click 'Hardware Config'



The 'Set System Configuration' dialog box is shown. It has a title bar with a gear icon and the text 'Set System Configuration Dialog...'. The dialog is divided into several sections:

- Test Process:** Includes checkboxes for [Process], [Master], [Slave], SMD F/T, PBA F/T, Calibration, Final Auto, Final Manual, IMEI Process, IMEI Write, IMEI Check, MDL+2nd Check, MDL Rework, IMEI Read, WLAN, Power Off-On before WLAN, and Bluetooth.
- Test Condition:** Includes a 'Calibration' section with 'Real CAL Cycle: on every' and a dropdown for '20' (default CALs), and a 'Calibration Mode' dropdown set to 'Dynamic'. It also has a 'Final' section with 'Supply RF Signal by' set to 'Conduction', a 'Test Signal Mode' dropdown set to 'Signaling', and a 'Developer Mode' checkbox.
- System Config.:** Includes 'Language' (English), 'Line Name' (LINE(temp)), 'Line Type' (Block Cell), '# of Phone' (1), 'Start Number of Jig' (1), and 'IP Address' (10.244.114.62).
- Operation Condition:** Includes a button labeled 'Operation Condition'.
- Model Information:** A vertical sidebar on the right with buttons for 'Model Information', 'Hardware Config' (highlighted with a red box), 'Signal Loss Config.', 'Channel Config.', 'MSTs Calibration', 'Setting End Band', and 'OK'.
- IMEI SVC&Repair Option:** A button at the bottom left.

9. Click 'Port Setting'



The 'Hardware Component Configuration' dialog box is shown. It has a title bar with a gear icon and the text 'Controller Type, IO Bus Type, Port Setting,...'. The dialog is divided into several sections:

- Phone:** Includes 'Count' (1), 'I/F - 1 Type' (Serial COM), 'I/F - 2 Type' (N/A), 'IF Jig Type' (AnyWayJig), and a checkbox for 'Use ID Check JIG'. A 'Port Setting' button is highlighted with a red box.
- MSTS Sharing Controller:** Includes 'Count' (0), 'Control Type' (N/A), 'I/F Type' (Serial COM), and buttons for 'Terminal' and 'Port Setting'.
- Robot / ShieldBox:** Includes 'Control Type' (N/A), 'I/F Type' (Serial COM), and a 'Port Setting' button.
- Power Supply:** Includes 'I/F Type' (GPIO) and a 'Port Setting' button.
- DBMS:** Includes 'Server' (HOME(GUMI)), 'Type' (Outside-Socket), and buttons for 'Port Setting'.
- Barcode Reader:** Includes 'Type' (N/A), 'I/F Type' (Serial COM), and a 'Port Setting' button.
- MES PN Sender:** Includes 'Type' (N/A) and a 'Port Setting' button.
- PBA F/T:** Includes 'Function Test Jig', 'NI-DAQ', 'Power Detector', and 'HDMI JIG', each with a 'Port Setting' button.
- SMD F/T:** Includes 'Type' (N/A), 'B'd Address' (5), and a 'Port Setting' button.
- MSTS:** Includes 'Count' (0), 'I/F Type' (GPIO), and a 'Port Setting' button.
- Buttons:** 'SAVE' and 'Cancel' buttons at the bottom right.

6. Level 1 Repair

10. Select Port Number and SAVE

Set IO BUS Configuration

Phone IO Bus Setting

Common

BaudRate: 115200
Data Bit: 8
Parity: No
Stop Bit: 1

No.	Port #1
1	1

SAVE

Cancel

11. Click OK to proceed

Set System Configuration

Set System Configuration Dialog...

Test Process

[Process]	[Master]	[Slave]
SMD F/T	☐	☐
PBA F/T	☐	☐
Calibration	☐	☐
Final Auto	☐	☐
Final Manual	☐	☐

IMEI Process

IMEI Write	☒	☐
IMEI Check	☒	☐
MDL +2nd Check	☐	☐
MDL Rework	☐	☐
IMEI Read	☐	☐

WLAN ☐
Power Off-On before WLAN ☐
Bluetooth ☐

Test Condition

Calibration
Real CAL Cycle: on every 20 default CALs
Calibration Mode: Dynamic

Final
Supply RF Signal by: Conduction

Test Signal Mode: Signaling

Developer Mode ☐

IMEI

Use RFSM ☐
Use Second PC ☐
Save ODS ☐

IMEI SVC&Repair Option

System Config.

Language: English
Line Name: LINE(temp)
Line Type: Block Cell
of Phone: 1
Start Number of Jig: 1
IP Address: 10.244.114.62

Operation Condition

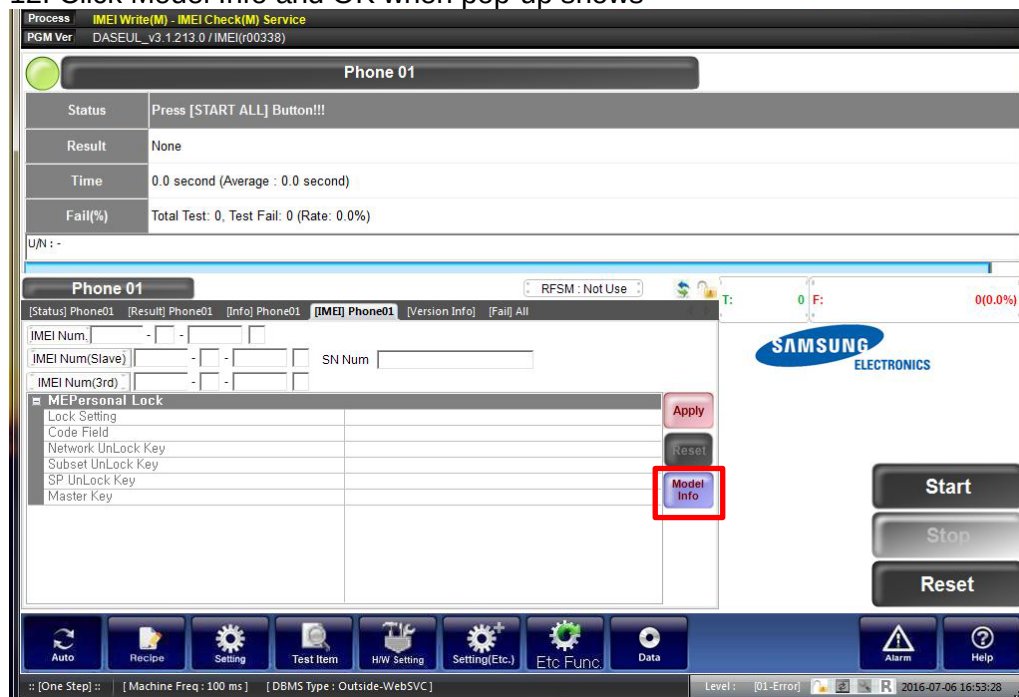
Operation Condition

Model Information
Hardware Config
Signal Loss Config.
Channel Config.
UART Calibration
Setting End Band

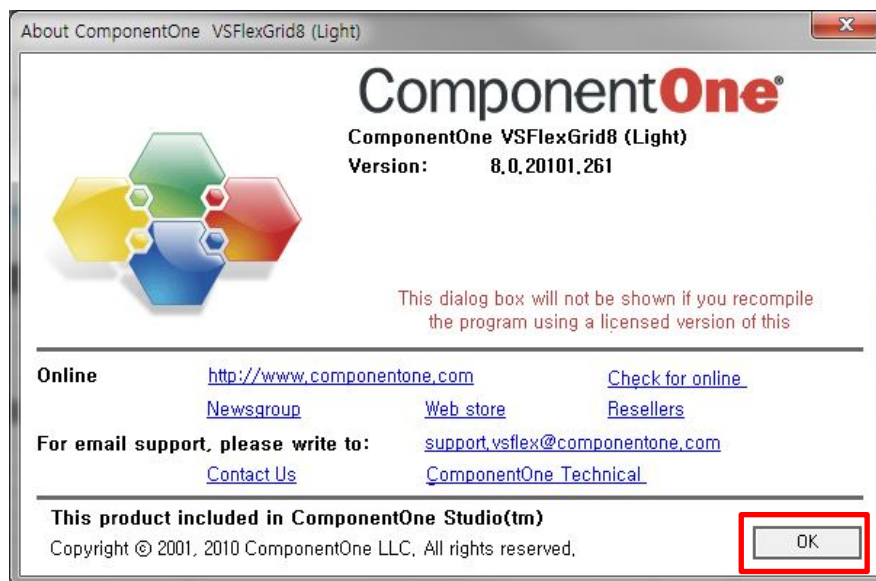
OK

6. Level 1 Repair

12. Click Model Info and OK when pop-up shows



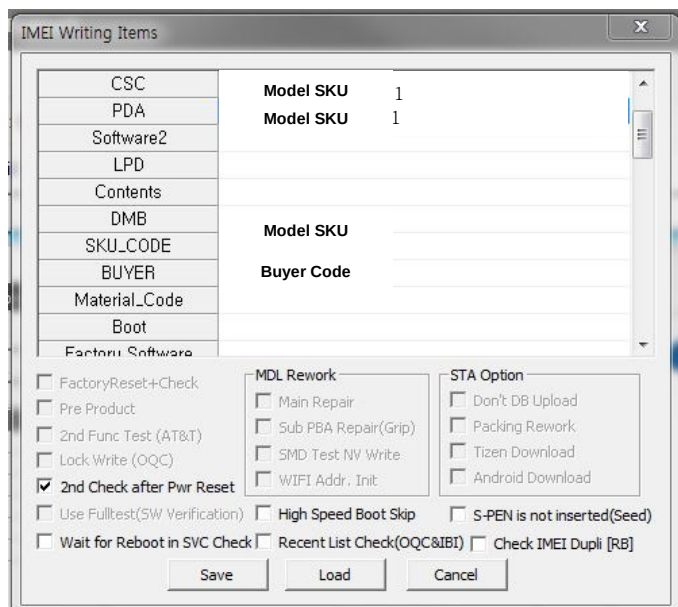
13. Click OK



6. Level 1 Repair

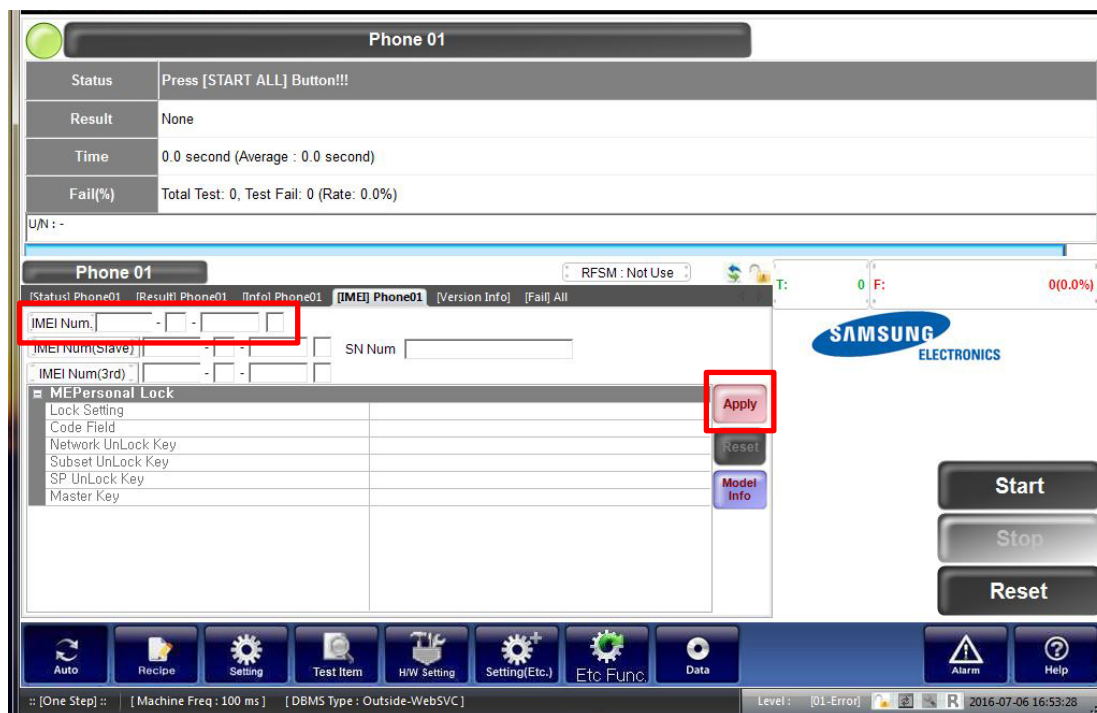
14. Input SKU_CODE and BUYER, then click Save button.

※ Refer to HHPsvc→IMEI Review to check SKU Code and buyer



The 'IMEI Writing Items' dialog box contains a list of items on the left and input fields on the right. The items listed are CSC, PDA, Software2, LPD, Contents, DMB, SKU_CODE, BUYER, Material_Code, Boot, and Factory Software. The input fields for Model SKU and Buyer Code are currently empty. Below the list, there are several checkboxes for various options: FactoryReset+Check, Pre Product, 2nd Func. Test (AT&T), Lock Write (OQC), 2nd Check after Pwr Reset (checked), Use Fulltest(SW Verification), Wait for Reboot in SVC Check, MDL Rework (Main Repair, Sub PBA Repair(Grip), SMD Test NV Write, WIFI Addr. Init), High Speed Boot Skip, Recent List Check(OQC&IBI), STA Option (Don't DB Upload, Packing Rework, Tizen Download, Android Download), S-PEN is not inserted(Seed), and Check IMEI Dupli [RB]. At the bottom are Save, Load, and Cancel buttons.

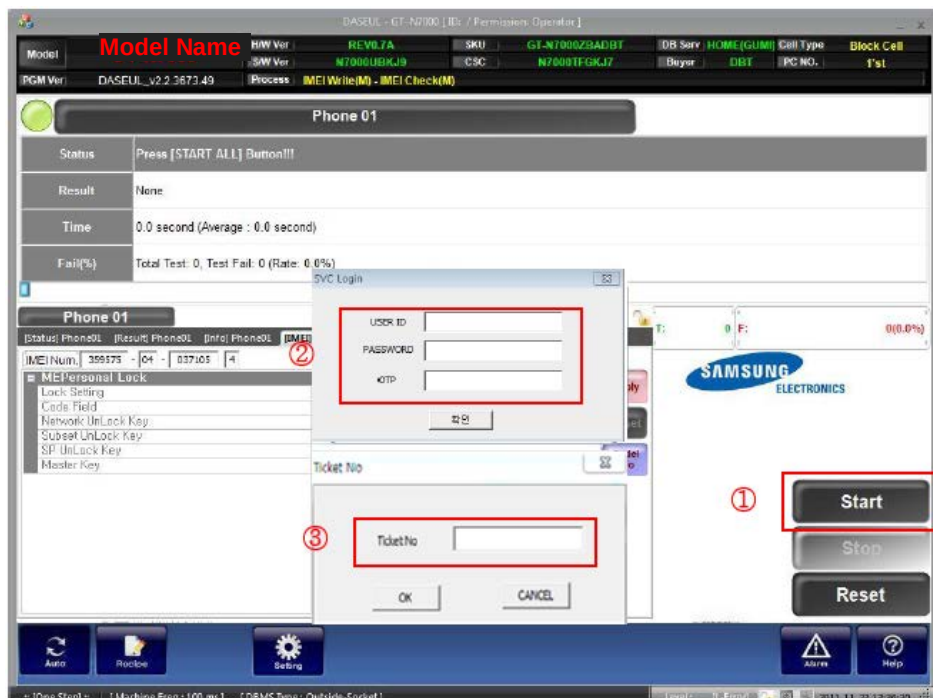
15. Input IMEI Number and click Apply



The screenshot shows the Samsung Electronics service tool interface. At the top, there's a 'Phone 01' header. Below it, a table shows Status (Press [START ALL] Button!!!), Result (None), Time (0.0 second), and Fail(%) (Total Test: 0, Test Fail: 0). The main area has tabs for Status, Result, Info, IMEI, Version Info, and Fail. The IMEI tab is active, showing fields for IMEI Num, IMEI Num(Slave), and IMEI Num(3rd). The IMEI Num field is highlighted with a red box. To the right of these fields is an 'Apply' button, also highlighted with a red box. Below the IMEI fields is a section for 'MEPersonal Lock' with various lock settings. At the bottom, there are buttons for Start, Stop, and Reset. The status bar at the very bottom shows 'Level : [01-Error]' and the date '2016-07-06 16:53:28'.

6. Level 1 Repair

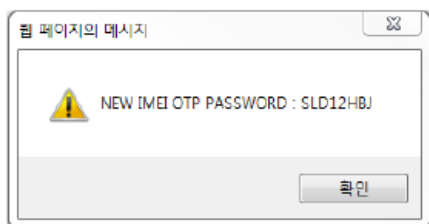
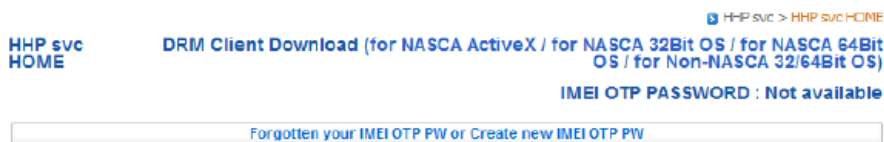
16. ① Click Start → ② Input IMEI writing ID and Password & OTP → ③ Input Ticket No



※ OTP(One time Password) : OTP is valid for 6 hours.

After that, you can get new OTP by click the “Forgotten your IMEI OTP PW or Create new IMEI OTP PW” button.

☞ OTP Location : GSPN → Knowledge → HHP svc → Home

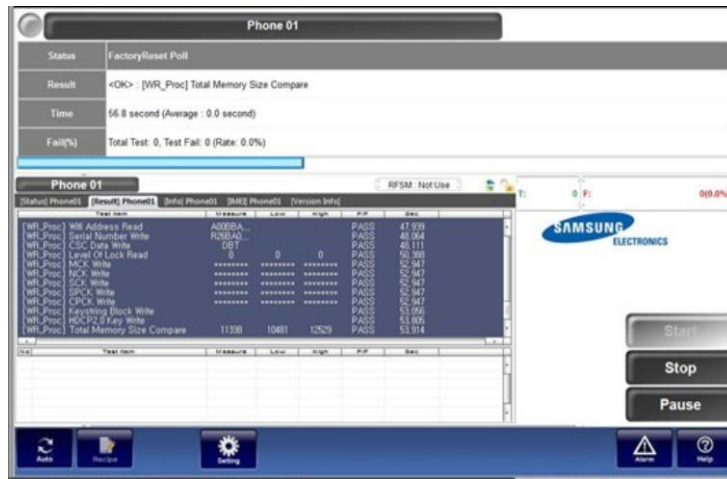


6. Level 1 Repair

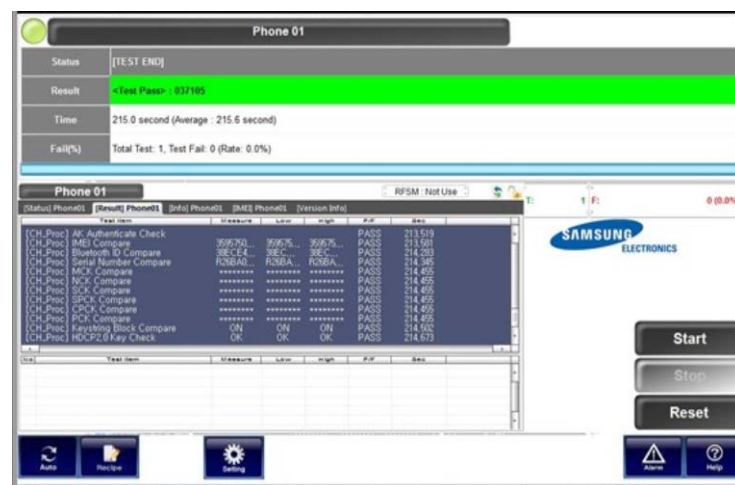
17. Connect the phone to Anyway JIG

- ※ When you connect the phone, the phone should be turned off.
After connecting the phone, the phone will be booted automatically.

18. IMEI Writing Proceeding



19. IMEI Writing Success



6. Level 1 Repair

6-4. RF Calibration

6-4-1. Required items in order to calibrate RF

- Installation program: RF Calibration Program
 - Daseul_Launcher_vx.x.xx.exe
 - Daseul_CAL_ALL_Runtime_x.x.xxx.x.CAB
 - Model File
- : **SM-xxxx_OPEN_CALIBRATION_Ver_x.x.xxx.x.CAB**

※ It is required to use the latest program.

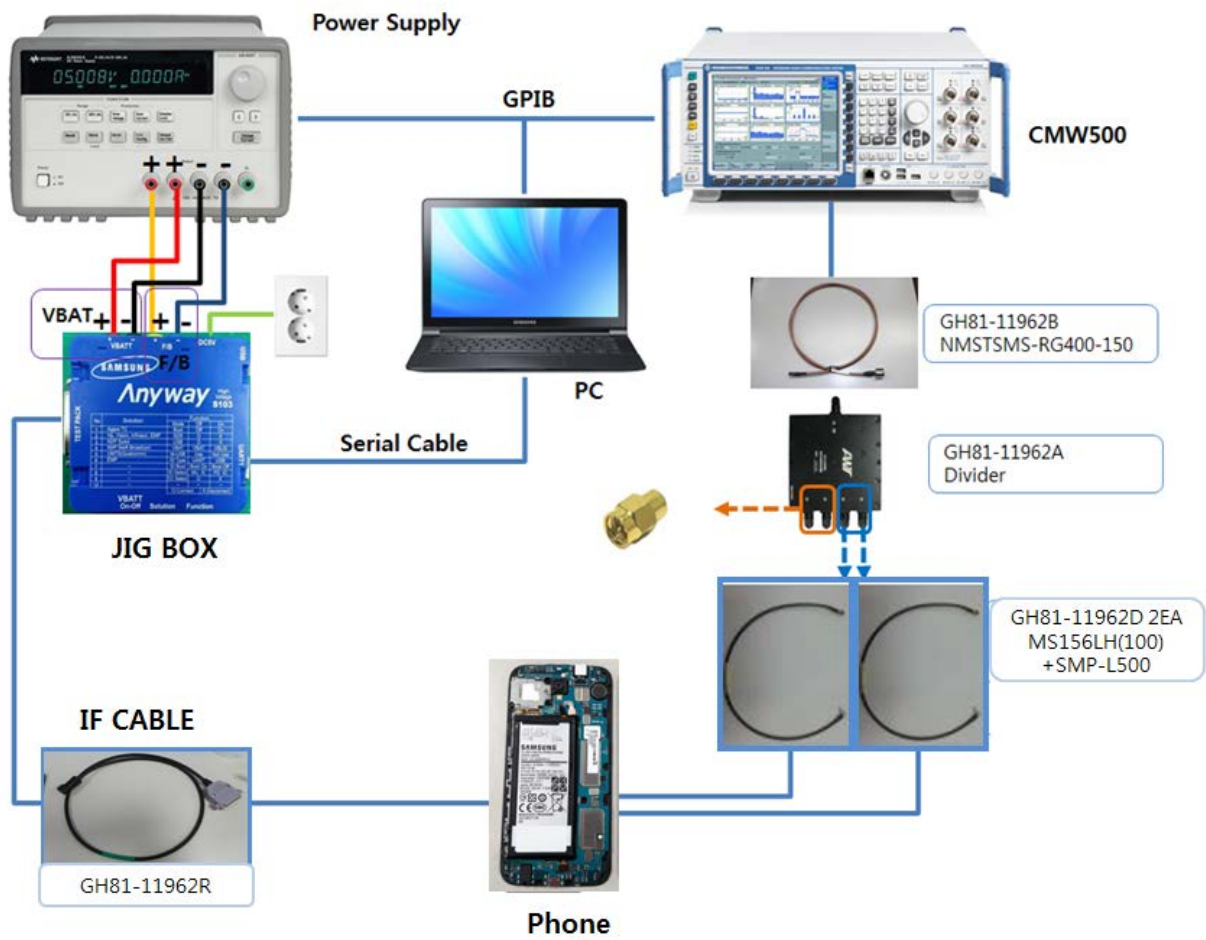
- Mobile Phone
- R&S CMW500
- E3632A Power Supply
- GPIB Cable (2ea)
- JIG BOX (S103)
- Adapter
- UART Serial Cable
- IF Cable (GH81-11962W)

❖ Table of test cables

RF Cable (Manual)	GH81-11962D 2EA		
	1.35T Short 		
4 Port Divider	GH81-11962A 	GH81-11962B 	GH81-11962E 2EA 
	Divider	Divider Cable	50Ω terminator

6. Level 1 Repair

❖ Setting



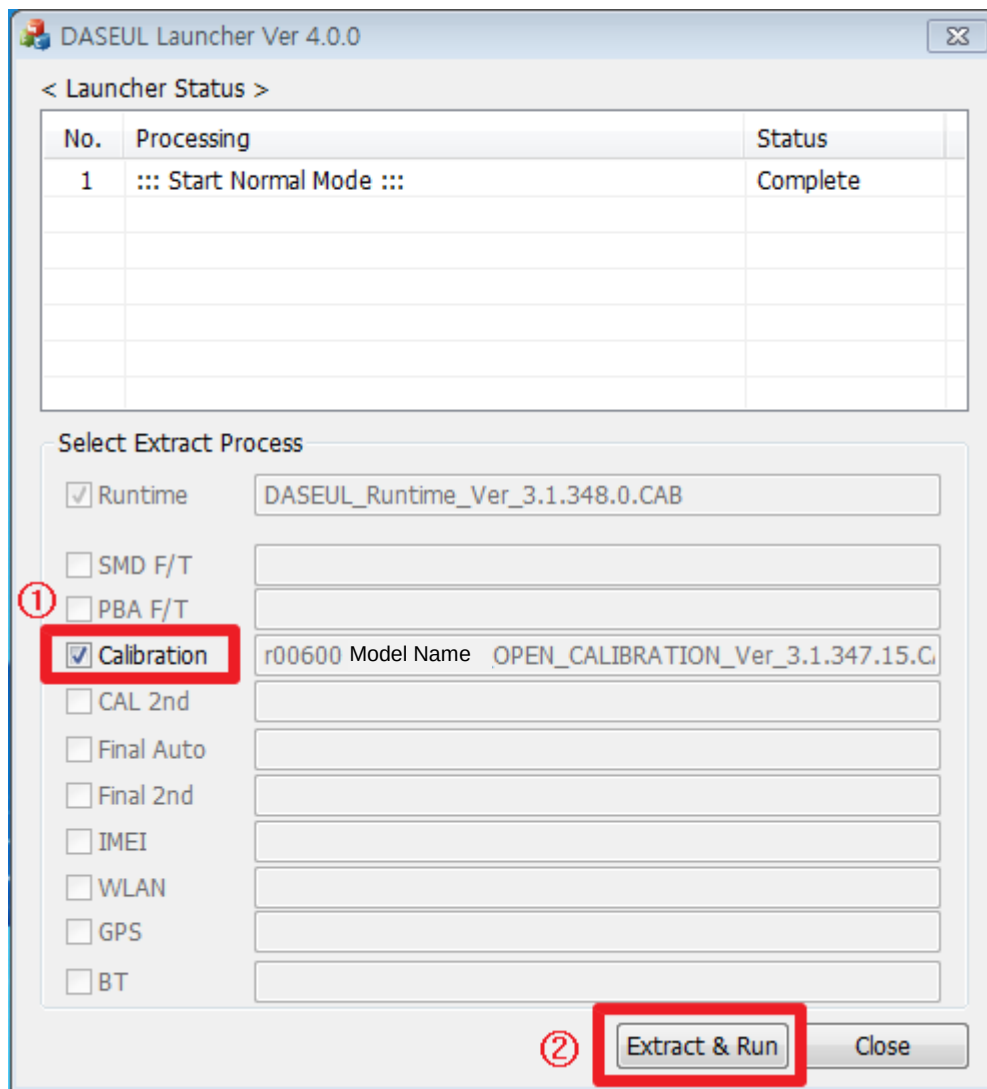
6. Level 1 Repair

6-4-2. RF Calibration Program

1. Run the RF Calibration Program Launcher, 'DASEUL_Launcher_vx.x.xx.exe'.

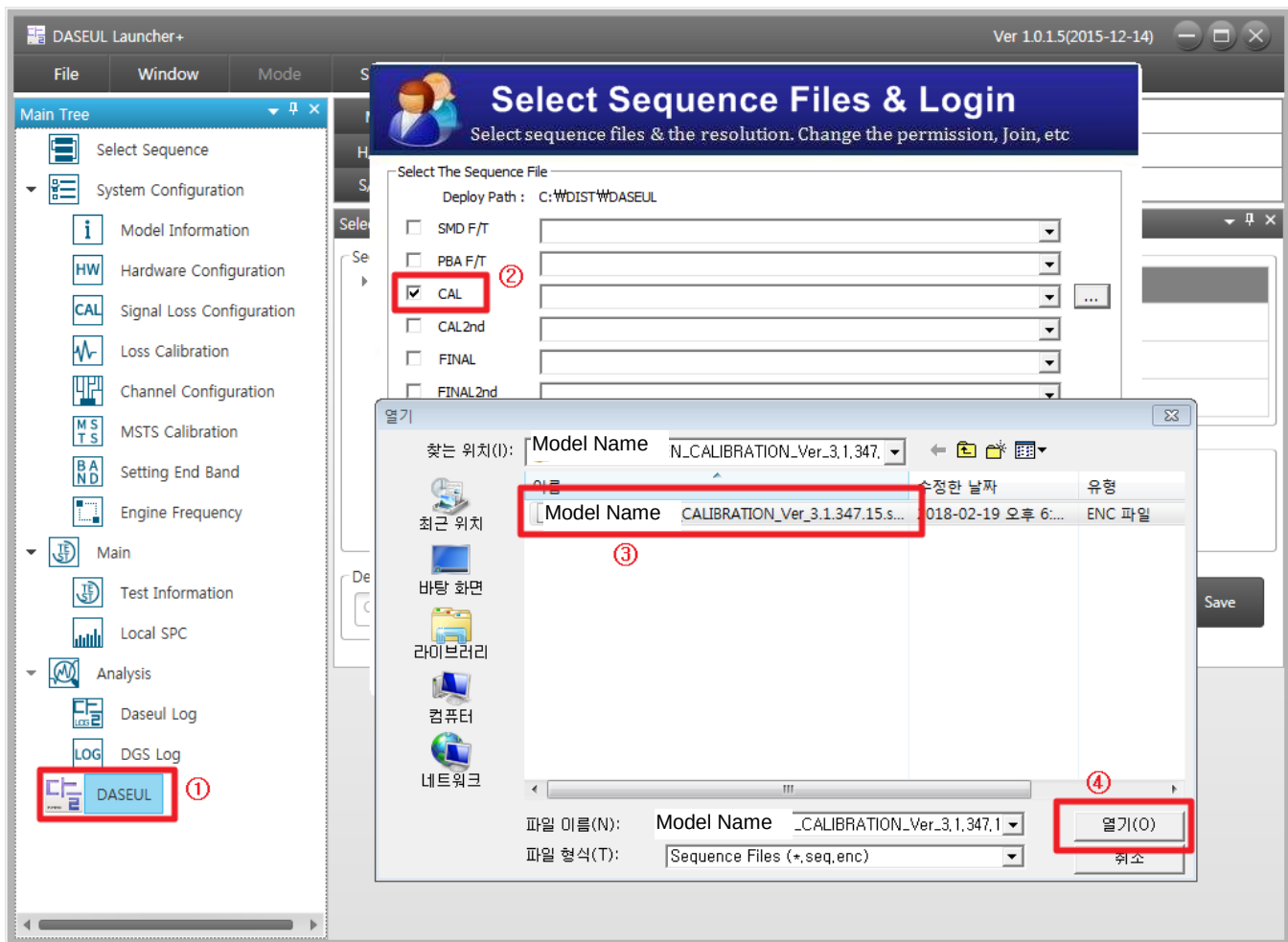
 DASEUL_CAL_ALL_Runtime_3.1.348.0_r00600.CAB
 DASEUL_Launcher_v4.0.0.exe
 Model Name PEN_CALIBRATION_Ver_3.1.347.15.CAB

2. Check the 'Calibration' option and Click 'Extract & Run'.



6. Level 1 Repair

3. Check the 'CAL' and open the [model file](#), then select 'Start' button.



6. Level 1 Repair

4. Change the Line Type to 'Block Cell' and disable 'Smart Cloud Cell'.

Set System Configuration
Set System Configuration Dialog...

Test Process

[Process]	[Master]	[Slave]
SMD F/T	☐	☐
PBA F/T	☐	☐
Calibration	☒	☐
Calibration 2ND	☐	☐
Final Auto	☐	☐
Final Auto 2ND	☐	☐
Final Manual	☐	☐
IMEI Write	☐	☐
IMEI Check	☐	☐
MDL+2nd Check	☐	☐
MDL Rework	☐	☐
IMEI Read	☐	☐
STA Write	☐	☐
STA Check	☐	☐
STA Reset	☐	☐
WLAN	☐	☐
GPS	☐	☐
BT	☐	☐
WLAN	☐	☐
Power Off-On before WLAN	☐	☐
Bluetooth	☐	☐
LCIA	☐	☐
Merge 2G3G Block Rad.	☐	☐

Test Condition

Calibration
Real CAL Cycle: on every 20 default CALs
Calibration Mode : FDT
CAL2nd Mode : FDT

Final
Supply RF Signal by Conduction

Reset Loss Correction Count

Test Mode : Signaling

WLAN
Test Mode : WLAN

IMEI
Use RFSM ☐
Use Second PC ☐
Save ODS ☐
Merge Felica Cal ☐
OQC Reset ☐
IBI Reset ☐
OQC SKD USER D/L ☐

System Config.

Language: Korean
Line Name: LINE(temp)
Line Type: Block Cell
☐ NP Cell ☐ Smart Cloud Cell
of Phone: 1
Start Number of UI: 1
Start Number of Jig: 1
IP Address: 10.60.157.50
SKD Mode ☐
MultiSharing(CMWS) ☐
Developer Mode ☐
Advanced Separating(ADS) ☐
SubpartsLife ☐

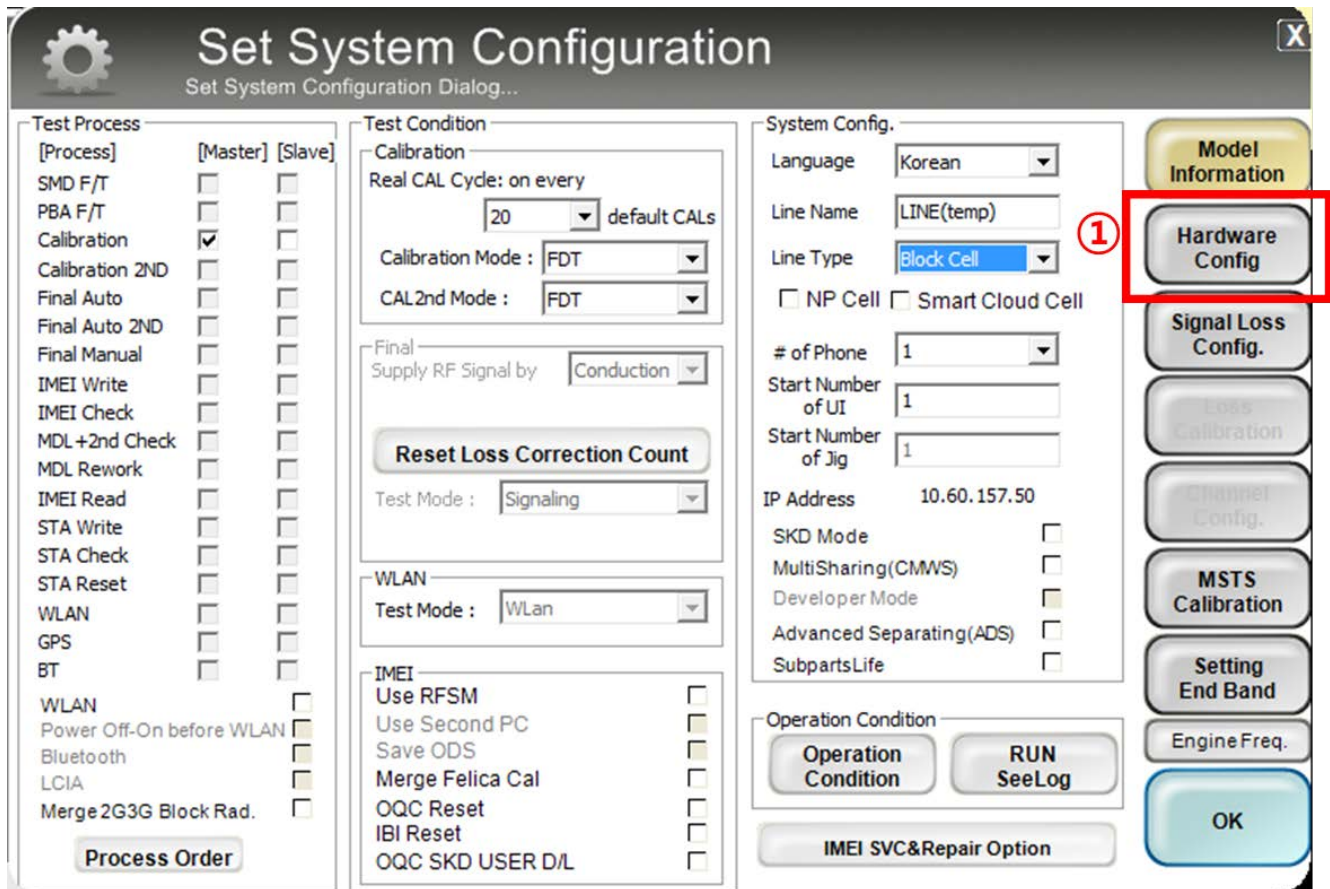
Operation Condition
Operation Condition **RUN SeeLog**

IMEI SVC&Repair Option

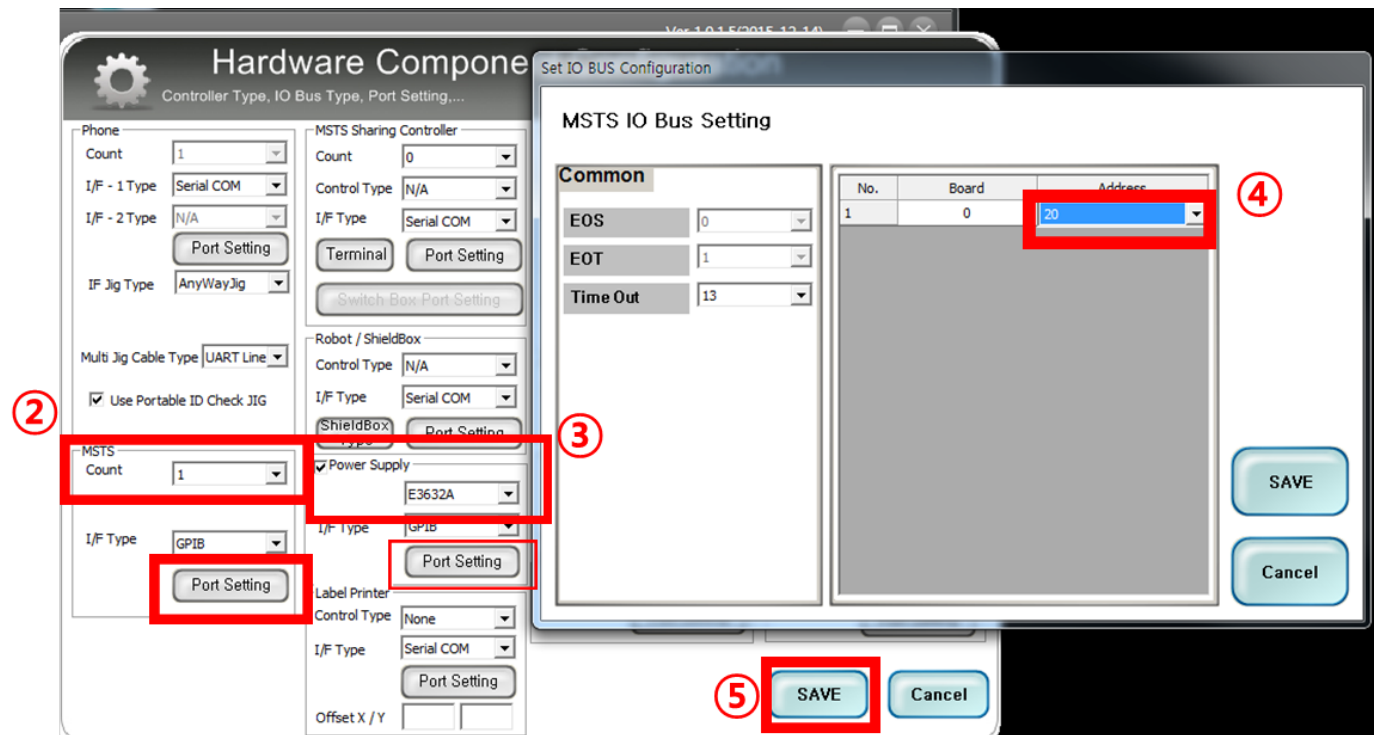
Model Information
Hardware Config
Signal Loss Config.
Loss Calibration
Channel Config.
MSTs Calibration
Setting End Band
Engine Freq.
OK

6. Level 1 Repair

5. Set the GPIB address of MSTS(CMW500) and Power Supply(E3632A) to enter 'Hardware Config' and 'Save'. (Check the GPIB address of equipments in advance)



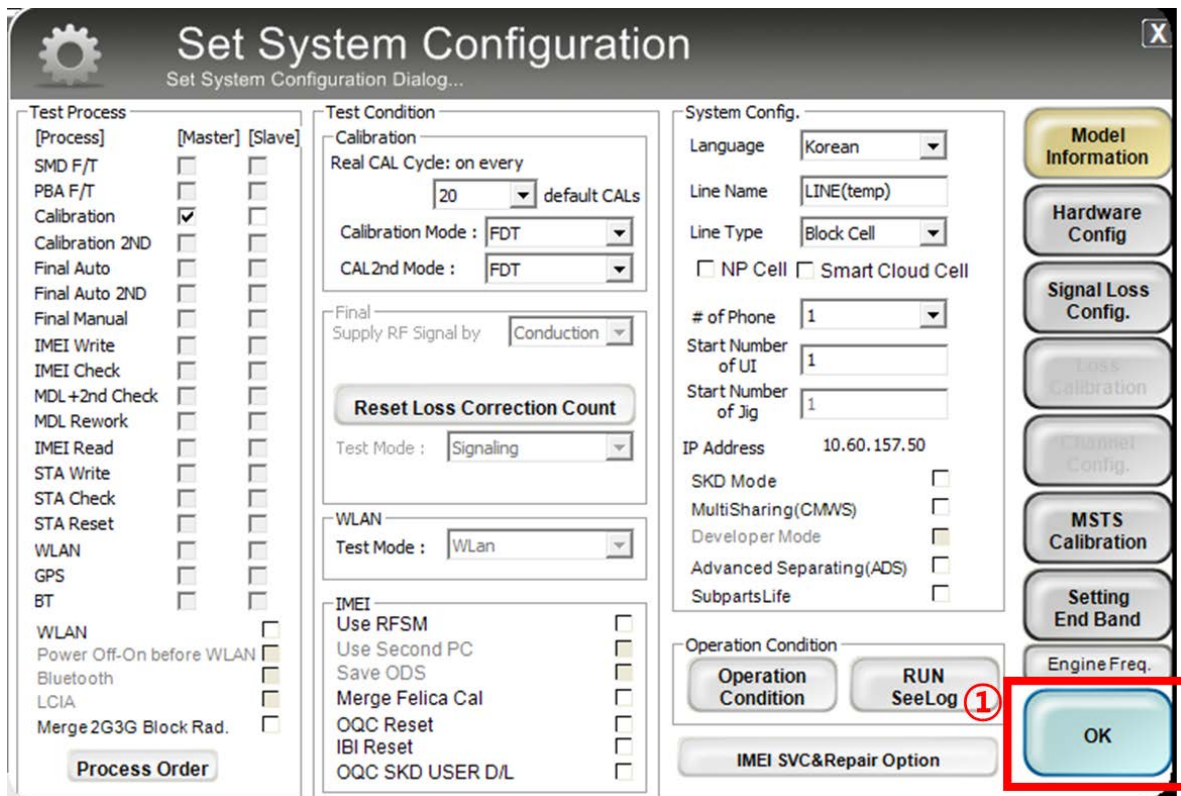
The 'Set System Configuration' dialog box is shown. It has several tabs: 'Test Process', 'Test Condition', 'System Config.', and 'Model Information'. The 'Model Information' tab is selected, and the 'Hardware Config' button is highlighted with a red box and a red circle with the number 1. Other buttons in the 'Model Information' tab include 'Signal Loss Config.', 'Loss Calibration', 'Channel Config.', 'MSTS Calibration', 'Setting End Band', 'Engine Freq.', and 'OK'.



The 'Hardware Component' dialog box is shown, with the 'MSTS IO Bus Setting' sub-dialog box open. The 'MSTS IO Bus Setting' dialog box has a 'Common' tab and a table with columns 'No.', 'Board', and 'Address'. The 'Address' column has a value of '20' highlighted with a red box and a red circle with the number 4. The 'SAVE' button is highlighted with a red box and a red circle with the number 5. The 'Hardware Component' dialog box has several sections: 'Phone', 'MSTS Sharing Controller', 'Robot / ShieldBox', and 'Label Printer'. The 'MSTS Count' is set to '1', and the 'Power Supply' is set to 'E3632A'. The 'I/F Type' is set to 'GPIB'. The 'Port Setting' button is highlighted with a red box and a red circle with the number 3. The 'SAVE' button is highlighted with a red box and a red circle with the number 5.

6. Level 1 Repair

6. Press 'OK' to start RF Calibration after completing all settings.



The 'Set System Configuration' dialog box is shown. It contains several sections: 'Test Process' with checkboxes for various tests; 'Test Condition' with settings for calibration cycles and modes; 'System Config.' with dropdowns for language, line name, and line type; and a 'Model Information' sidebar with buttons for different configuration options. The 'OK' button is highlighted with a red box and a red circle with the number 1.

Test Process

[Process]	[Master]	[Slave]
SMD F/T	☐	☐
PBA F/T	☐	☐
Calibration	☒	☐
Calibration 2ND	☐	☐
Final Auto	☐	☐
Final Auto 2ND	☐	☐
Final Manual	☐	☐
IMEI Write	☐	☐
IMEI Check	☐	☐
MDL+2nd Check	☐	☐
MDL Rework	☐	☐
IMEI Read	☐	☐
STA Write	☐	☐
STA Check	☐	☐
STA Reset	☐	☐
WLAN	☐	☐
GPS	☐	☐
BT	☐	☐
WLAN	☐	☐
Power Off-On before WLAN	☐	☐
Bluetooth	☐	☐
LCIA	☐	☐
Merge 2G3G Block Rad.	☐	☐

Test Condition

Calibration
Real CAL Cycle: on every 20 default CALs
Calibration Mode : FDT
CAL2nd Mode : FDT
Final Supply RF Signal by : Conduction
Reset Loss Correction Count
Test Mode : Signaling
WLAN Test Mode : WLAN
IMEI Use RFSM
Use Second PC
Save ODS
Merge Felica Cal
OQC Reset
IBI Reset
OQC SKD USER D/L

System Config.

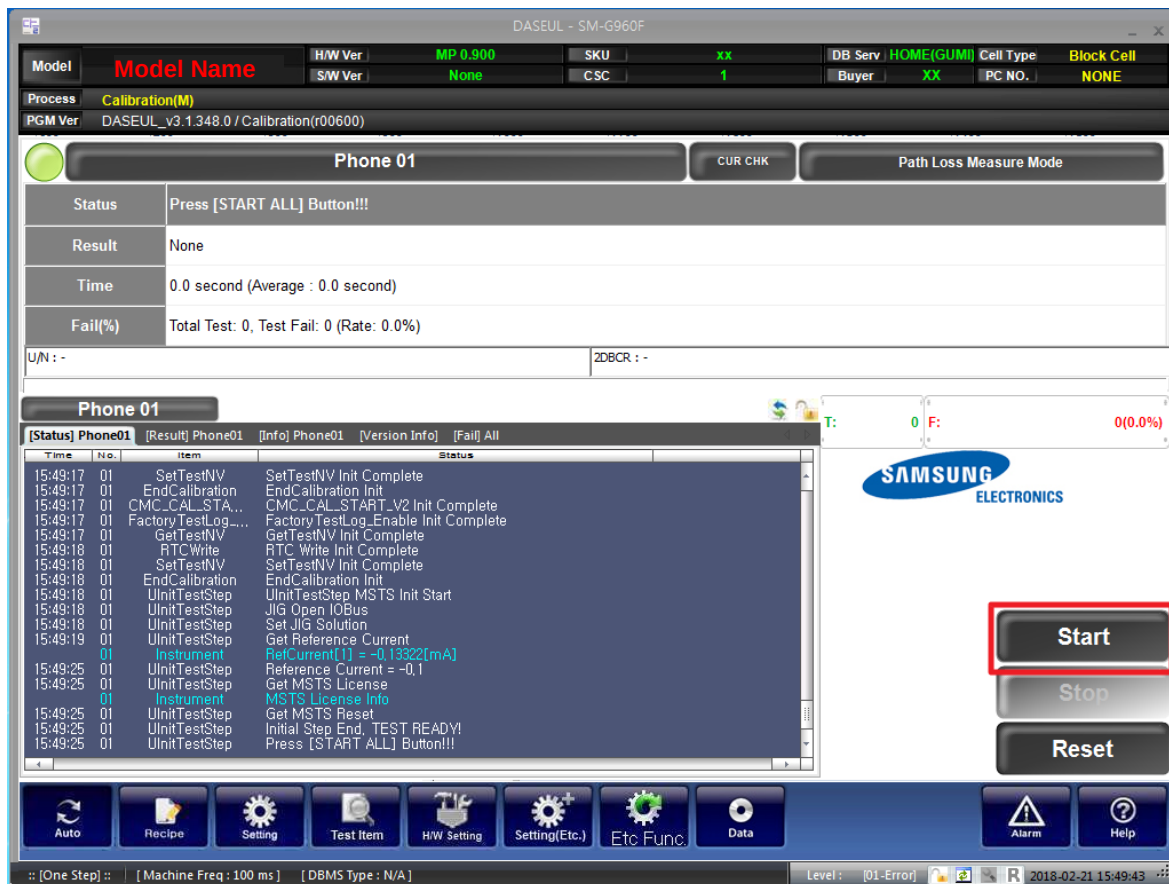
Language : Korean
Line Name : LINE(temp)
Line Type : Block Cell
☐ NP Cell ☐ Smart Cloud Cell
of Phone : 1
Start Number of UI : 1
Start Number of Jig : 1
IP Address : 10.60.157.50
SKD Mode
MultiSharing(CMWS)
Developer Mode
Advanced Separating(ADS)
SubpartsLife

Operation Condition

Operation Condition RUN SeeLog 1 OK

Model Information

Hardware Config
Signal Loss Config.
Loss Calibration
Channel Config.
MSTS Calibration
Setting End Band
Engine Freq.



The 'DASEUL - SM-G960F' calibration screen is shown. It displays various status information and a list of test steps. The 'Start' button is highlighted with a red box.

Model Model Name H/W Ver MP 0.900 SKU xx IDB Serv HOME(GUMI) Cell Type Block Cell
SW Ver None CSC 1 Buyer XX PC NO. NONE

Process Calibration(M)
PGM Ver DASEUL_v3.1.348.0 / Calibration(r00600)

Phone 01 CUR CHK Path Loss Measure Mode

Status Press [START ALL] Button!!!
Result None
Time 0.0 second (Average : 0.0 second)
Fail(%) Total Test: 0, Test Fail: 0 (Rate: 0.0%)
U/N : - ZBCR : -

Phone 01

[Status] Phone01 [Result] Phone01 [Info] Phone01 [Version Info] [Fail] All

Time	No.	Item	Status
15:49:17	01	SetTestNV	SetTestNV Init Complete
15:49:17	01	EndCalibration	EndCalibration Init
15:49:17	01	CMC_CAL_STA...	CMC_CAL_START_V2 Init Complete
15:49:17	01	FactoryTestLog...	FactoryTestLog_Enable Init Complete
15:49:17	01	GetTestNV	GetTestNV Init Complete
15:49:18	01	RTCWrite	RTC Write Init Complete
15:49:18	01	SetTestNV	SetTestNV Init Complete
15:49:18	01	EndCalibration	EndCalibration Init
15:49:18	01	UnitTestStep	UnitTestStep MSTS Init Start
15:49:18	01	UnitTestStep	JIG Open IOBus
15:49:18	01	UnitTestStep	Set JIG Solution
15:49:19	01	UnitTestStep	Get Reference Current
15:49:25	01	UnitTestStep	RefCurrent[1] = -0.13322[mA]
15:49:25	01	UnitTestStep	Reference Current = -0.1
15:49:25	01	UnitTestStep	Get MSTS License
15:49:25	01	UnitTestStep	MSTS License Info
15:49:25	01	UnitTestStep	Get MSTS Reset
15:49:25	01	UnitTestStep	Initial Step End, TEST READY!
15:49:25	01	UnitTestStep	Press [START ALL] Button!!!

SAMSUNG ELECTRONICS

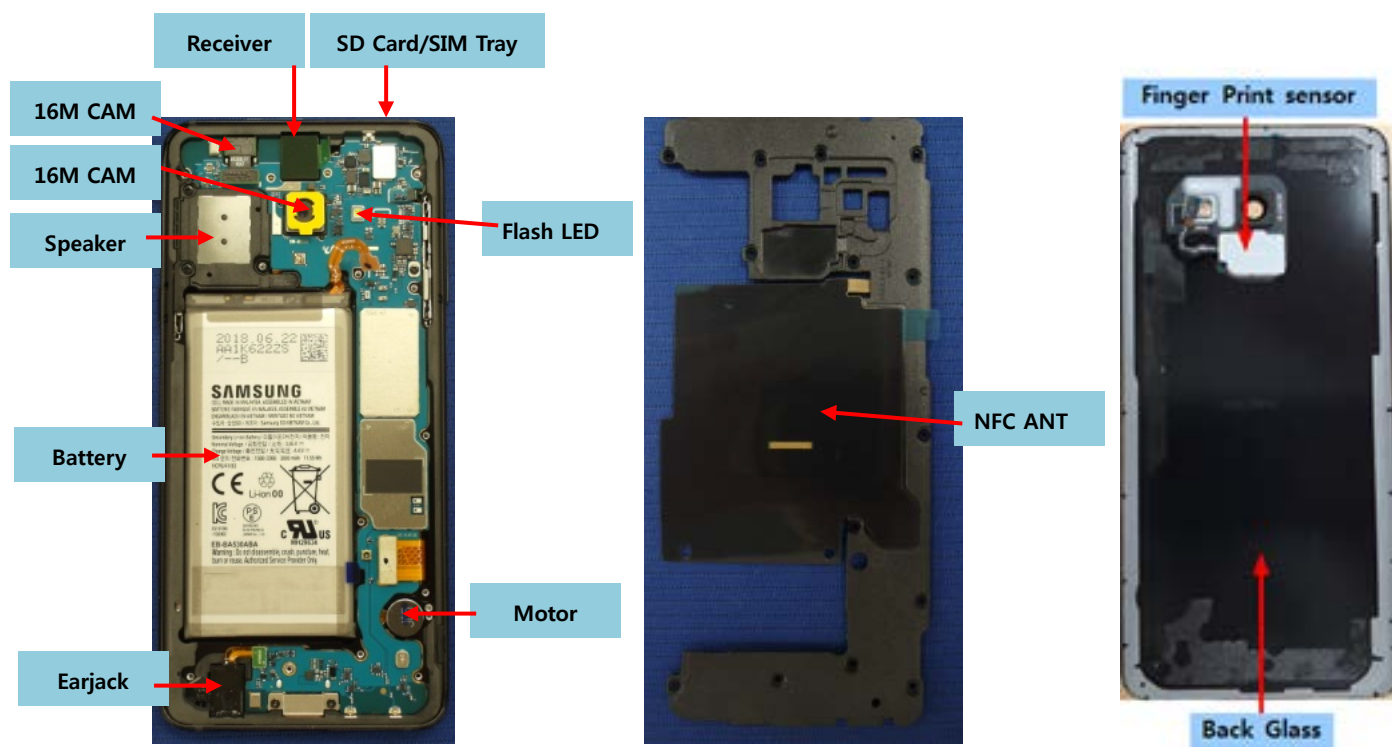
Start Stop Reset

Auto Recipe Setting Test Item H/W Setting Setting(Etc.) Etc Func. Data Alarm Help

Level : [01-Error] 2018-02-21 15:49:43

7. Level 2 Repair

7-1 Components on the Rear Case



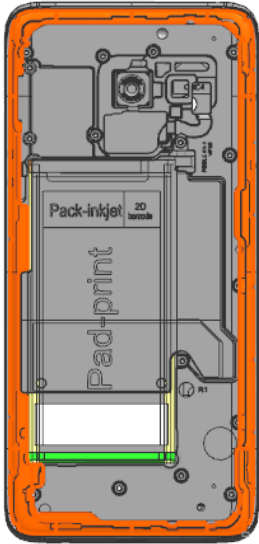
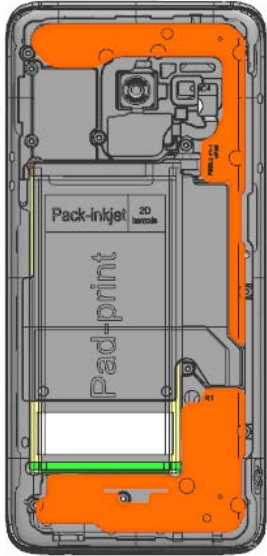
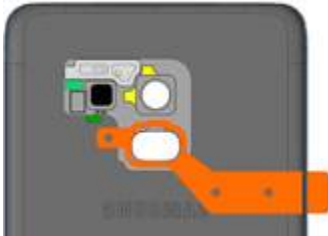
7. Level 2 Repair

7-2. Pre-requisite

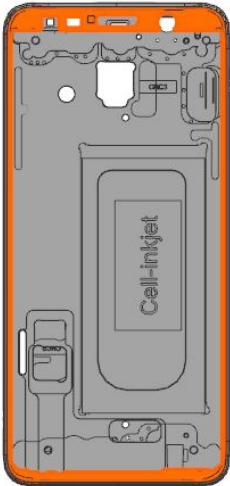
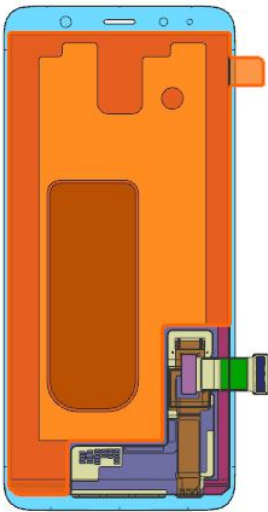
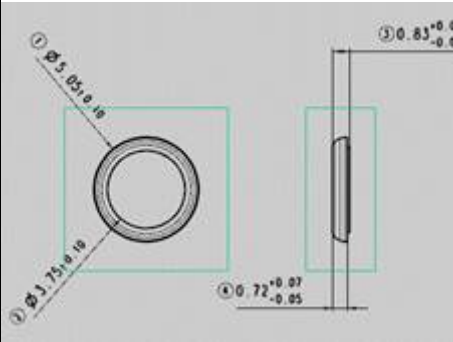
	
Tweezers / Disass'y Sticks / Screw Driver	Anti-static Gloves
	
Anti-static Mat	Hot Plate
	
A OCTA Disassembly Holder	OCTA Disassembly Upper
	
Ethyl Alcohol	Cotton Swab

7. Level 2 Repair

7-3 Parts which must be changed after repair

BOM description & part code	Kit code	Image	Remarks
A/S-TAPE BG OUT [GH02-16854A]	A/S-REWORK KIT [GH02-17945A]		Replace for Back Glass repair
A/S-TAPE BG IN [GH02-17170A / GH02-17171A]			
TAPE WP FINGER [GH02-15669A]			Replace for Back Glass Or Finger print sensor repair

7. Level 2 Repair

WINDOW TAPE [GH02-16852A]			
SVC HR TAPE (Attached OCTA. Don't use JIG) [GH81-16215A]			Replace for OCTA
ORING [GH81-15238A] (Need Replace When the dispensing is damaged)			Replace for I/F Connector

7. Level 2 Repair

TAPE WPC [GH02-07406A]			For waterproof test (mic sealing)
TAPE WPI [GH02-15865A]	A/S-REWORK KIT [GH02-17945A]		For waterproof test (air vent hole sealing)

7. Level 2 Repair

7-4. Disassembly

1 1) Put the device in the chamber or Hot plate as below heating condition - SOC 68%↓: 70°C/10~20min - SOC 68%↑: 60°C/10~20min ※ Please confirm the heating condition released lastly, and follow it. 2) Detach the Backglass.	2 Disassemble SD Card SIM Tray from device. Unscrew 14Points.
※ Caution Be careful not to damage set and OCTA.	※ Caution Be care of scratch and molding damage.
3 Disassemble the REAR. Rear TOP -> MID -> BOT By using guide hole disassemble Rear	4 Unscrew 3 Points front PBA
※ Caution Be care of damage to components.	※ Caution Be careful not to damage the PBA.

7. Level 2 Repair

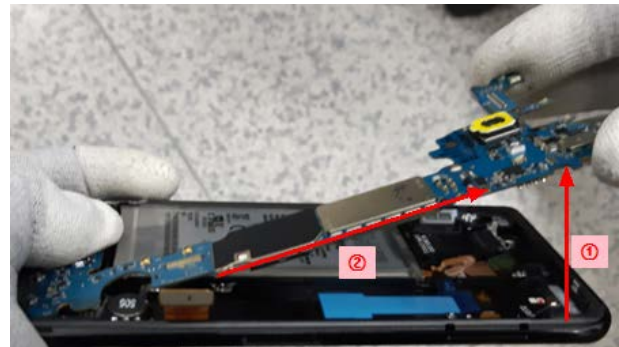
5

Disassemble connector 5 Points.



6

Disassemble PBA from Front.
(like ①② image)



※ Caution

Be care of damage to connector

※ Caution

Be care of damage to side contact point

7

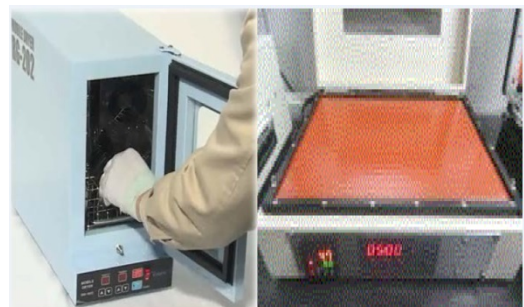
Disassemble another items



8

To separate the OCTA, heat hot chamber or hot plate

* Temperature : $70^{\circ}\pm 5$, time : 15 minutes



※ Caution

Be care of damage/ scratch

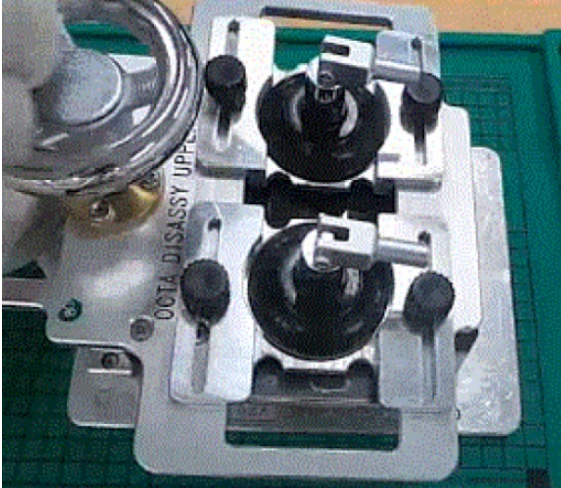
※ Caution

Be careful not to damage set and OCTA

7. Level 2 Repair

8

Separate the OCTA



※ Caution

Be careful not to damage set and OCTA

7. Level 2 Repair

7-5. Assembly

1

Assembly Main CAM, RCV, Motor at PBA, Front.
Screw 1Point for fix earjack.
* Torque value : 1.1 Kgf•cm

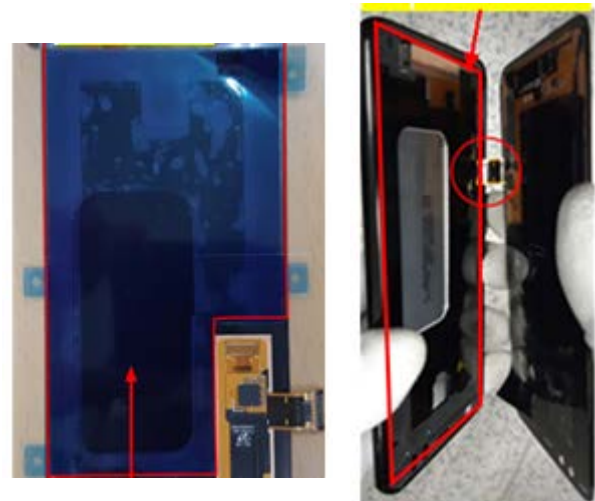


※ Caution

Be care of strong pushing

2

Attach OCTA.

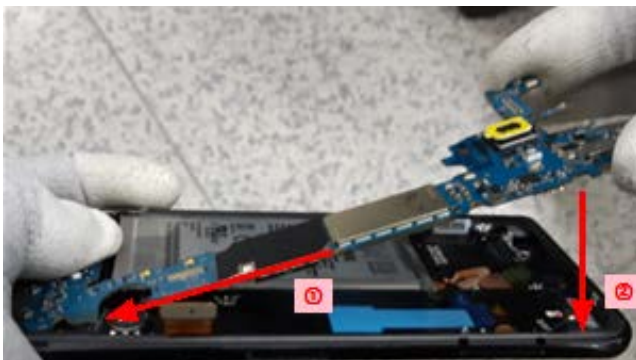


※ Caution

Be care of FPCB damage

3

Assemble PBA on Front.

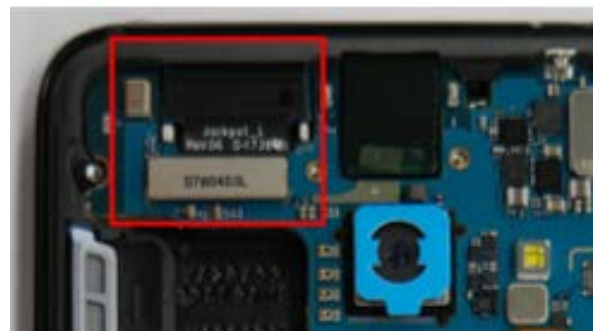


※ Caution

Be care of FPCB damage.PBA Damage

4

Assemble connector of sub CAM.



※ Caution

Be careful not to FPCB damage and tilt, strong pushing

7. Level 2 Repair

5

Assemble Battery connector and Screw
4Point.



6

Screw 3Point of PBA.
* Torque value : 1.2 Kgf·cm±7%



※ Caution

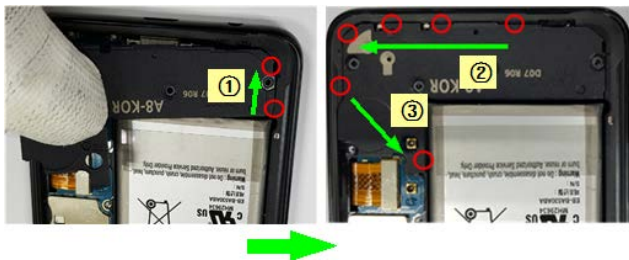
1) Be care of connector damage

※ Caution

Be care of PBA damage

7

Assemble Rear BTM.



8

Assemble Rear MID.



※ Caution

Be care of REAR damage

※ Caution

Be care of REAR damage

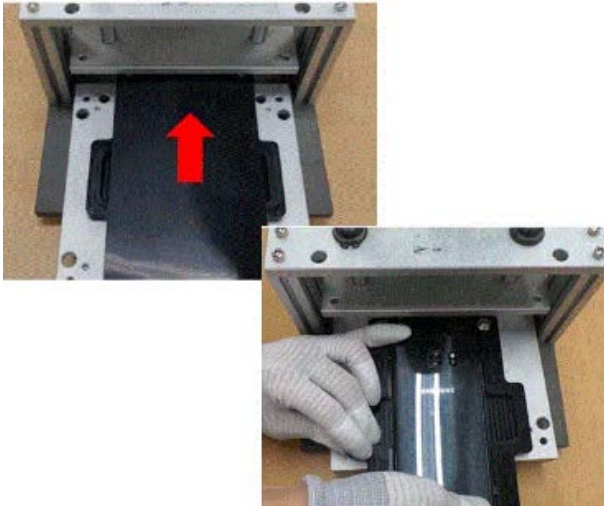
7. Level 2 Repair

9 Assemble Rear TOP. 	10 SCREW 14POINT. * Torque value : 1.35 Kgf•cm 
※ Caution Be care of REAR damage	※ Caution Be care of REAR scratch
11 Attach finger sensor in Rear. 	12 Attach Back Glass. 
※ Caution Be care of REAR scratch	※ Caution Be care of REAR scratch

7. Level 2 Repair

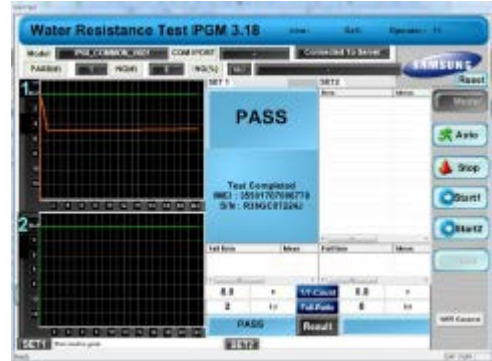
13

Press the device using pressing jig.
- Force : 1N
- Time : 1minute



14

Test WRT.



※ Caution

Please confirm the pressing condition released lastly, and follow it.

※ Caution

Be care of REAR scratch

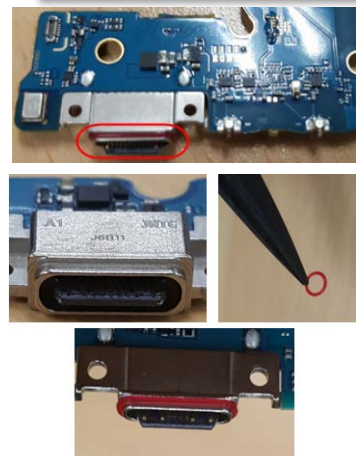
15

If WRT test happen fail, check the reason.
(sealing or etc)



16

The reason is IF connector,
Disassemble PBA from Front and change O-
Ring



※ Caution

Be care of REAR scratch

※ Caution

Be care of PBA damage

7. Level 2 Repair

17

Test WRT.



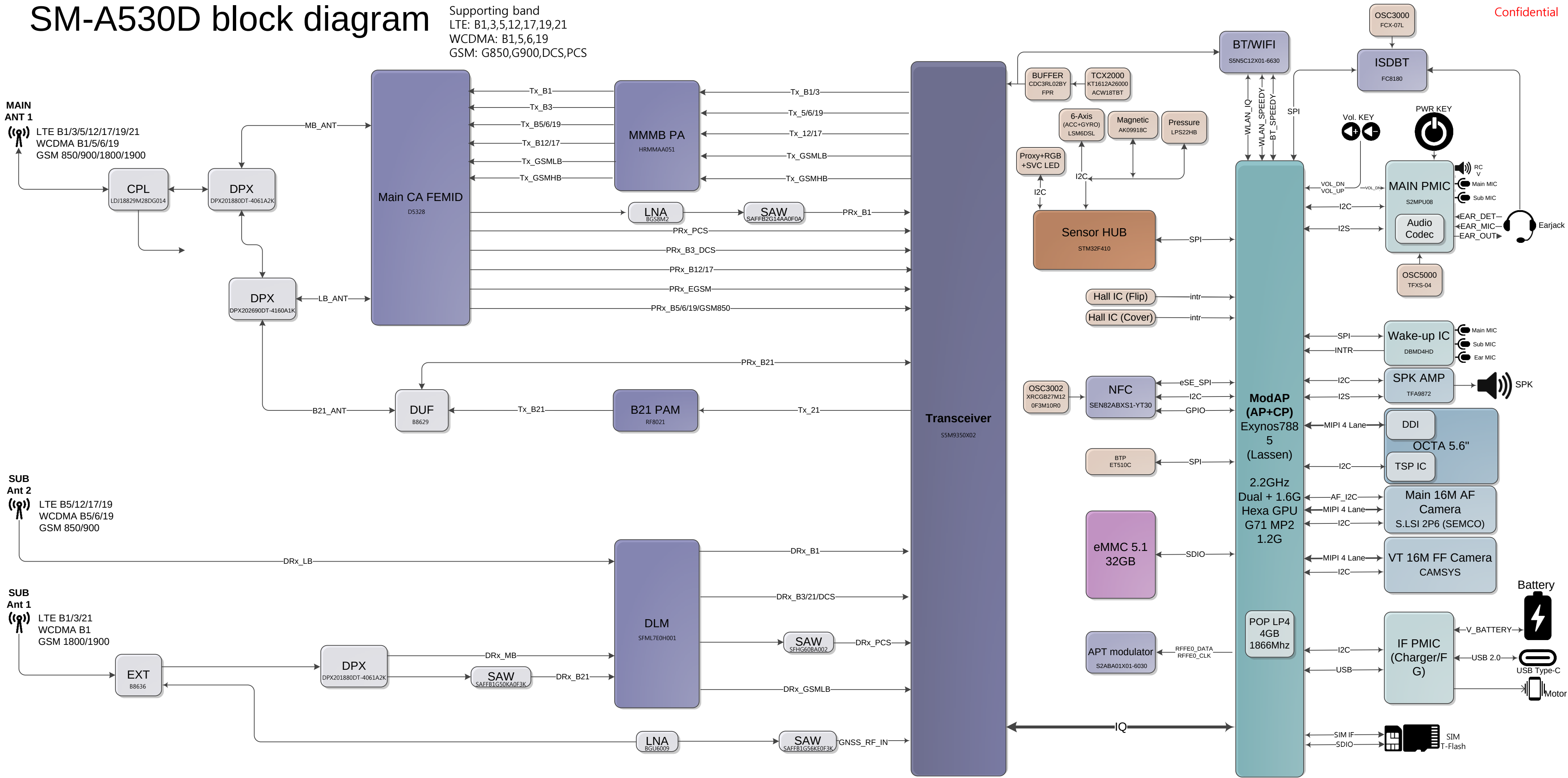
※ Caution

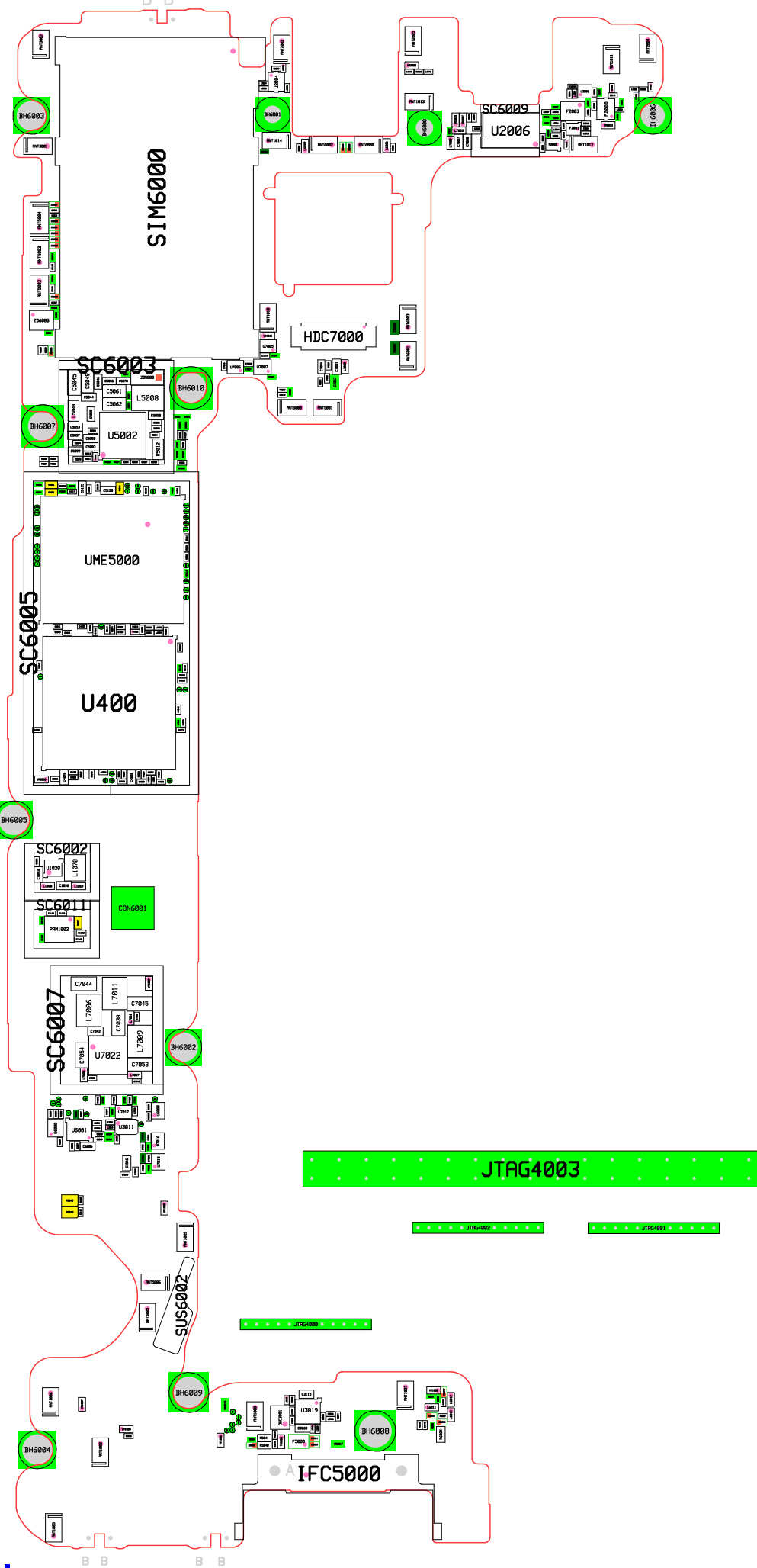
Be care of REAR scratch

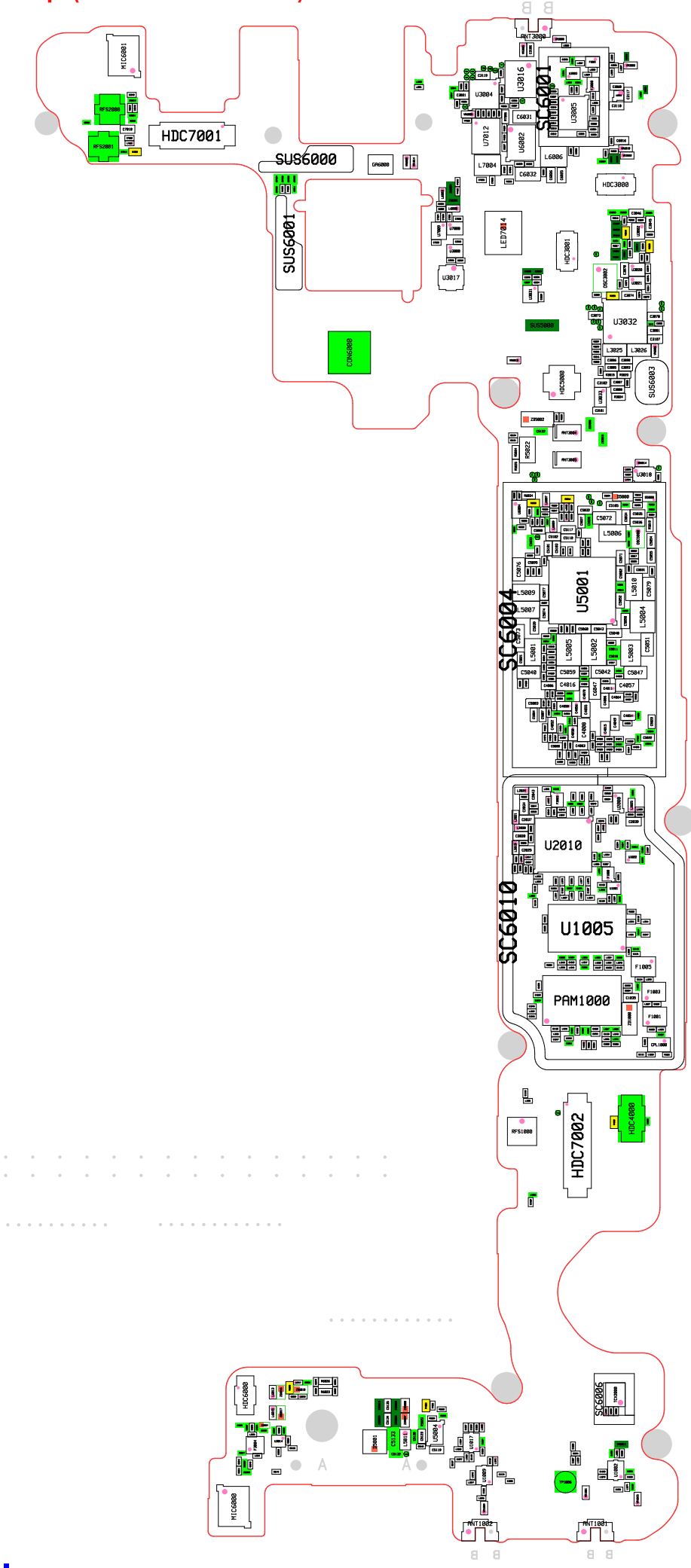
SM-A530D block diagram

Supporting band
LTE: B1,3,5,12,17,19,21
WCDMA: B1,5,6,19
GSM: G850,G900,DCS,PCS

Confidential

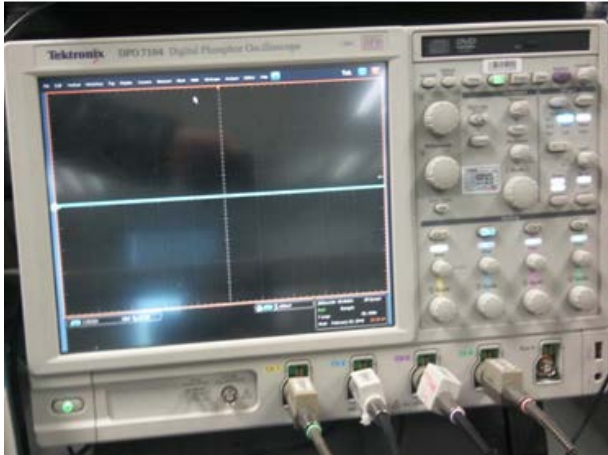






8. Level 3 Repair

8-3. Flow chart of Troubleshooting.



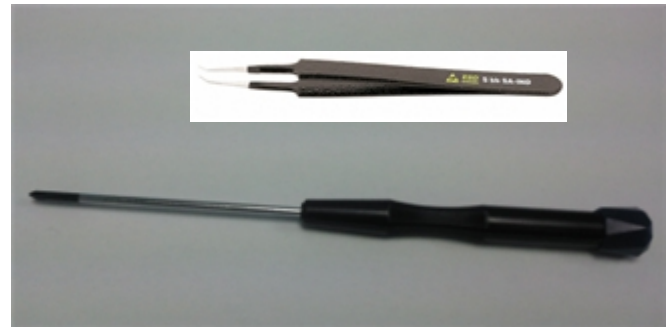
Oscilloscope



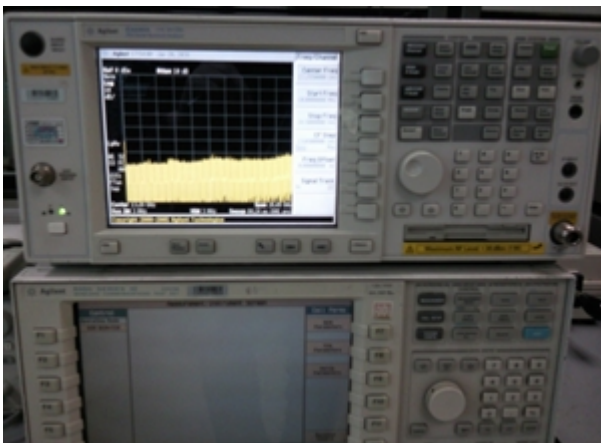
Digital Multimeter



Power Supply



+ driver, ESD Safe Tweezer



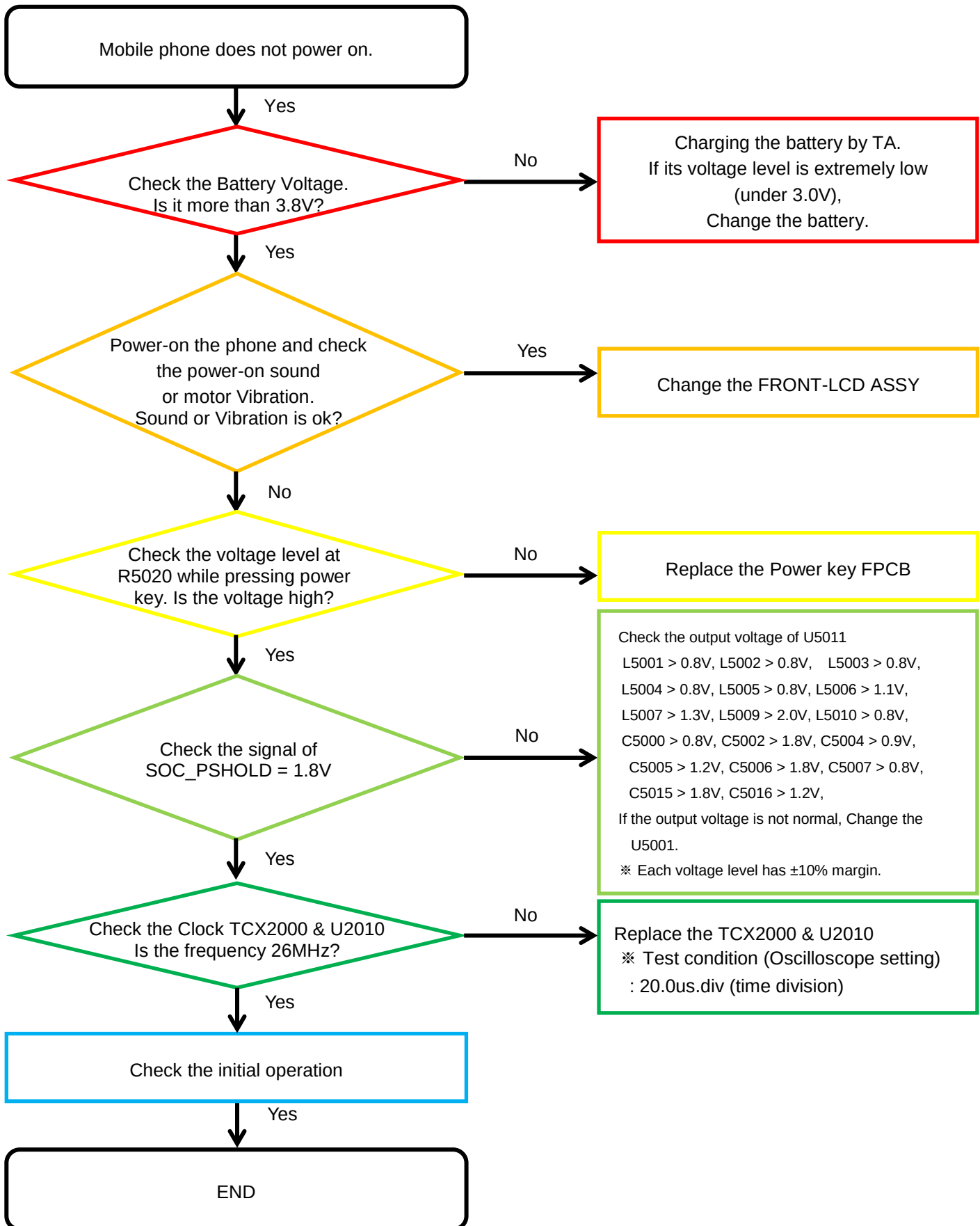
8960 & Spectrum Analyzer



Soldering iron

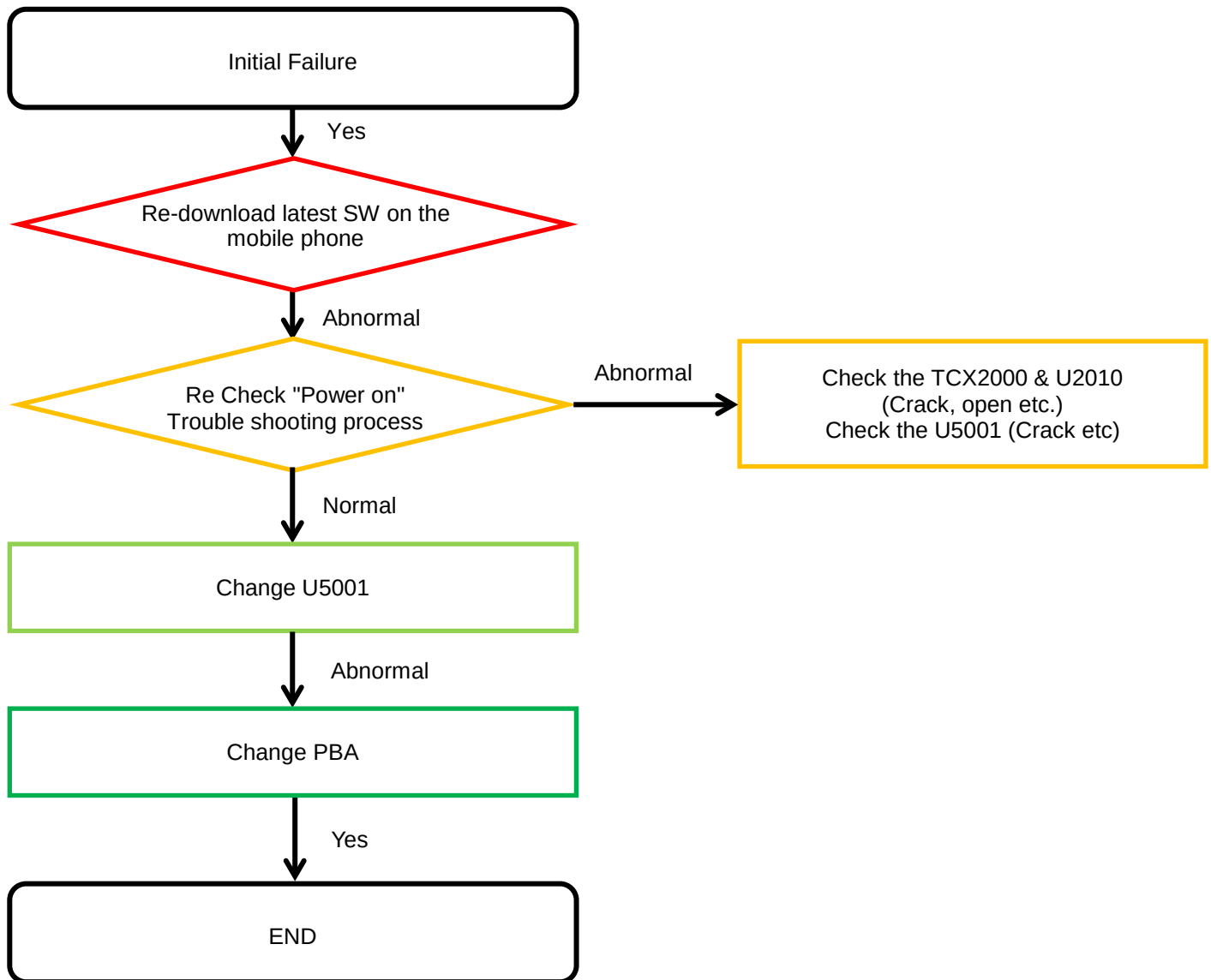
8. Level 3 Repair

8-4-1. Power On



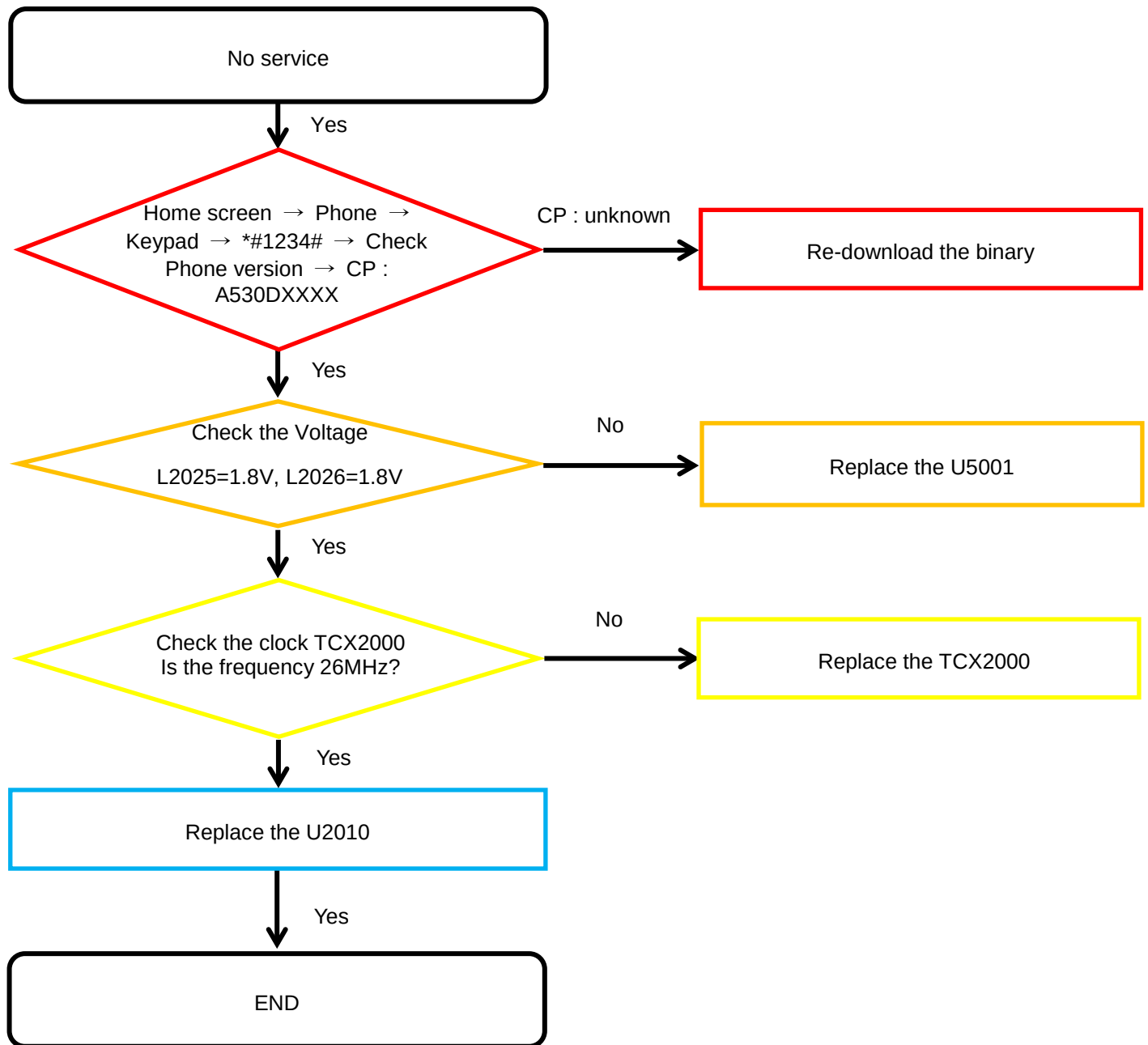
8. Level 3 Repair

8-4-2. Initial



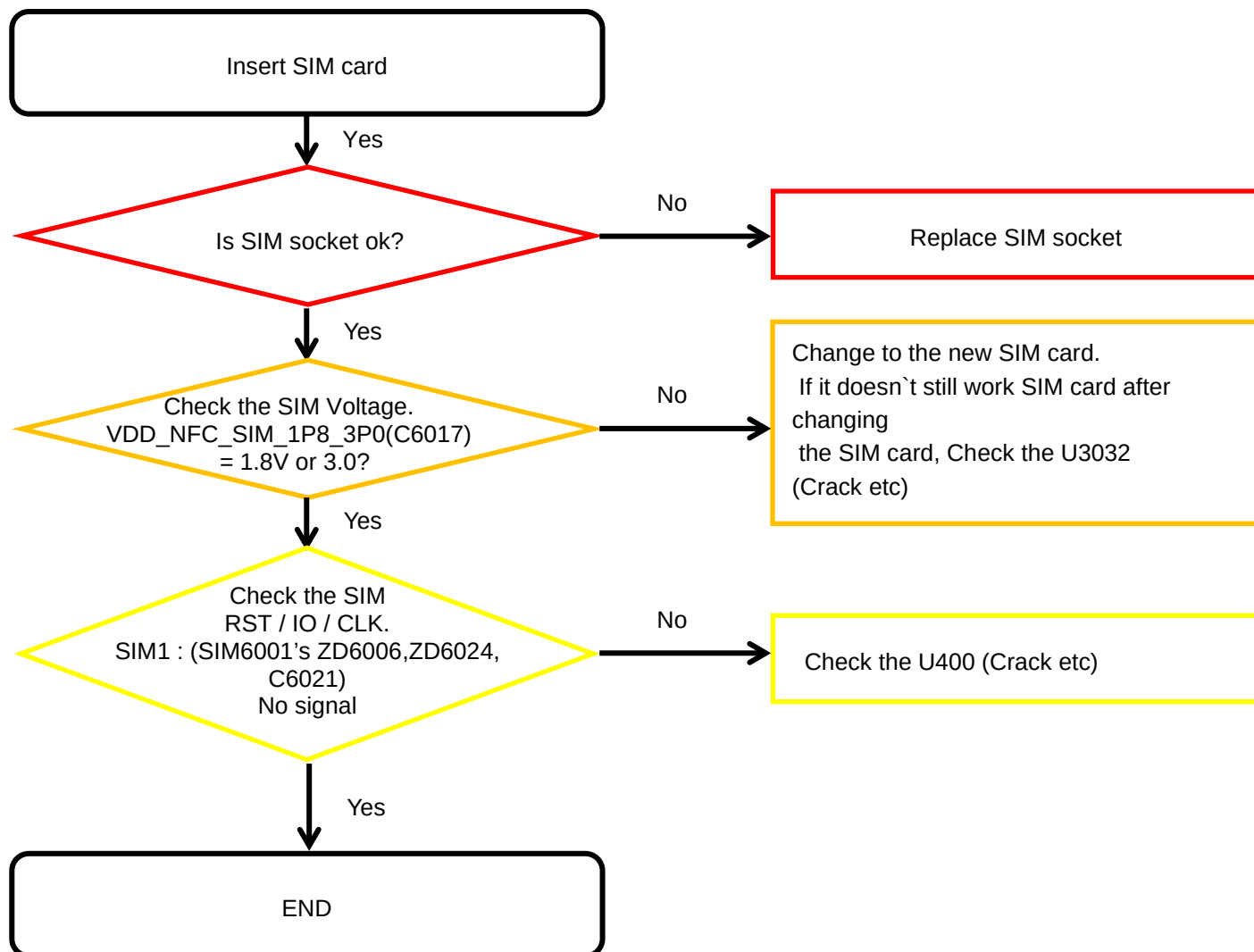
8. Level 3 Repair

8-4-3. No Service



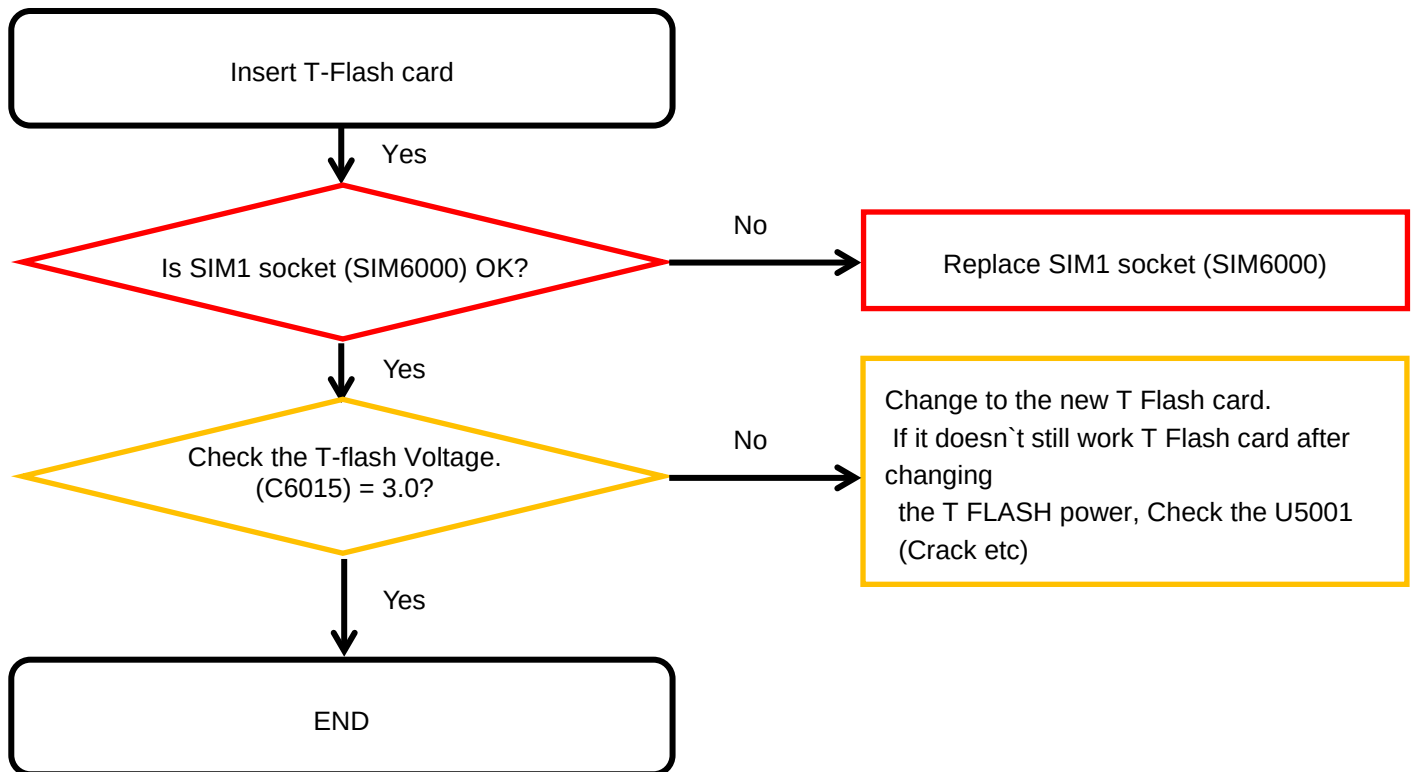
8. Level 3 Repair

8-4-4. SIM Part



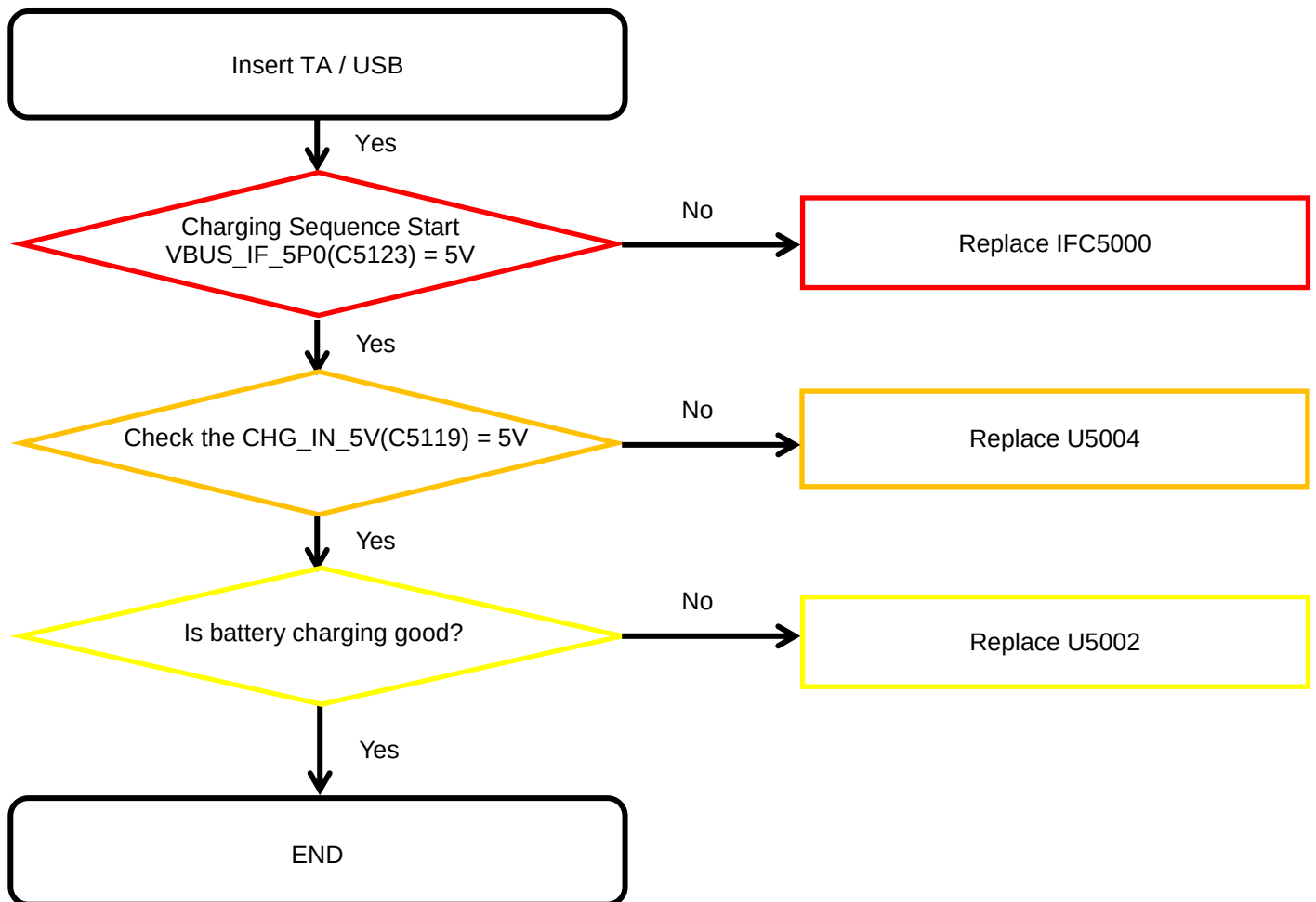
8. Level 3 Repair

8-4-5. T-Flash Part



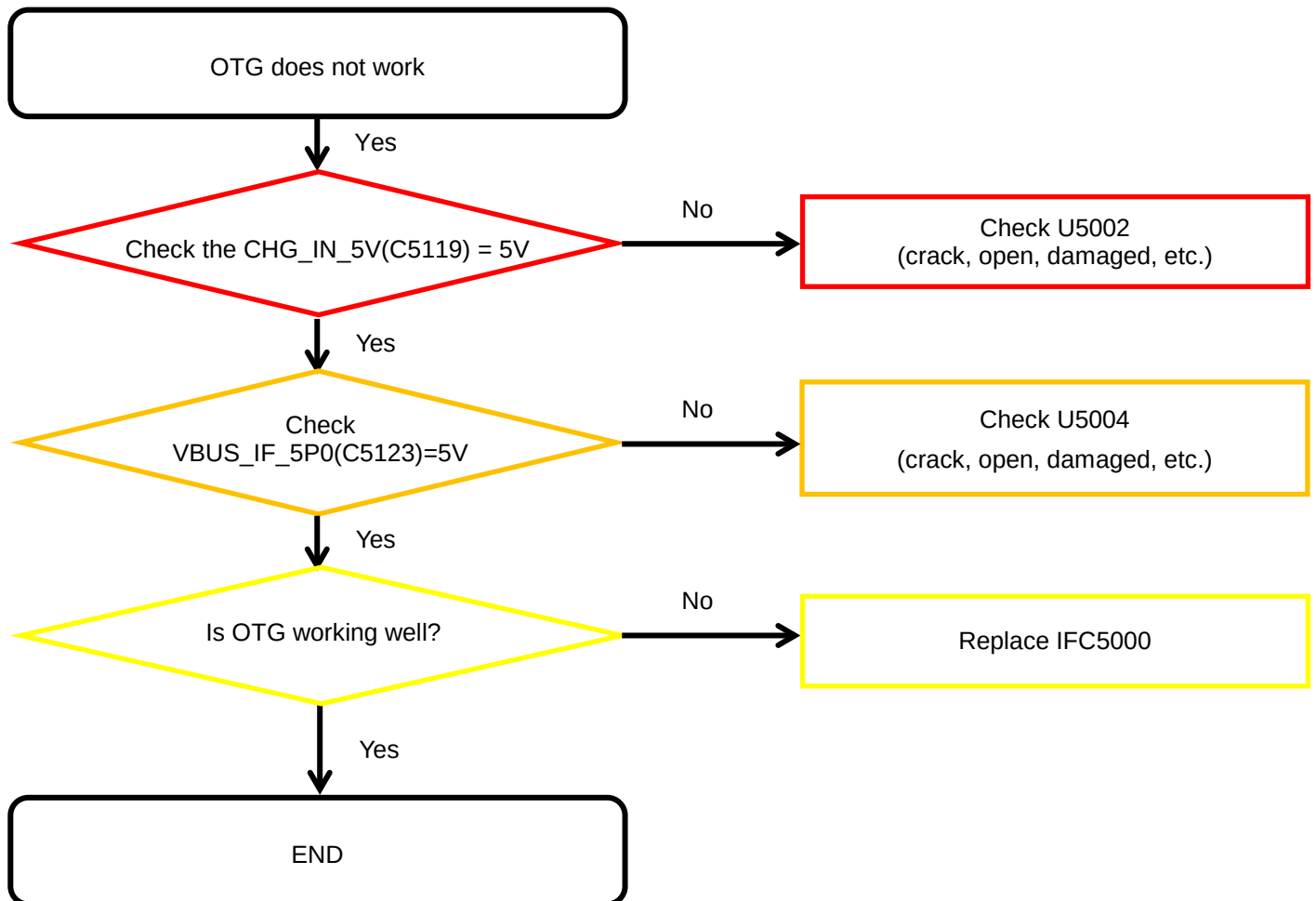
8. Level 3 Repair

8-4-6. Cable Charging Part



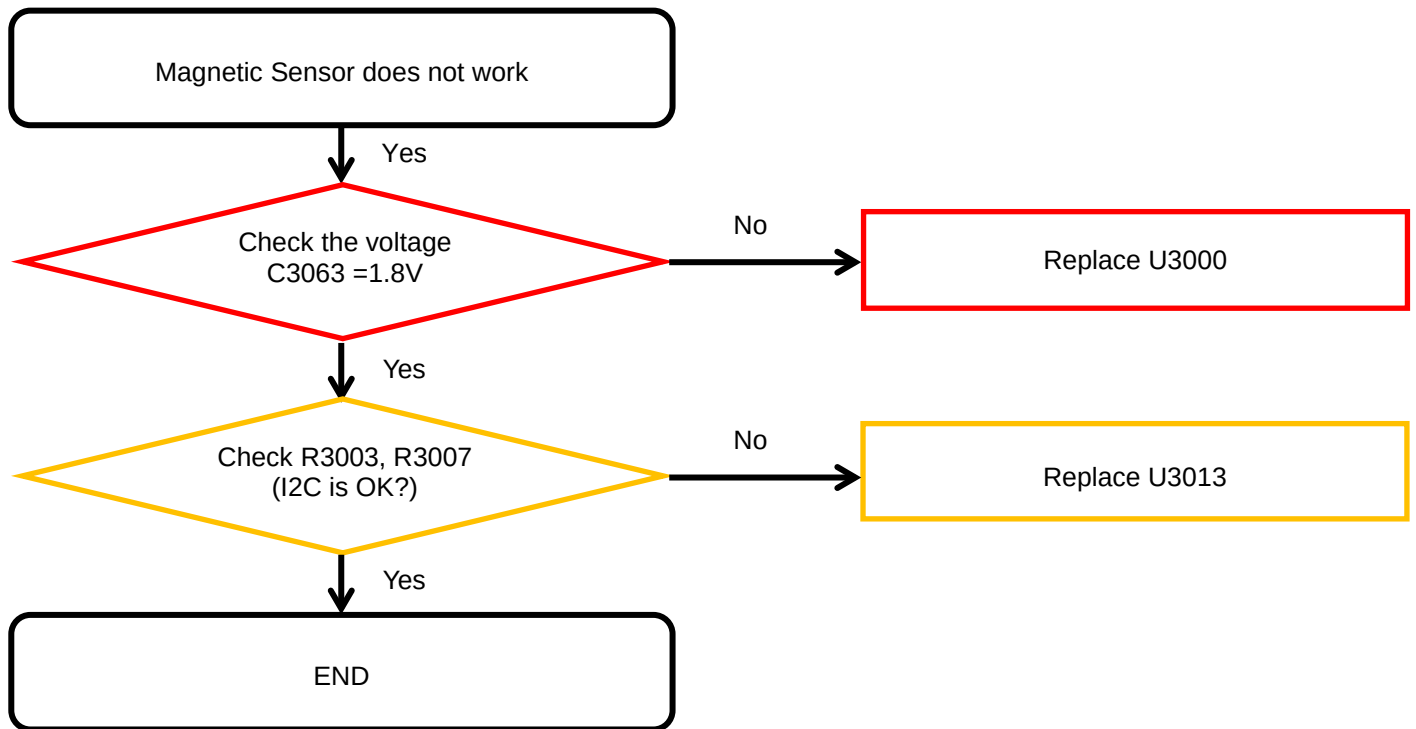
8. Level 3 Repair

8-4-7. OTG



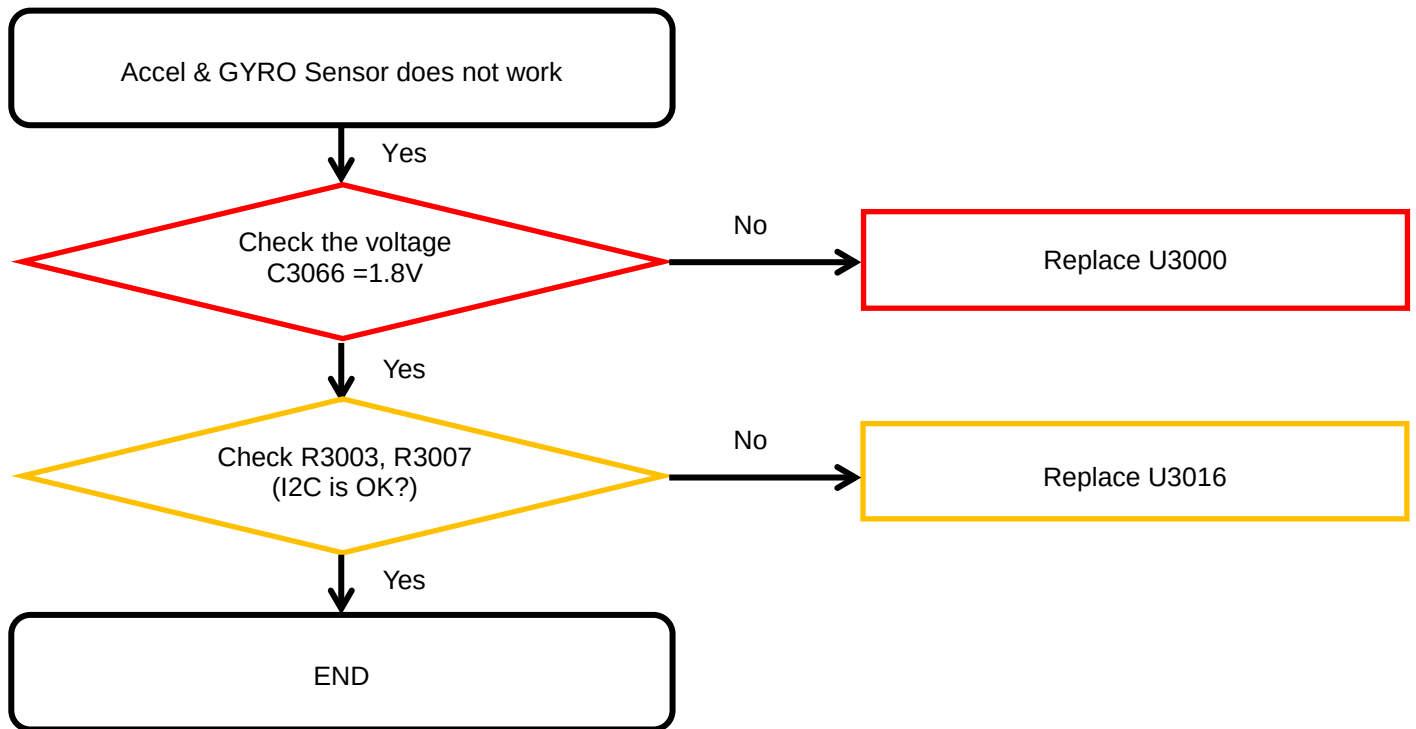
8. Level 3 Repair

8-4-8-1. Magnetic Sensor



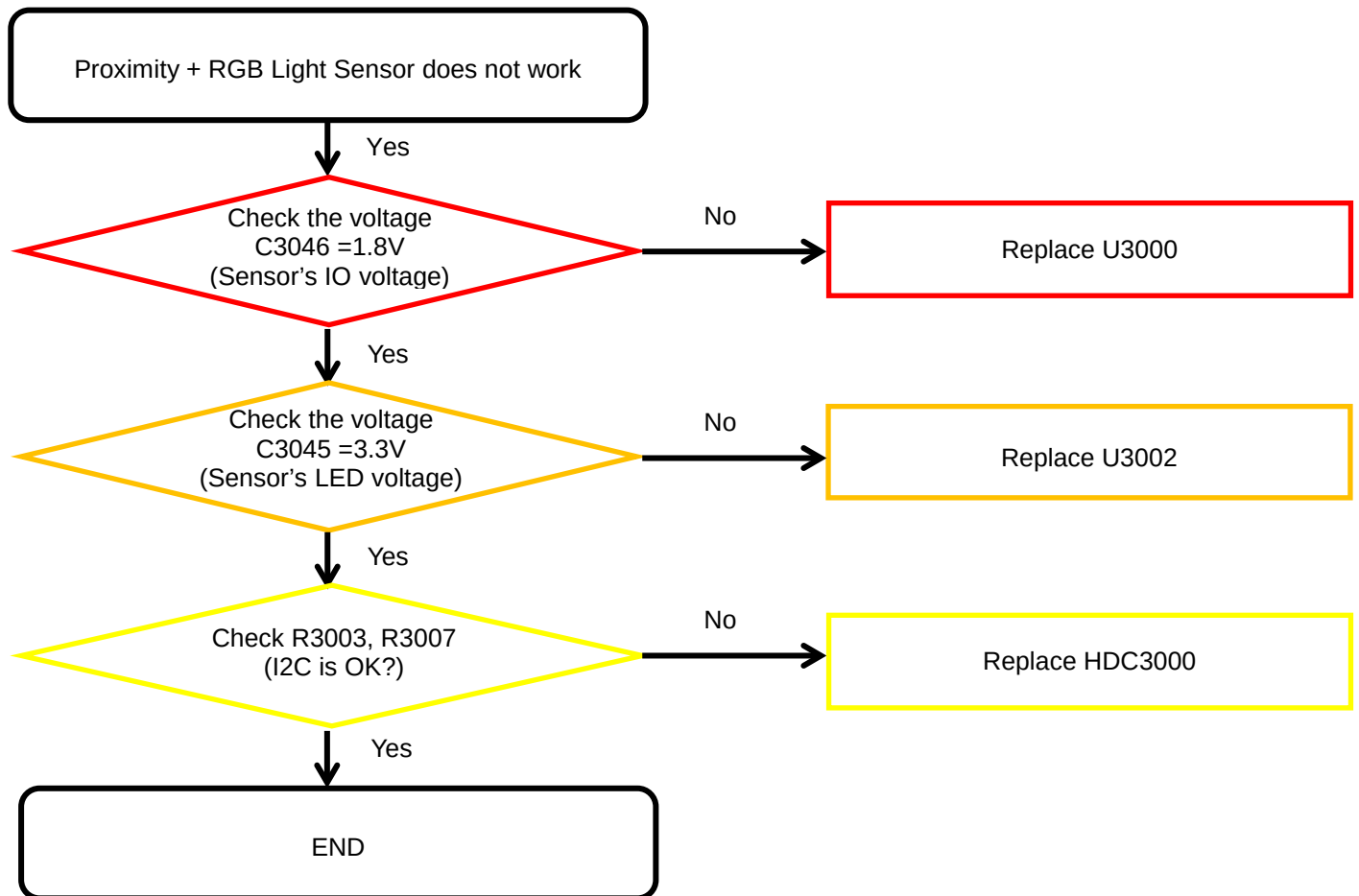
8. Level 3 Repair

8-4-8-2. Accel & GYRO Sensor



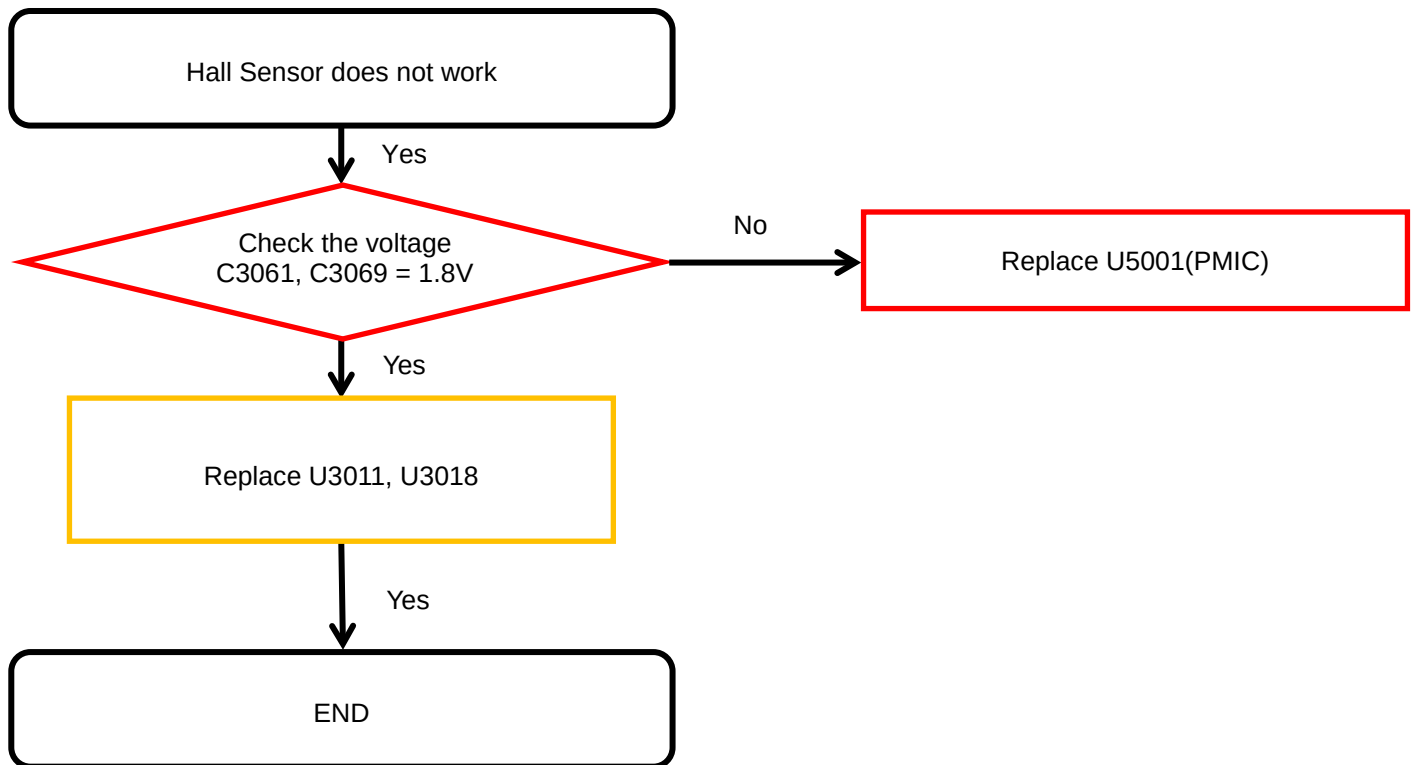
8. Level 3 Repair

8-4-8-3. Proximity + RGB Light sensor



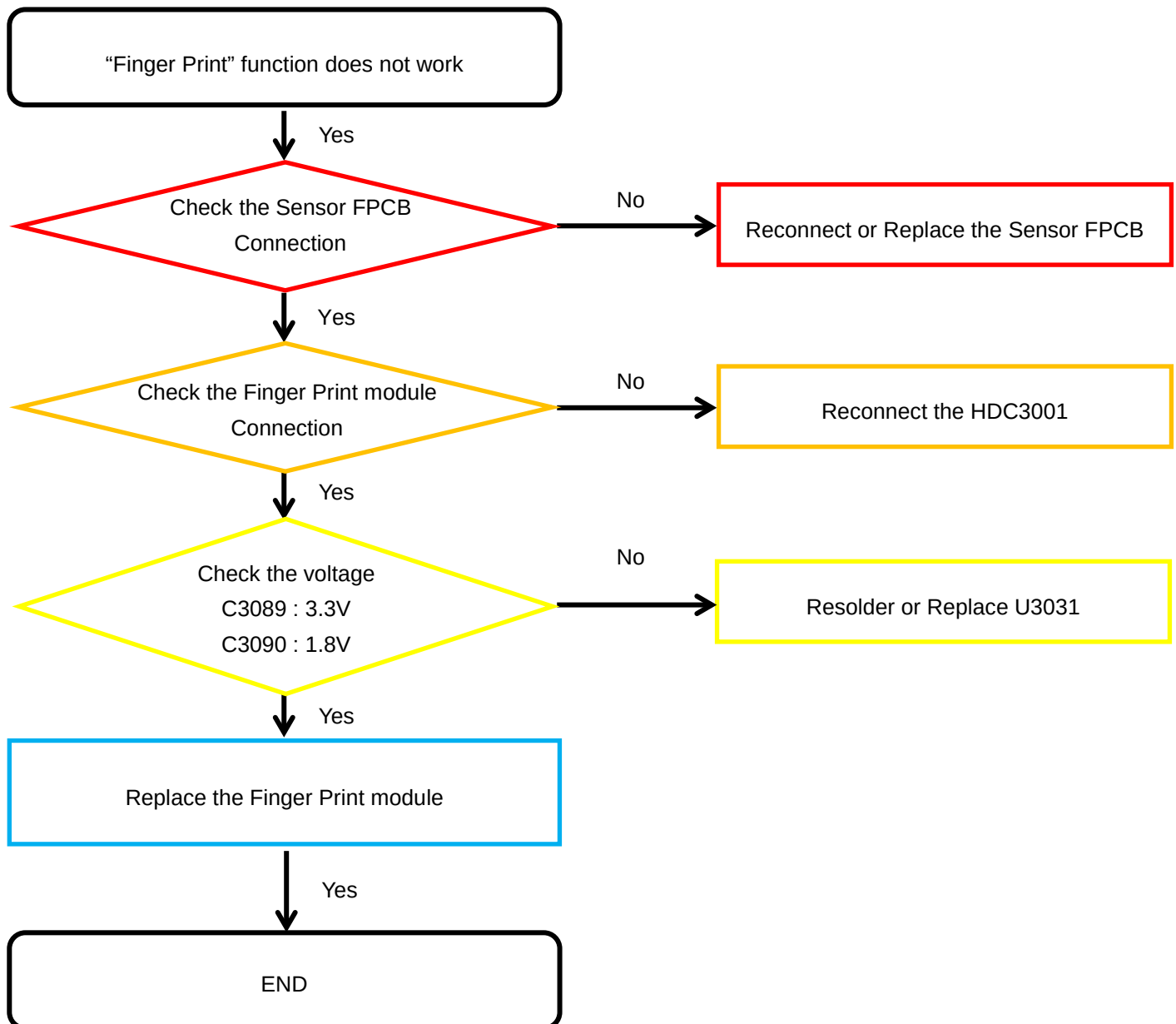
8. Level 3 Repair

8-4-8-4. Hall Sensor



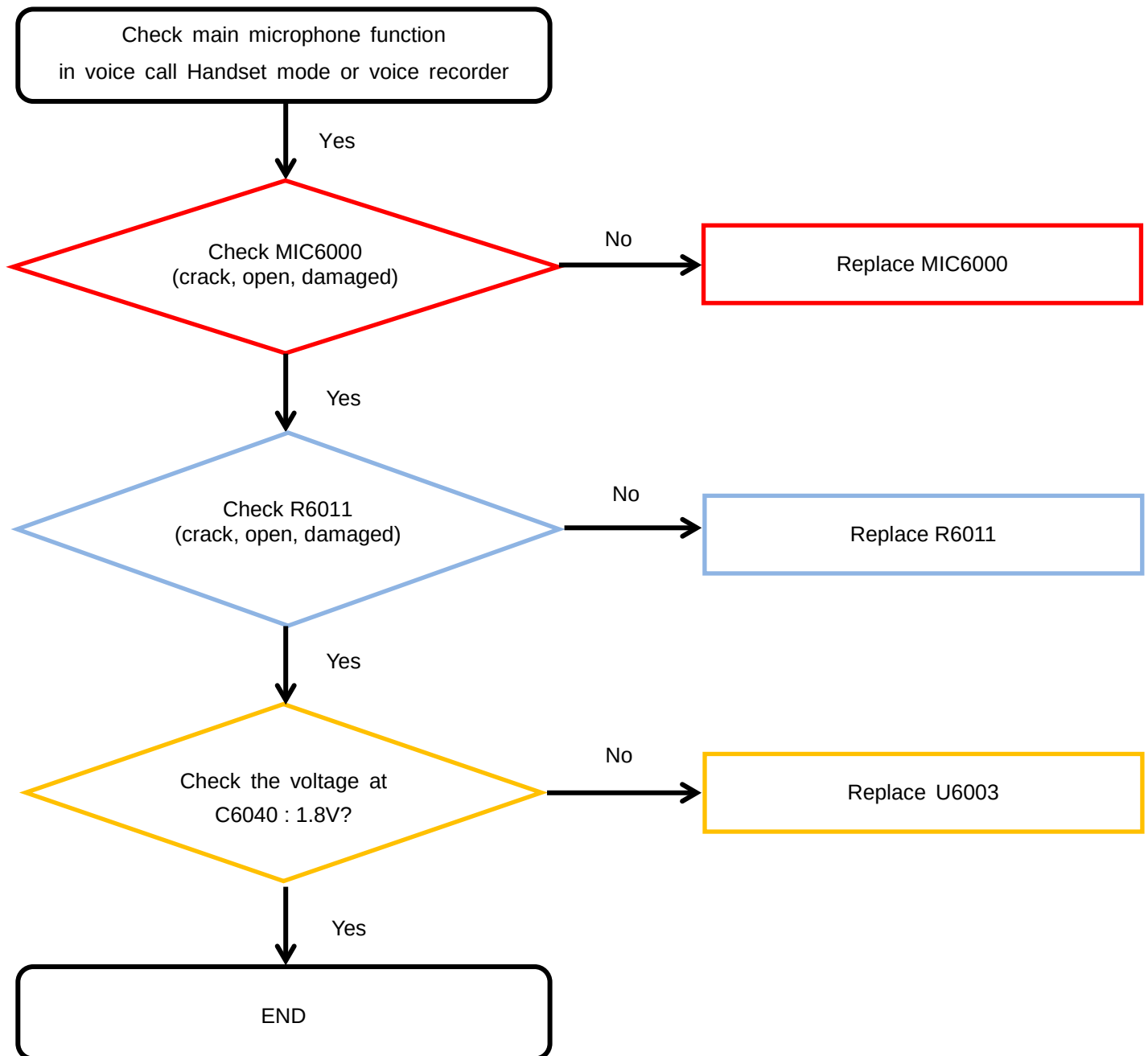
8. Level 3 Repair

8-4-8-5. Fingerprint Sensor



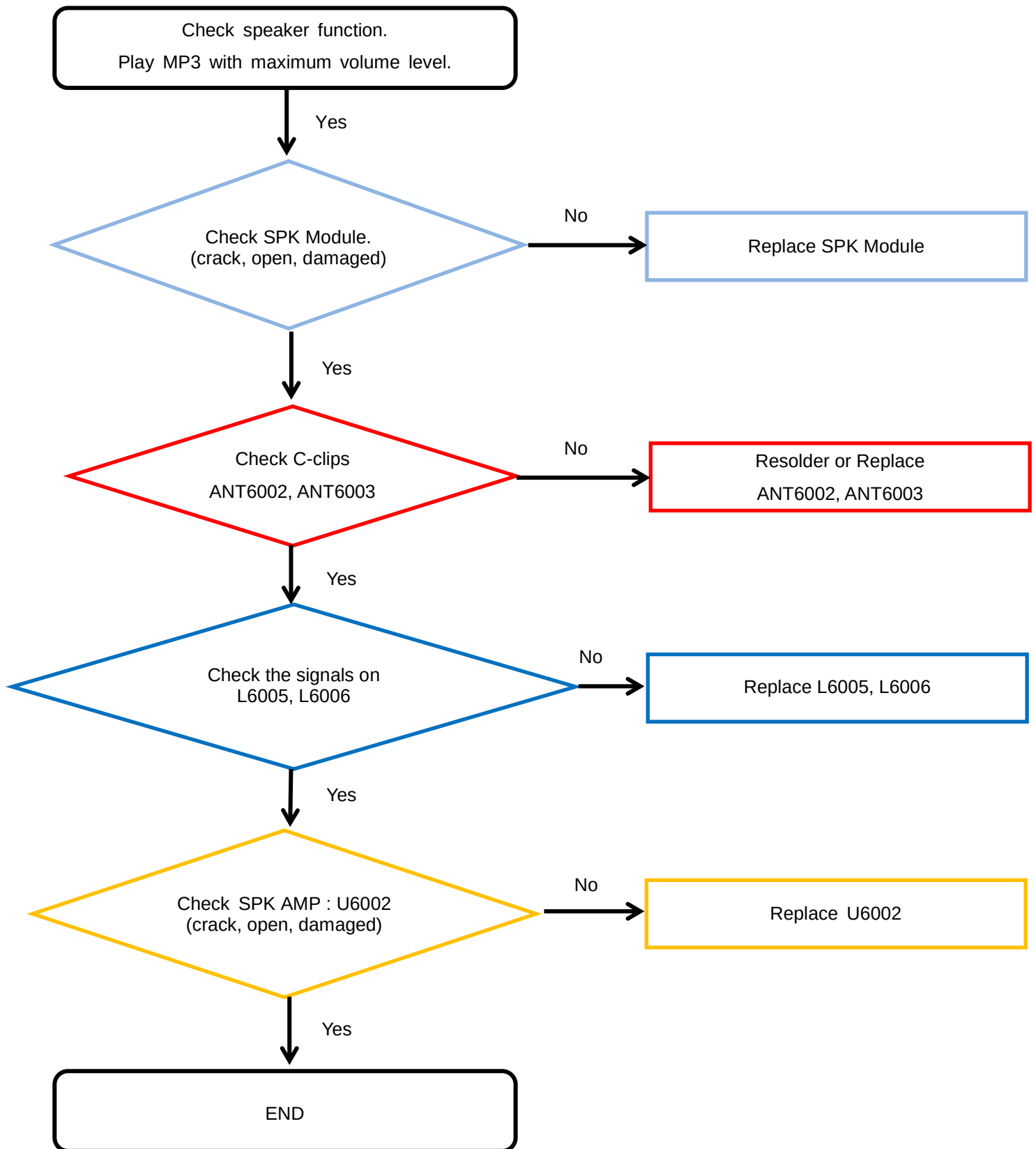
8. Level 3 Repair

8-4-9. Microphone Part



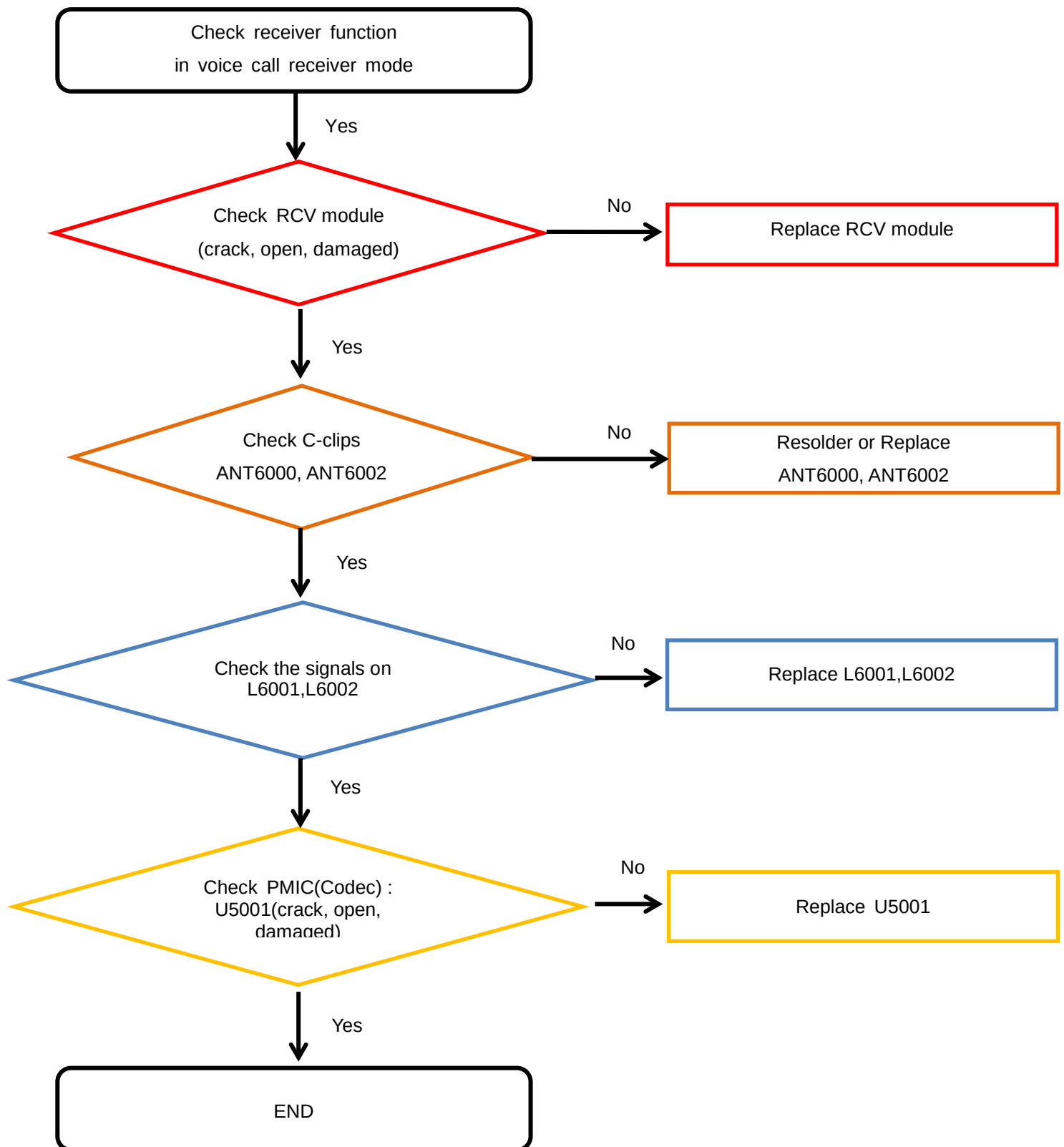
8. Level 3 Repair

8-4-10. Speaker Part



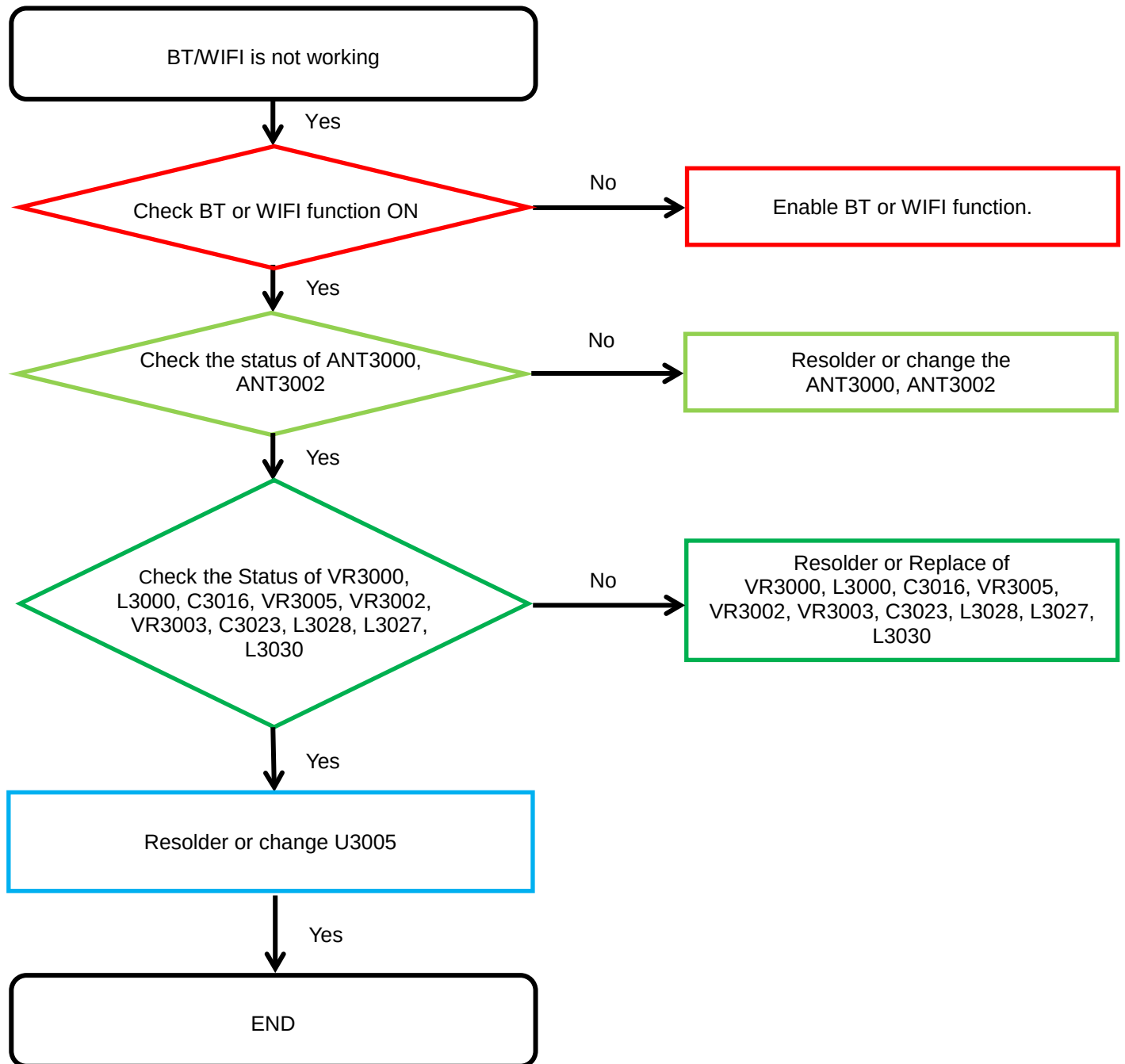
8. Level 3 Repair

8-4-11. Receiver Part



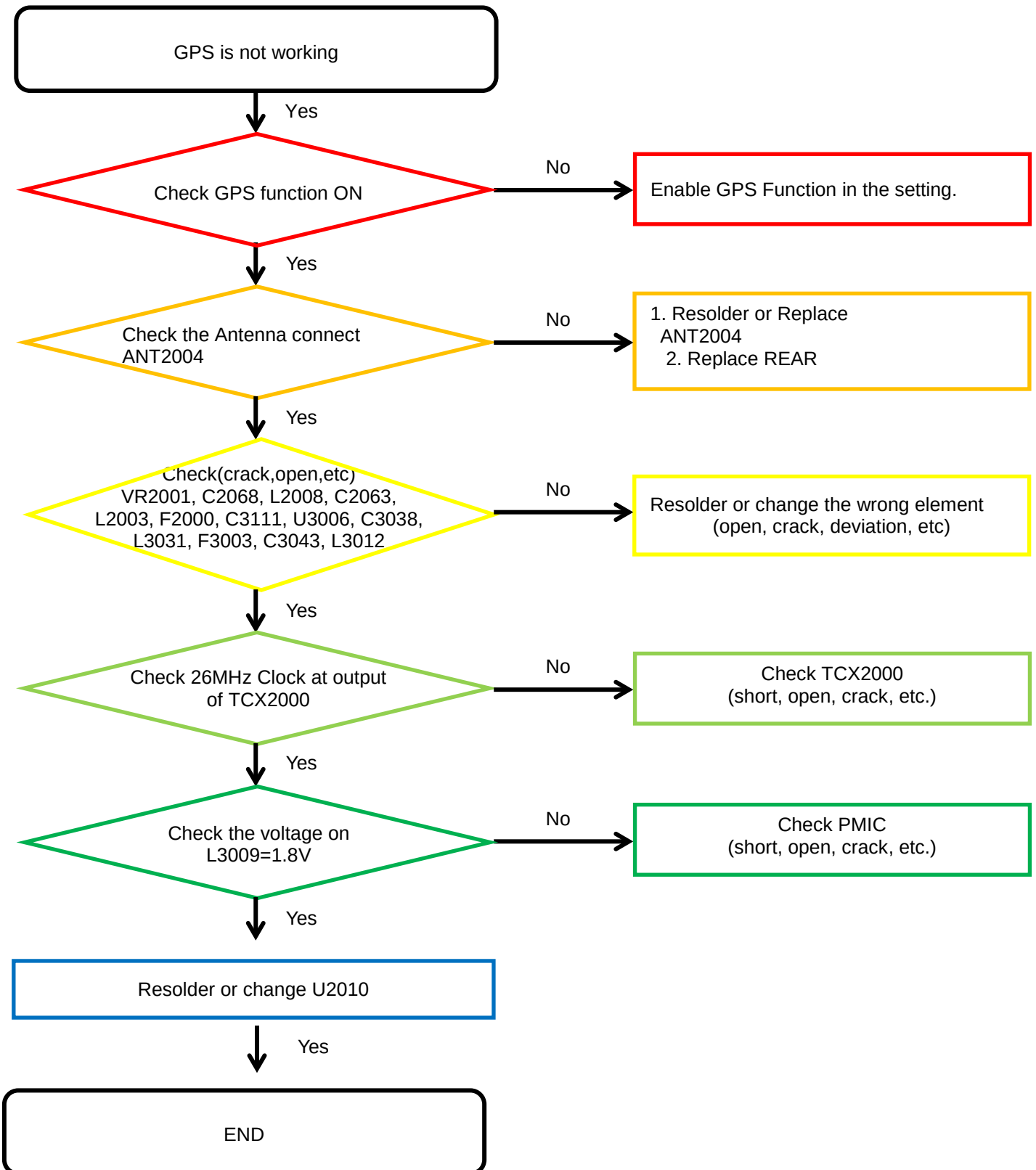
8. Level 3 Repair

8-4-12. BT/WIFI



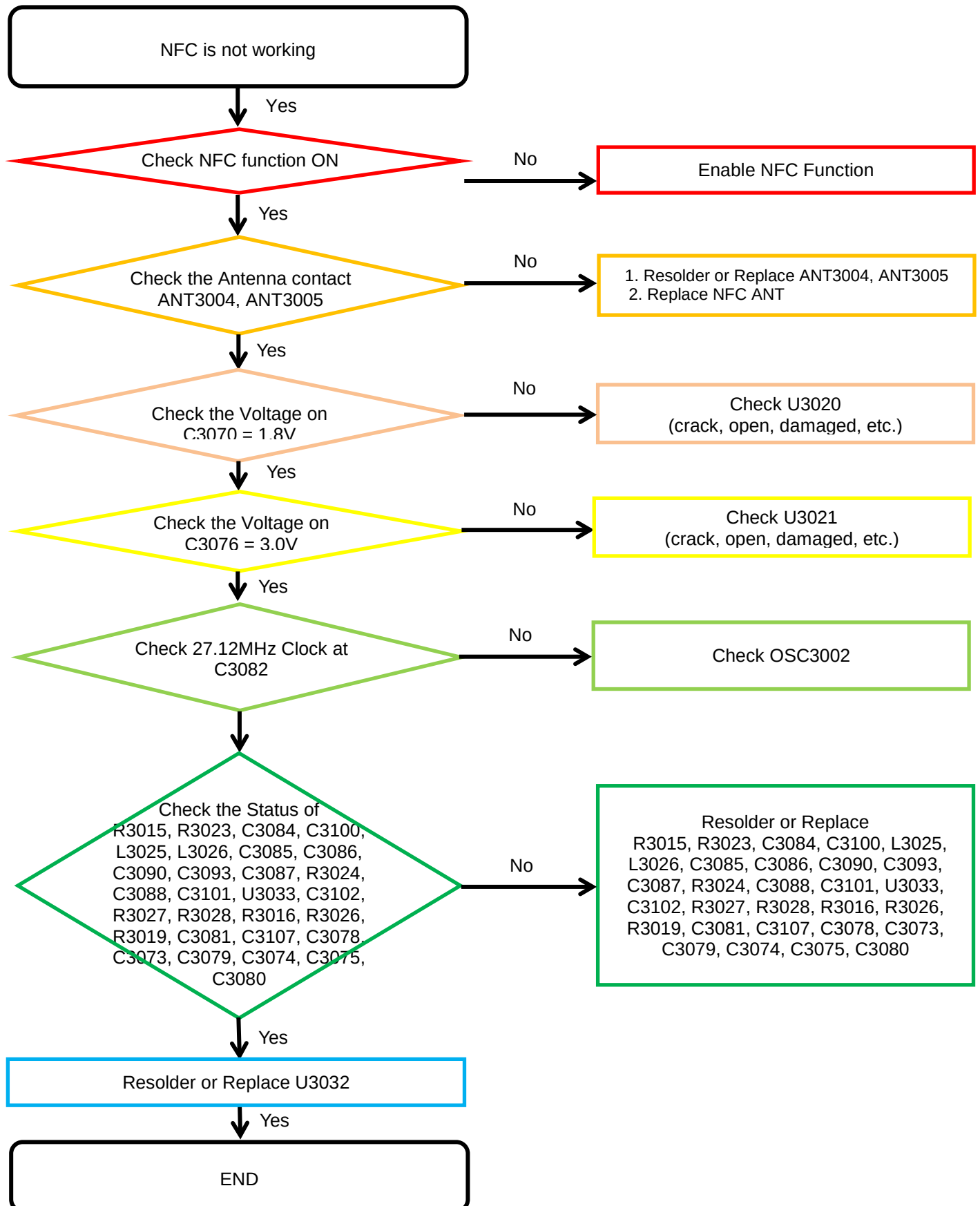
8. Level 3 Repair

8-4-13. GPS



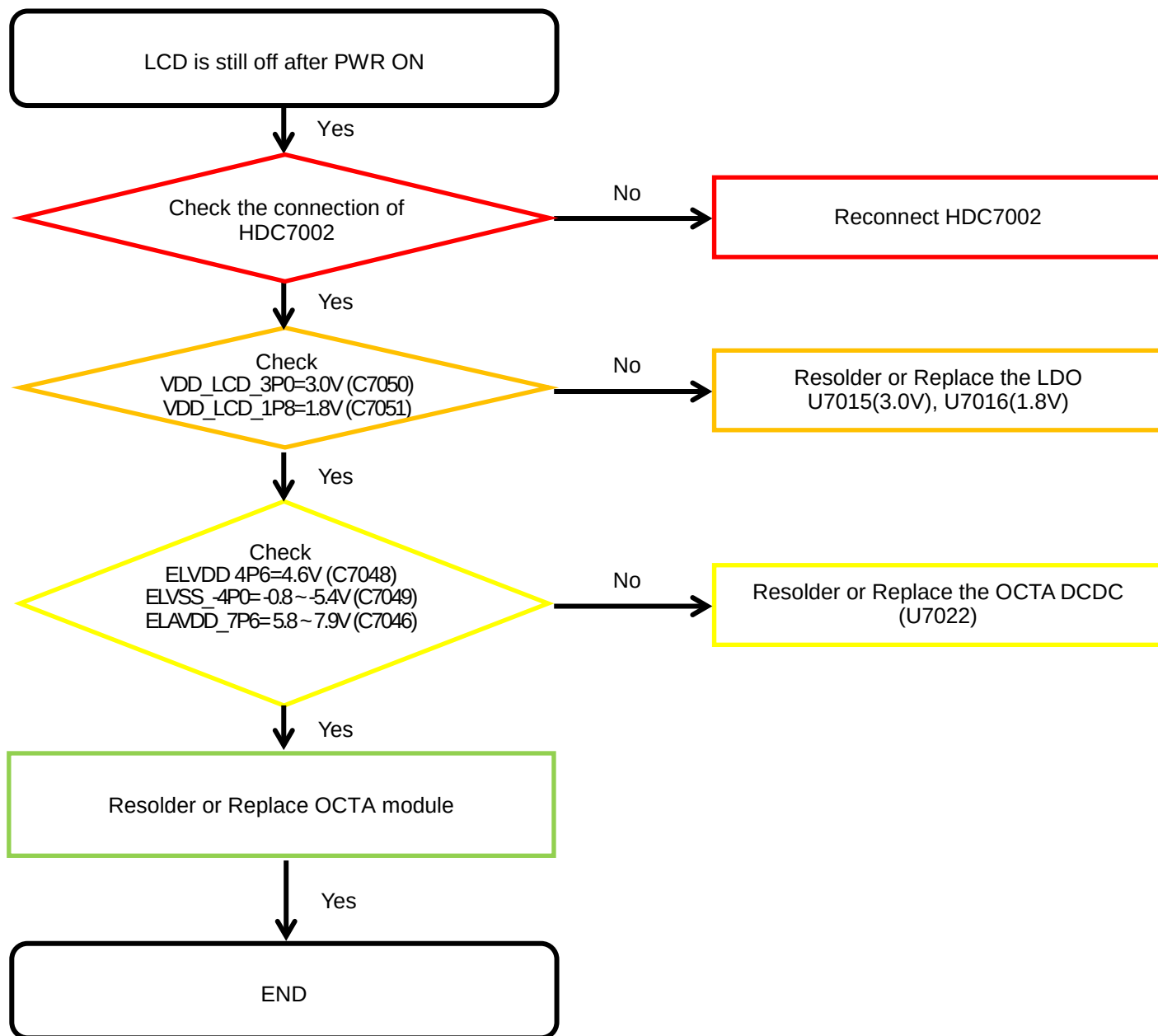
8. Level 3 Repair

8-4-14. NFC



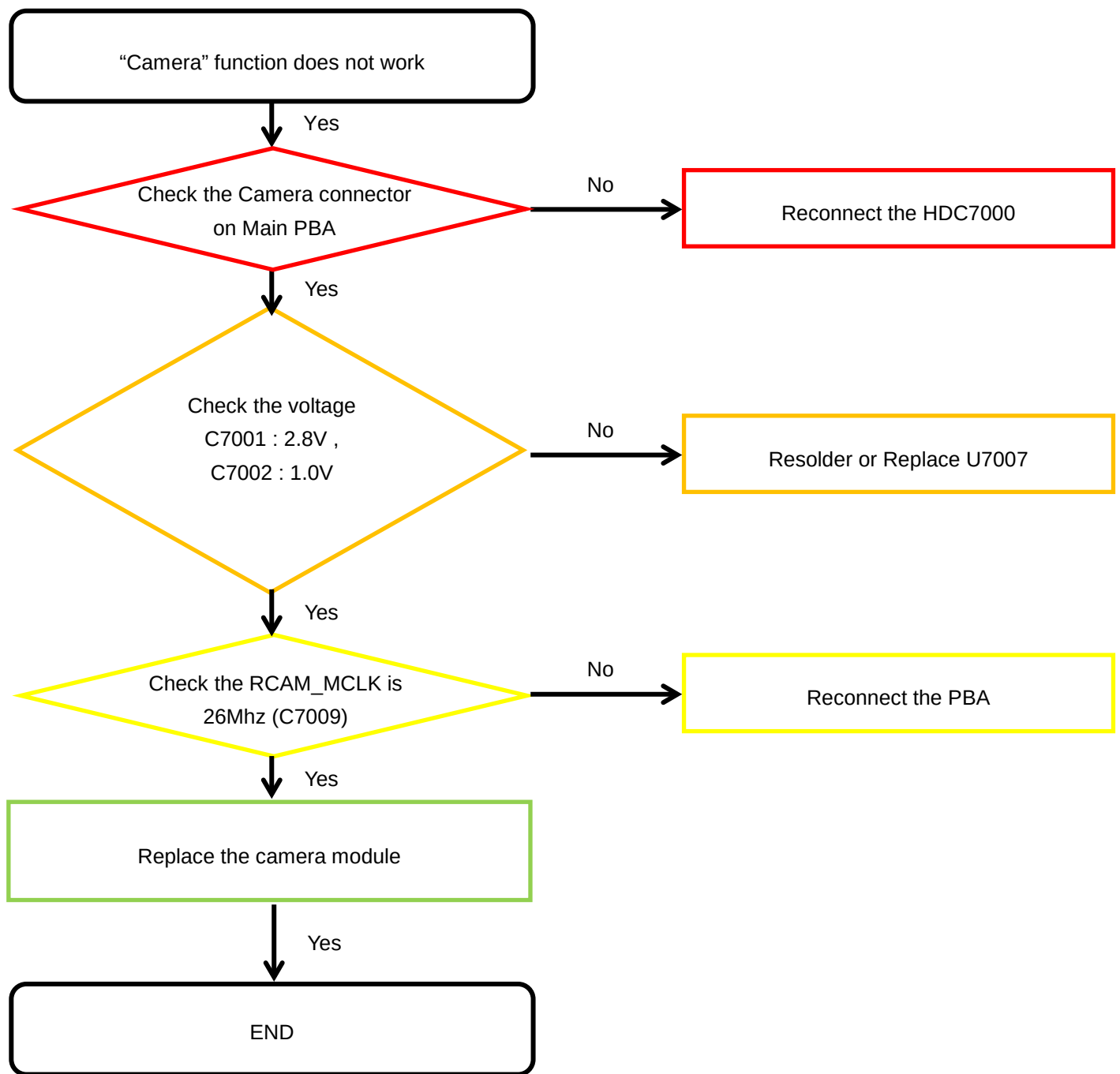
8. Level 3 Repair

8-4-15. Display



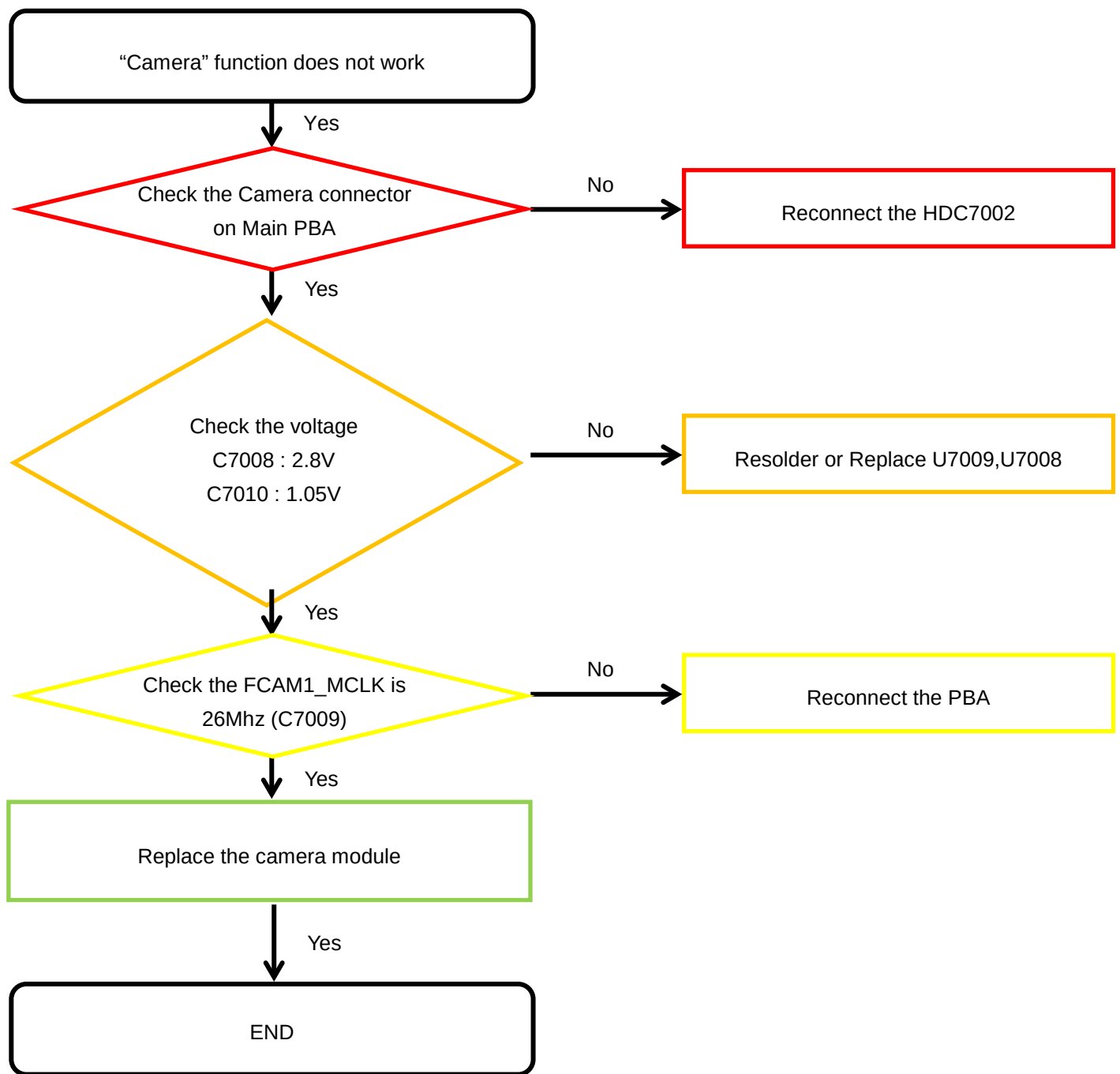
8. Level 3 Repair

8-4-16. REAR CAMERA



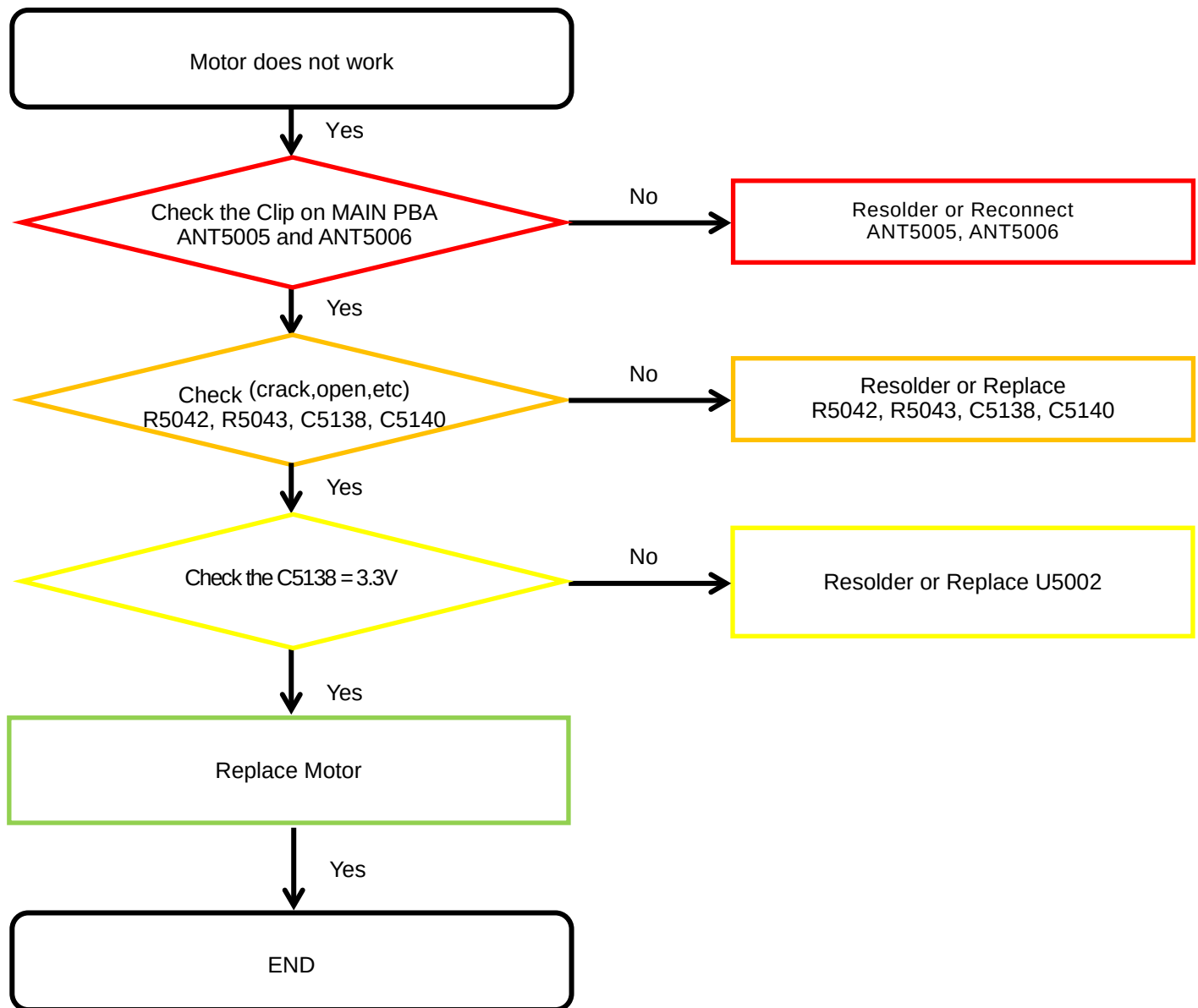
8. Level 3 Repair

8-4-17. FRONT CAMERA



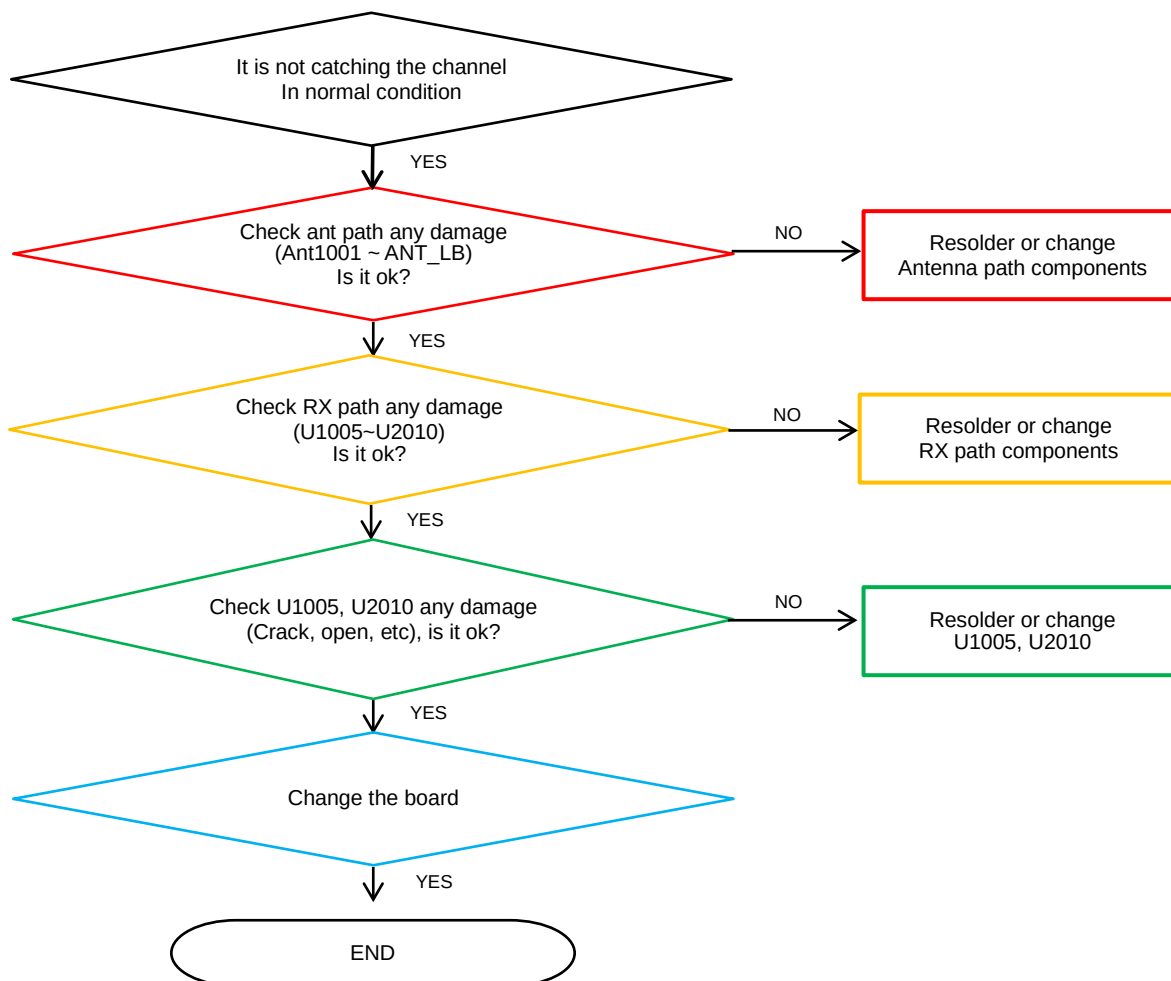
8. Level 3 Repair

8-4-18. Motor



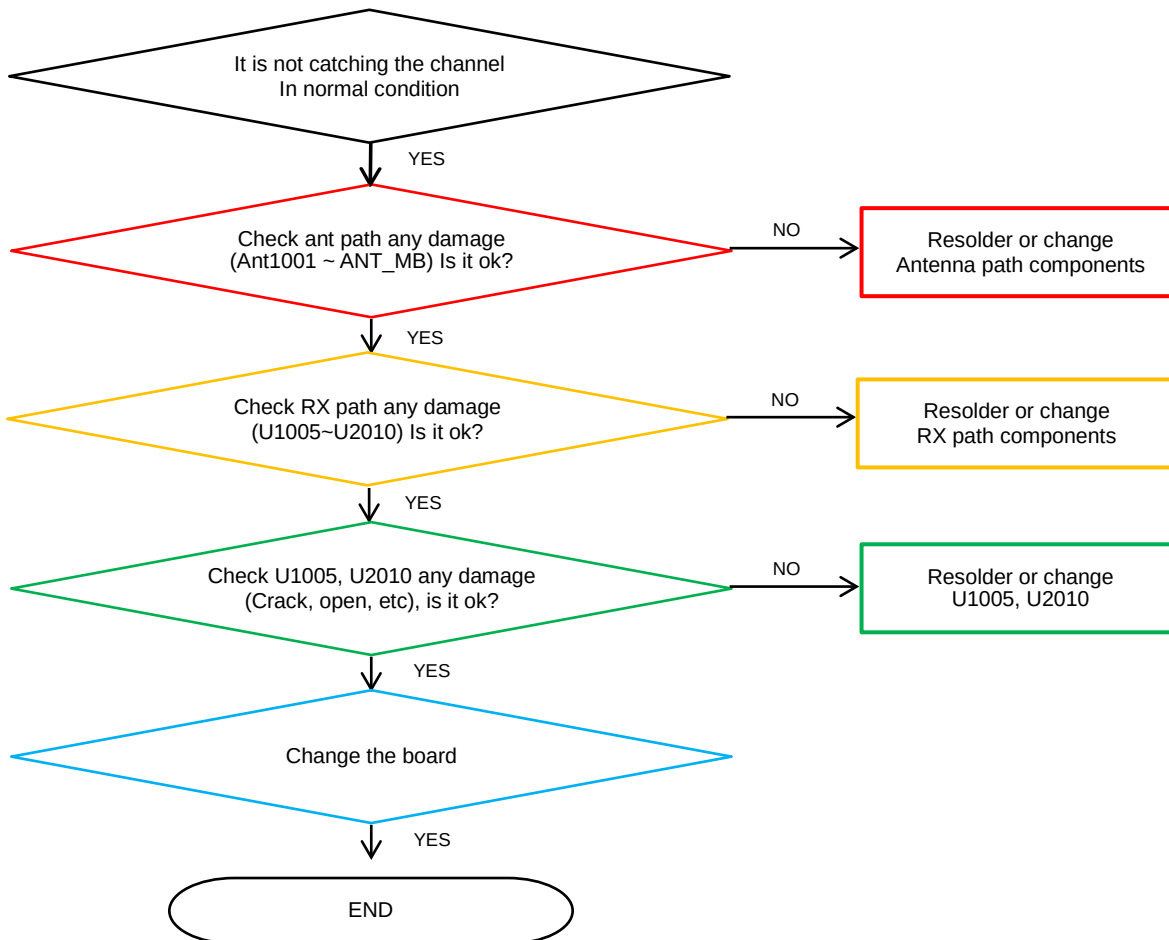
8. Level 3 Repair

8-4-19. GSM 850/900, WCDMA B5/6/19, LTE B5/12/17/19 Rx



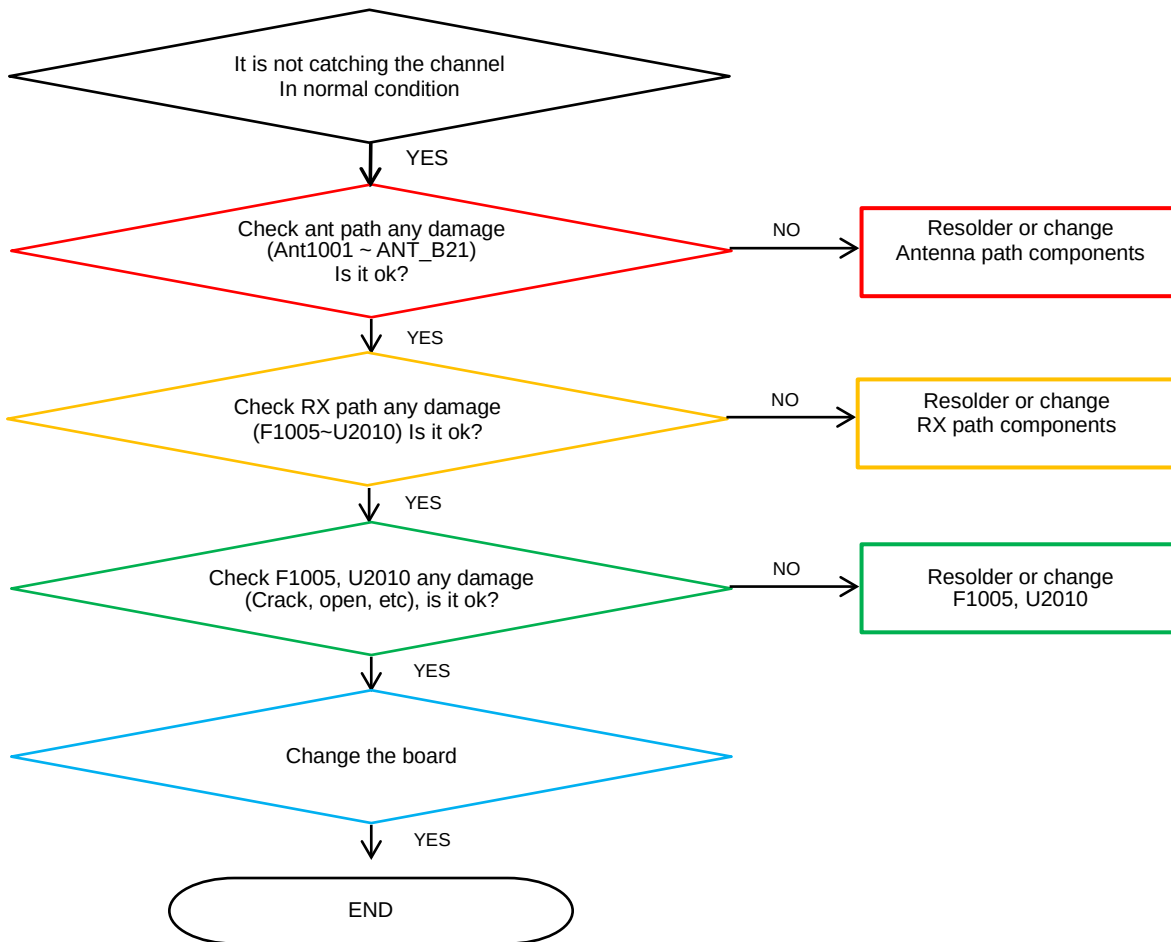
8. Level 3 Repair

8-4-20. GSM DCS/PCS, WCDMA B1, LTE B1/3 Rx



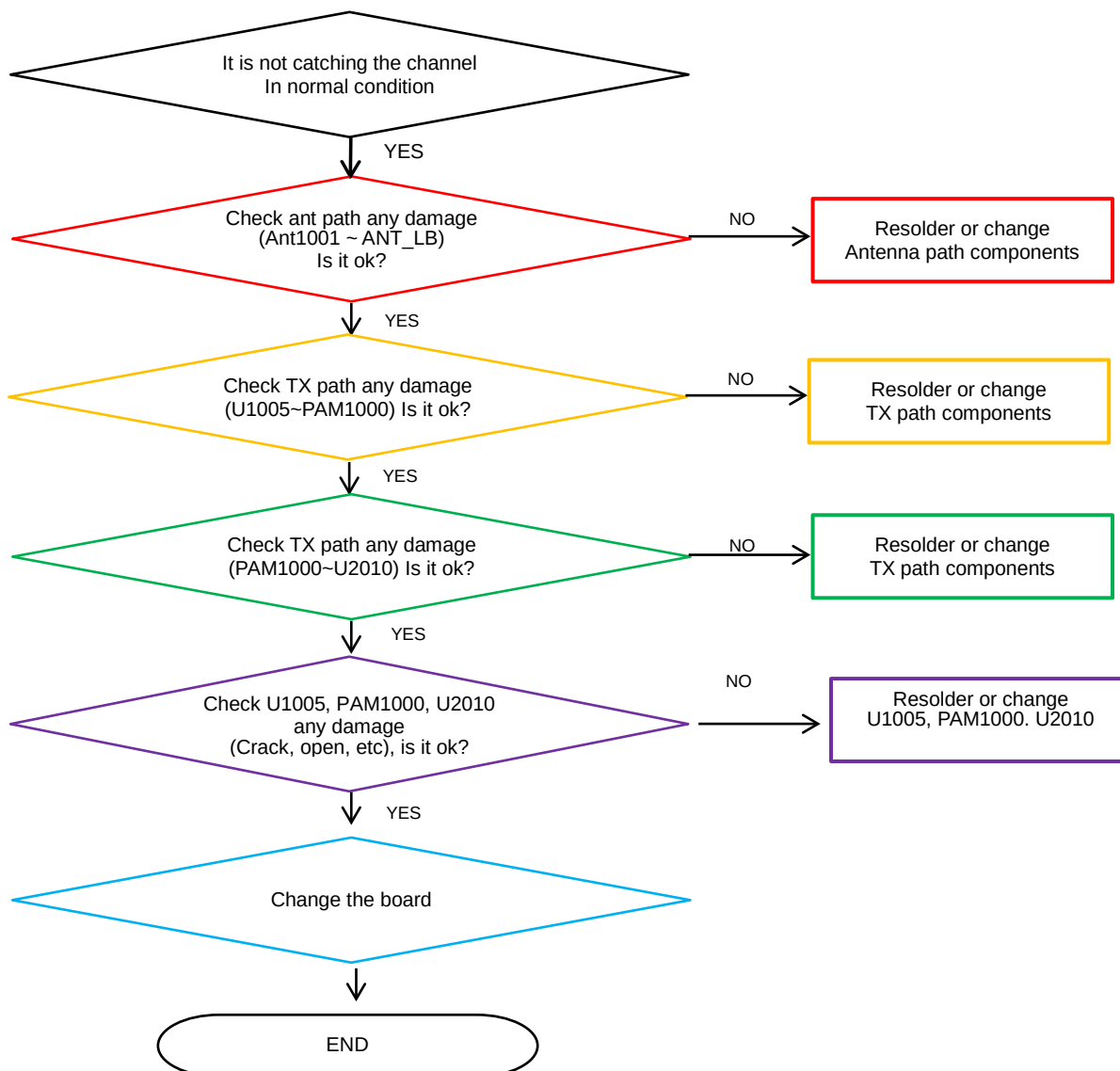
8. Level 3 Repair

8-4-21. LTE B21 RX



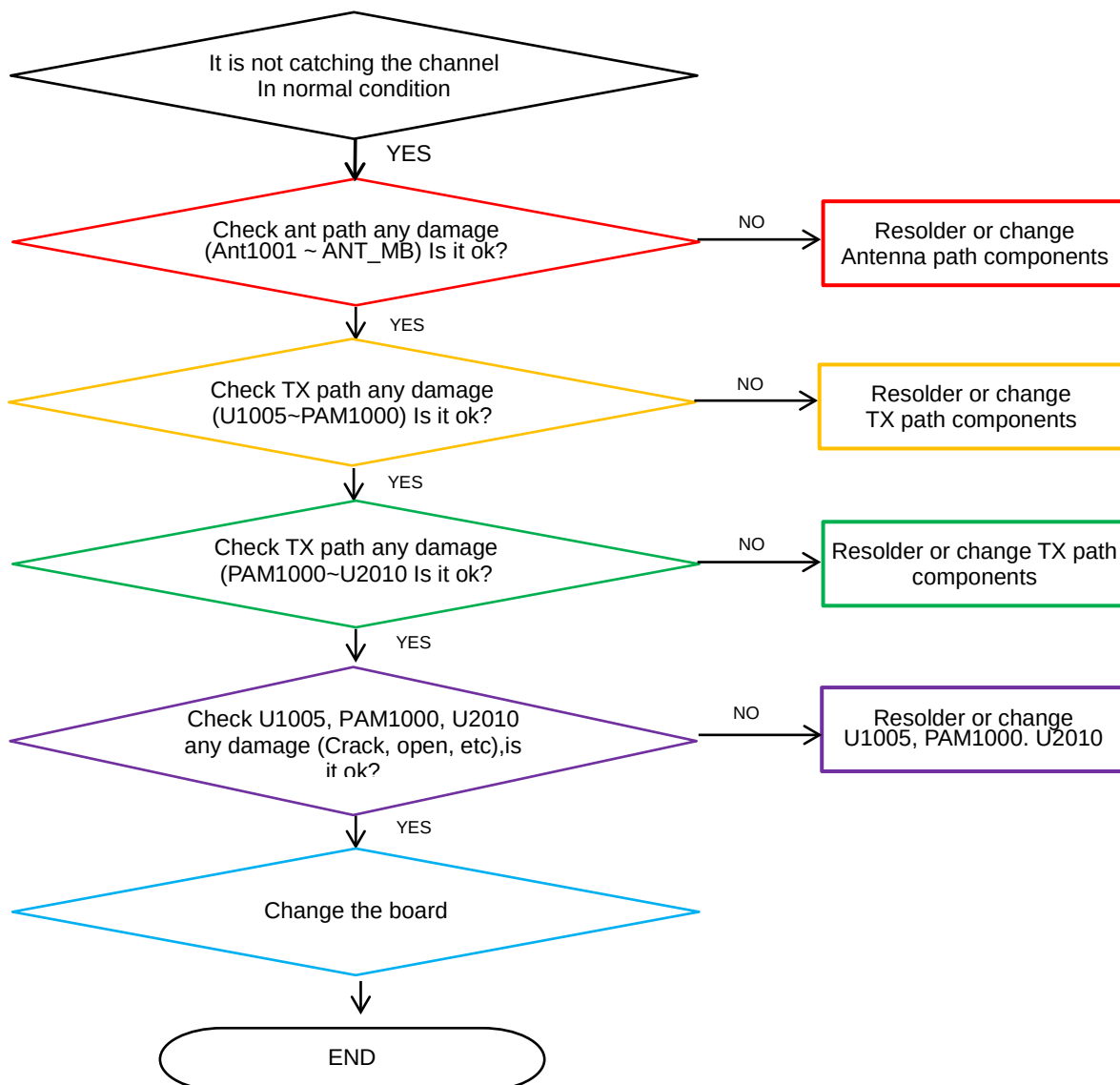
8. Level 3 Repair

8-4-22. GSM 850/900, WCDMA B5/6/19, LTE B5/12/17/19 Tx



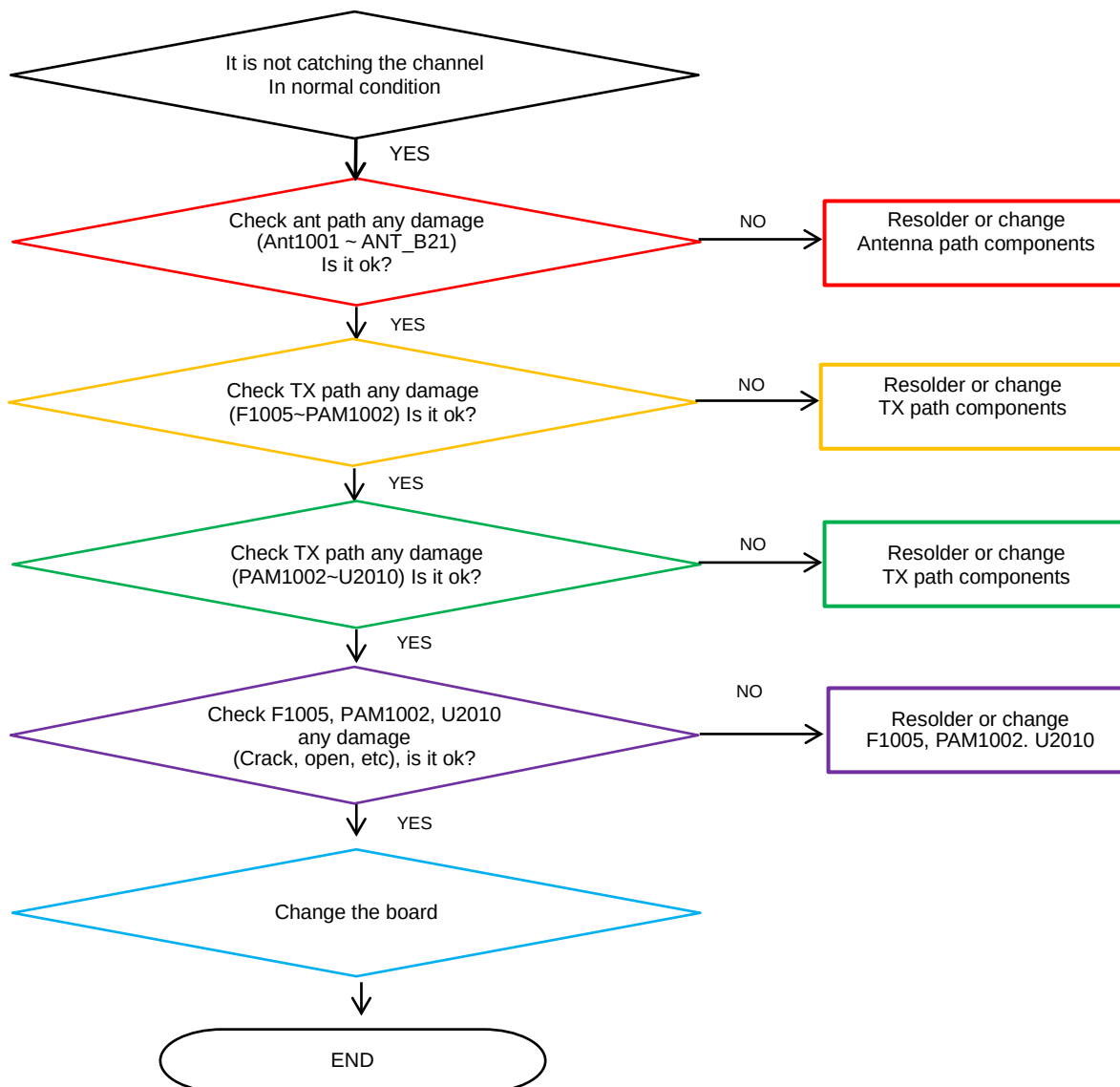
8. Level 3 Repair

8-4-23. GSM DCS/PCS, WCDMA B1, LTE B1/3 Tx



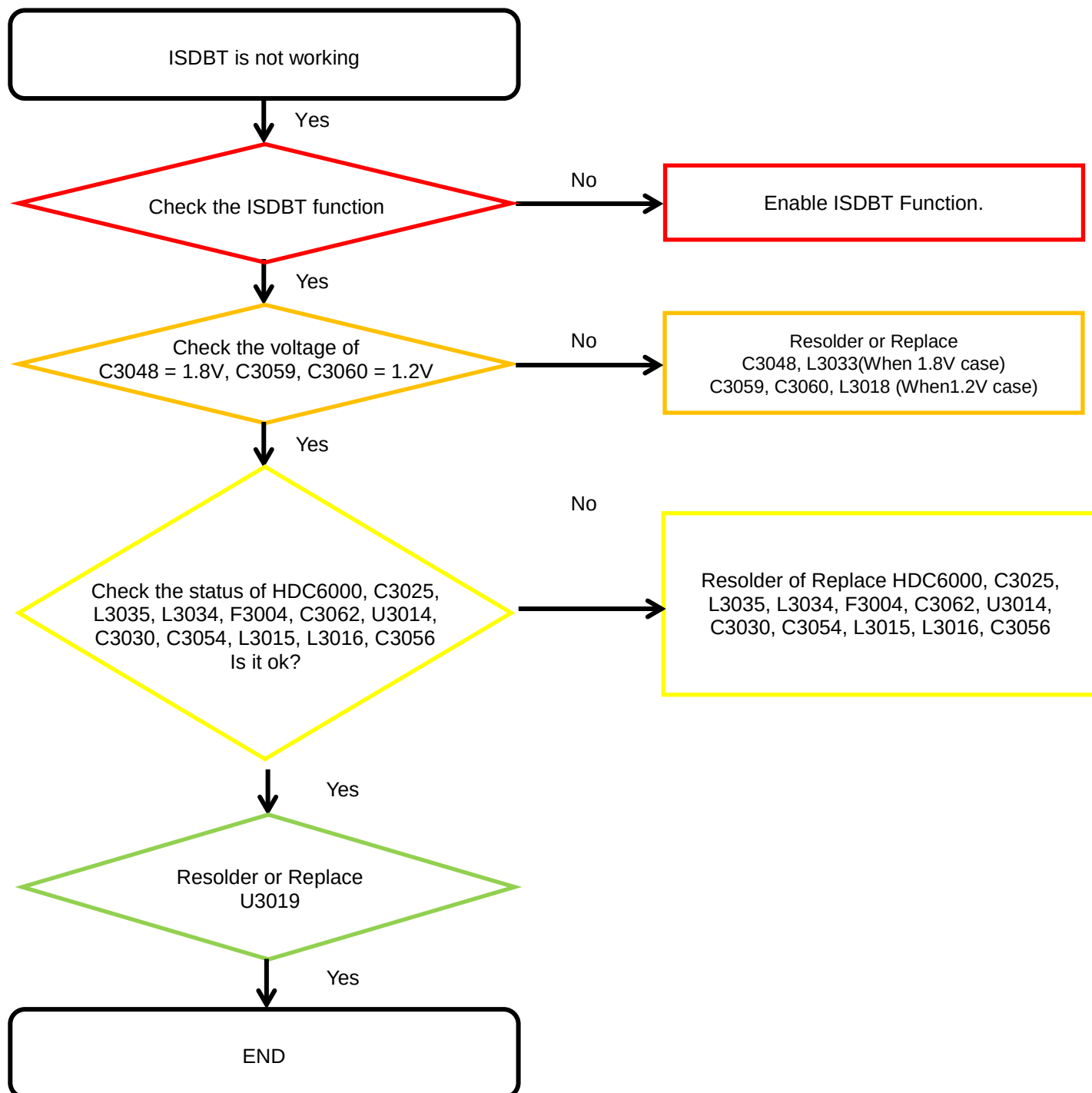
8. Level 3 Repair

8-4-24. LTE B21 TX



8. Level 3 Repair

8-4-25. ISDBT part



9. Reference Abbreviation

Reference Abbreviation

- **AAC**: Advanced Audio Coding.
- **AVC** : Advanced Video Coding.
- **BER** : Bit Error Rate
- **BPSK**: Binary Phase Shift Keying
- **CA** : Conditional Access
- **CDM** : Code Division Multiplexing
- **C/I** : Carrier to Interference
- **DMB** : Digital Multimedia Broadcasting
- **EN** : European Standard
- **ES** : Elementary Stream
- **ETSI**: European Telecommunications Standards Institute
- **MPEG**: Moving Picture Experts Group
- **PN** : Pseudo-random Noise
- **PS** : Pilot Symbol
- **QPSK**: Quadrature Phase Shift Keying
- **RS** : Reed-Solomon
- **SI** : Service Information
- **TDM** : Time Division Multiplexing
- **TS** : Transport Stream

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